GTPA v2.2.1 — core edition and Part H, the human sector

Version 2.2.1, published edition (a correction). Zenodo concept DOI 10.5281/zenodo.17636280; version DOI 10.5281/zenodo.23374852. C. Valdepeñas. Text of 2026-10-14; published 2026-10-14. It supersedes GTPA v2.2 (2026-10-13, 10.5281/zenodo.23352473), which superseded GTPA v2.1 (2026-10-06, 10.5281/zenodo.23195763), which superseded GTPA v2.0 (2026-10-05, 10.5281/zenodo.23172305), which superseded GTPA v1.2 (2025-12-20, 10.5281/zenodo.17994303).

What changes, and why. v2.2.1 corrects v2.2. No result changed in its hypotheses, conclusion or level (Result H.4, which cites without restating what Part H takes from the core, gains two citations in its row on Model 4.5), and no statement is added, removed or renumbered. After v2.2 was published, an assurance pass by independent AI contexts read its record again (see "How it was checked"). It found no false result. It confirmed points in the citations, the summaries, some wording, the notation table and the typesetting, and these are corrected here, each in its place, with the fixes of two reads of the corrections:

- Citations and levels.
  - Model 4.5, the prefactor π of the mean time spent in the well: v2.2 cited Berglund 2013 for it as a standard fact, but Berglund states the Eyring–Kramers law with 2π, for a stop past the saddle. The π form comes from the closed formula by Laplace's method (the maximum of J on $[q^*, q_s]$ is at the end point, so half a Gaussian counts); it is now `PROVED` in the note (AB-5 (c)) · level 3, and Berglund 2013, eqs. (1.8) and (3.15), is cited for the 2π form.
  - References, Berglund 2013: they now say what was checked (its bibliographic record on 2026-10-13; eqs. (1.8) and (3.15) against its arXiv version on 2026-10-14).
  - Model 4.11, the fixed-subspace variant: its heading cites IAC-1 to IAC-3, as Model 4.6 and the References do (v2.2: IAC-2, IAC-3).
  - Model 4.6, composed self-models: the two-worlds sentence now gives level 3 for general N (AC-9 (a)–(c), as the adjudication read it), and it states the separating record for $N = 2$, 3, where it was checked (level 2); v2.2 gave no level for general N.
- Summaries.
  - This header's description of v2.2 (below) and the version note: the wells result holds in a declared one-dimensional gradient well with constant noise, and without noise the least effort to leave is approached but not attained (v2.2 said "a well of the base").
  - This header's description of v2.2 (below): v2.1 had collectives through coherent intents (CT-09 to CT-11) as well as through a joint landscape (v2.2 said "only through a joint landscape").
  - This header's description of v2.2 (below): three formulas of v2.1's PDF were set wrongly, counted by formula as elsewhere (v2.2's header counted four here).
  - The version note: each statement carries its hypotheses and a status, and each result its check level (v2.2 gave every statement a level; definitions and models carry none).
  - Model 4.5, the well bullet: it opens on "a declared well" (v2.2: "a well").
- Wording that could be misread.
  - Model 4.5, the bullet without noise: the infimum $4\lambda_{\text{effort}}\,\Delta J$ is over all paths of all durations together (v2.2: "over paths of every duration", which could be read as at each fixed duration).
  - Model 4.5: U-13 (b)'s hypotheses fail because an equilibrium, q_s, is on ∂D and because D is unbounded (v2.2 named only the first).

  - Model 4.6, two phases: the third phase is a continuous function of time, as in the note.
  - Model 4.11, its head: it declares its target and two levels of information on top of Definition 2.4 and adds no definition or result to §§1–3 (v2.2: "It adds to Definition 2.4 a target" and "it is not part of the core", against the reading rule that the core is §0–§5).
  - Model 4.11, the Scope: φ and the gain g are written out for a fixed subspace and for no restriction.
  - Model 4.11, information over time: the window values are summed without the letter k, which names the action dimension, and $\frac{1}{2}e^{-\varsigma t/8}$ is called half the Bhattacharyya coefficient (v2.2 called it the coefficient).
  - Part H, Result H.4's row on Model 4.5 and Annex H-A row H-26: without noise, leaving a well needs added control, at effort at least $4\lambda_{\text{effort}}\,\Delta J$ on the declared well of Model 4.5 (in the Result H.4 row); with noise, the first-order base leaves that well without control (v2.2 gave only the first half, without the effort). Part H's lists of note IDs now include AB-.
- The notation table (§0.3).
  - The Model 4.5 row: q_s is the barrier top, a strict local maximum of J that the text calls the saddle as the metastability literature does, not a saddle in U-12 (c)'s sense.
  - The Model 4.11 row: Part H's action gain g, S_1, S_2, the proof-local χ of Result H.14 and the dimension p of Model H.10, and Model 4.5's length scale ℓ, are named as other objects; the notations of the research notes are given note by note, with those v2.2 left out, among them the opposite sign of D in IAC- and CCI-, which is immaterial in the variants and the Scope.
  - The τ row: Part H's time change $\tau(t)$ is named as another object.
- Typesetting.
  - `RESEARCH-NOTES-v2.2.1.pdf`: the display of u_{ind} and u_{common}, and AL_F in a table, are set as mathematics (v2.2: code font), and the matrix sizes print $\times$ (v2.2: an italic x).
  - `BLOCK-C-OBJECTS.pdf`: one displayed equation is set as mathematics (v2.1's PDF: code font).
  - The three PDFs built for this record: each bookmark lands on its heading's page (five landed a page early in v2.2's two built PDFs, and two in v2.1's `BLOCK-C-OBJECTS.pdf`); the two block PDFs copied from v2.1 keep eight such bookmarks.
- The deposited notes. Their cover, §0.2, the References and "Proofs" below say that the copies are as the adjudication read them, that one note also omits a closing remark on internal use, and that their status lines, levels and what they say GTPA contains predate the adjudication, so this text governs where it differs. In acoplamiento-entre-dos, a pointer to an essay now names it and says that it is unpublished. Nothing else in the copies changes.
- Two quotations in `BLOCK-C-OBJECTS`. In the question quoted in §3.10 (EG-09.5), "[the author]" replaces the author's first name, a marked editorial substitution, and its opening record no longer quotes the words of a go-ahead; its status block records both, and its PDF is rebuilt.

Two points the pass confirmed in deposited files are left as published: a local folder name in a comment of the Lean file, which stays byte for byte v2.1's, and the shortened condition in the summary sentence of informacion-y-accion-colectiva, whose exact form is in AC-5 (b) of automodelo-colectivo, also deposited, and in Model 4.6.

What v2.2 added (2026-10-13), as v2.2's header described it, with the corrections above. v2.2 is a focused edition. It adds a few exact results on declared models to four worked models of §4, under three themes, and it fixes two omissions. Each addition comes from a research note and was

adjudicated result by result before it entered.

- Information and action (Model 4.11, the value of calibrated information). v2.1 left out a restricted action, a random action coupling and information that arrives over time. Four declared variants now cover them: a fixed action subspace, with the exact part of the value of finer information that it loses and when it loses none; a menu of subspaces chosen on the information; an uncertain coupling, where acting on the mean coupling is not enough; and a hidden mode read through a noisy channel, where, when the mode matters for the action ($\kappa_F > 0$), the value of a new window of observation tends to zero if the mode is fixed and stays above an explicit positive bound if it switches. The Scope bullet now gives the exact net value of the option to observe at a known cost.
- Wells and terrain (Models 4.5 and 4.1). v2.1 said only that leaving a well needs an added control (CT-08 (b)). In a declared one-dimensional gradient well with constant noise ($A = 0$, $G = 1$, $\Sigma = \sqrt{(2\varepsilon)}$), as the noise level ε tends to zero the barrier sets the exponential rate of the mean time spent in the well; the curvature at the bottom sets only the return rate and the variance, and the curvatures enter the mean time only through the prefactor. Without noise, the least effort to leave ($4\lambda_{\text{effort}}\,\Delta J$, approached but not attained) is set by the barrier alone. Redesigning the base, paid once, is set against holding against it, paid at every instant: in a declared model the redesign is cheaper exactly past the horizon $T_{\text{red}} = C_{\text{red}}/c_{\text{hold}}$; and cancelling part of a landscape gives the same motion as removing it.
- Coupling between two and collectives (Model 4.6). v2.1 had collectives through coherent intents (CT-09 to CT-11) and through a joint landscape (CO-10, CO-11). A common order is a fixed subspace of actions, with a witness where it halves the value of information. Calibrated self-models of the members compose into the group's calibrated model exactly when, given all the records, the members' parameters are independent with those laws; otherwise the composition can do worse than no action, and neither the members' models nor the law of their records fixes the group's model. Two coupled phases are a base motion, with the condition to lock, the common rhythm and what a third phase that does not respond can and cannot break. Part H gains only pointers to these bullets: two rows of its sector signature and one row of its annex.
- Two fixes. §0.2 names the ID families of the research notes and says that they are never renumbered (two notes share AC-, so those IDs are cited with their note). The References list every research note that the text uses, among them the VIDA note that §4 cites and v2.1 did not list, and five more cited works.

No result of v2.1 changed. Of its declarations, only the Scope bullet of Model 4.11 is rewritten, for its variants, §0.2's status rule and the opening paragraph now say that some proofs are in research notes, which are deposited with this record, §0.1 and Definition H.1 name Part H as added in v2.1, and two rows of Definition H.3 gain their pointers; §0.3 gains rows for the new symbols and extends its rows on Model 4.11 and on τ. Three formulas that the PDF of v2.1 set wrongly ($M_d\,\ddot{y} = -\nabla V_{\text{eff}} - \Gamma_d\,\dot{y}$ in Model 4.2, $B_i\,u_i$ in Model 4.6, $F_{\text{req},i}$ in Model 4.11's first bullet) are now written so that the PDF sets them as intended; they do not change. Every new result holds for its declared model only.

Base. The base motion is that of UFD v2.0 (2026-10-05, 10.5281/zenodo.23171454), unchanged since GTPA v2.0. This text cites UFD by the IDs of its working nucleus `ufd/UFD-NUCLEO.md`: the convention C1–C6 and the statements U-01 to U-15 are reproduced unchanged in UFD v2.0 (its Sections 3 and 4); "UFD §0", "UFD §8" and "UFD §9" below are UFD v2.0's Sections 2, 7 and 8.

Scope. The core: base, agent, control, efficacy and worked models. Part H: the human sector, a class of declared models of the core read under written mapping assumptions; a typed candidate sector, not admitted (Definition H.3). Nothing here is physics, nothing here is about any person, and nothing here is a claim of the author's claim ledger: admitted ledger records are cited by C-ID, read only, with their own status. No result of the author's mathematical corpus (MATH) is used as support for GTPA.

How it was checked. Check levels, strongest first: (1) formally checked in Lean; (2) the decisive calculation recomputed by machine; (3) read by an independent AI context; (4) written only. Every cited result carries its level (§0.2, §H.0). The text was drafted by AI systems (Claude, by Anthropic) working for the author; the first form of Model 4.11 and the two short additions in §2 were adapted by the theory lead from research notes produced by Codex, and the additions of v2.2 were written for the theory lead from research notes produced by Codex and by Claude.

The core, as in v2.1: the decisive calculations in UFD and in the cited block files were recomputed by machine on 2026-09-26 with 0 disagreements; the remark in Model 4.4 and the expansion in Model 4.11 were recomputed on 2026-10-05 and agree. The items with no machine computation (B-10 (e), (f); B-11 (b), (d); CT-04 (c); the converse in CT-08 (a); CO-02 (c); CO-12 (c); Model 4.11's conditioning step), the readings and the cited facts stay at level 3. The core was read three times by independent AI contexts of the same provider: the text and its declarations (2026-09-26: three real errors, fixed; eight minor points, applied), the additions of 2026-10-05 (one real error, fixed; five minor points, applied) and the v2.0 publication copy (release read of 2026-10-05: one real error, fixed; seven minor points, applied). Since v2.0, the core of Result 2.5 ($Z \succeq \lambda_{\text{effort}} I$ and its invertibility whatever the rank of B, the unique minimizer with its quadratic expansion, and $\langle v, u^* \rangle = 0$ on $\ker B$) is formally checked in Lean (level 1; `GtpaControlLaw.lean`, deposited with this record): 18 theorems, no `sorry`, only Lean's standard axioms, compiled again by an independent context (PASS). The gradient and Newton forms and the bound on $|u^*|$ stay at level 2, and no other statement is at level 1.

The additions of v2.2 (2026-10-13): seven research notes were adjudicated result by result by three independent AI contexts of the same provider, none of which wrote the notes it judged. They gave 61 verdicts (two results were judged under two themes): 46 enter (28 as they stood, 18 with a correction), 14 are left for a later edition, and one is not a mathematical statement and enters no theory text. The contexts recomputed the decisive calculations by machine on 2026-10-13 with their own scripts, not the notes' checkers: over 20,000 symbolic, exact and numeric checks, with no disagreement with any note (level 2 where the text says so). The general arguments marked level 3 in Models 4.1, 4.5 and 4.6, and the conditioning steps, the martingale property and the comparison principle of Model 4.11, were read by those contexts (level 3). They found no false result in the notes; the corrections concern hypotheses, the wording of statements and proofs, notation, references and which part of a result enters.

The text of these additions and the v2.2 publication copy were then read by two more independent AI contexts of the same provider (2026-10-13), which recomputed the decisive calculations with their own scripts, with no disagreement with any number. A mathematics read checked every added or changed statement for truth, complete hypotheses, status, level and notation: four real errors and thirteen minor points. A publication read checked that copy against the living texts and the metadata, and the PDF on its rendered pages: five real errors and twelve minor points. Two real errors were found by both. The well bullet of Model 4.5 opened by saying that the barrier, not the curvature, sets the time spent in a well, which is false at a given noise level (at equal barrier and ε the mean time is proportional to $1/J''(q^*)$); it now says that the barrier sets the exponential rate as $\varepsilon \to 0$. And §0.2's status rule did not cover the proofs that rest on research notes. The others: in the composed self-models of Model 4.6, the shared sign was not said to be independent of the parameters, and the numbers given hold for $F_{\text{req}} = 1$; in two phases, the return of the offset was stated for every locked pair and holds for $|\Delta\omega| < K_1 + K_2$; the header, the version note and the metadata dropped the hypothesis $\kappa_F > 0$; the Spanish summary said things the text does not say; and the PDF set four formulas wrongly, three of them as the PDF of v2.1 did. All are fixed, and each fix that changes a claim was recomputed by machine on 2026-10-13. Of the 25 minor points, 23 are applied, one in part (the notes were said not to be deposited with the record; the theory lead then decided to deposit them, see Proofs below) and one is not (the paths of the notes stay as written, as provenance).

The corrections of v2.2.1 (2026-10-14): after v2.2 was published, independent AI contexts of the same provider read its record again area by area (the mathematics of Models 4.1, 4.5, 4.6 and 4.11, the records, the PDFs, the summaries, the deposited copies and the texts that use GTPA, and three areas

that a further context found unchecked), and one or two more contexts checked each finding, trying to refute it: 75 findings were confirmed and 3 refuted. They found no false result. The corrections listed above are the confirmed findings that concern this text and its files, but for the two left as published, and one side effect of a converter fix for the notes, the display of `BLOCK-C-OBJECTS` set as mathematics; the other findings concern working records and surfaces outside this record. Of the findings corrected here, three were real errors (the citation of the π prefactor, the summaries of the wells result, and the description of v2.1's collectives) and the others minor. To check the corrected citation, the Laplace step of Model 4.5 was recomputed by machine on 2026-10-14 (the half and the whole Gaussian symbolically, and the closed formula by quadrature on two wells, whose ratio to the π form tends to 1; the text keeps level 3 for the step), and Berglund's equations were checked against its arXiv version.

These corrections and this copy were then read by two more independent AI contexts of the same provider (2026-10-14). A mathematics and wording read checked every changed sentence against its finding and recomputed by hand the mathematics the corrections touch: three real errors and eight minor points. A publication read checked this package against the living texts, the sources of the notes and the records: three real errors and fourteen minor points. One real error was found by both, so there were five. Three are in this record: Result H.4's row gave the least effort to leave a well for the first-order base in general, where it holds on Model 4.5's declared well only (with the mobility ½ the reversed path costs $2\lambda_{\mathrm{effort}}\,\Delta J$); §0.3 said that every use of D in Model 4.11 is even in D, which fails in the proof of its first bullet and holds in the variants and the Scope; and this header pointed to an AC-5 (b) in a note that has none. The other two concern working records outside this record. All five are fixed, and the first two were recomputed by machine on 2026-10-14. The minor points that concern this record are applied, among them the separating record of Model 4.6, now stated for $N = 2$, 3, where it was checked, but one in part: the quoted words of the go-ahead are dropped from `BLOCK-C-OBJECTS`, whose PDF is rebuilt, and stay in the two block files whose PDFs are v2.1's, until those are rebuilt. The fixes were re-read once by one more independent AI context of the same provider (2026-10-14), which found no real error and nine minor points, all applied; these nine edits were not re-read.

Part H: the decisive calculations of Results H.6, H.7, H.8 and H.11, the identity of Model H.9, the two equivalences of Definition H.3 and two small computations of Annex H-A (H-36, H-42) were recomputed by machine on 2026-10-05: 33 symbolic and numeric checks, 0 disagreements (level 2). For Result H.14 (2026-10-06): 23 symbolic checks, the exact isolation of g^* at four rates and at one more instance, and 31 interval-certified or exact signs, 0 disagreements (level 2); the argument of its step (1) off the stable set, the estimates of its step (3) and the argument of its step (4) were checked by hand by independent readers (level 3). The dispositions of Annex H-A are a reading (level 4). Part H was read once by an independent AI context of the same provider (2026-10-05): it recomputed the decisive calculations and the annex counts with no disagreement and found four real errors (overclaims or source mismatches) and nine minor points, all fixed. Result H.14 was read once (2026-10-06) in two lenses, its proof and a recomputation from scratch: no disagreement with any number, no real error, and thirteen minor points, twelve applied in the text and the thirteenth (a stale roadmap row) applied outside it. In v2.2 Part H changed only in three pointers to Model 4.6 and in Definition H.1, which now names it Part H of GTPA (added in v2.1), and in v2.2.1 only in two rows that point to Model 4.5 (one of them in Result H.4, which gains two citations) and in its list of note IDs; no result of Part H changed in its hypotheses, conclusion or level.

The release read of the v2.1 publication copy (2026-10-06), by one more independent AI context of the same provider, recomputed the table and the closed forms of Result H.14 with no disagreement and checked by hand the argument of its step (1) off the stable set (now level 3). It found two real errors, both fixed: the PDFs set the formulas as code and dropped or misplaced some bars, dots and tildes over Greek letters (they are now compiled with LaTeX), and a sentence of the Spanish summary, which is not part of this record, misstated the Lean check. It found ten minor points, all applied (one in part: a later pass over the Reino's own pointers is left to the Reino).

None of the fixes of these reads was re-read. Only the core of Result 2.5 is checked in Lean, and no AI system of another provider has checked any of this text.

Proofs. The core's proofs are cited by ID (the short proofs of Model 4.11 are written out in place). They are in three files published with this record: `BLOCK-B-ANSWERS.pdf` (B-nn), `BLOCK-C-THEOREMS.pdf` (CT-nn) and `BLOCK-C-OBJECTS.pdf` (CO-nn). Their texts are unchanged since v2.0, except, in `BLOCK-C-OBJECTS`, the quotation of §3.10, the words of a go-ahead its opening record quoted, and the status line that records them; `BLOCK-C-OBJECTS.pdf` is rebuilt with the build of this record, and the other two PDFs are those of v2.1. The additions of v2.2 marked "`PROVED` in the note" rest on the proofs of the five research notes named with them (revisable notes, four in Spanish and one in English, not part of this text). Public copies of the five are deposited with this record as `RESEARCH-NOTES-v2.2.1.pdf`, as the adjudication read them, omitting their provenance line, their pointers to unpublished working files and, in one note, a closing remark on internal use; their status lines, check levels and what they say GTPA contains are those of the notes before the adjudication (their base is GTPA v2.1), and where this text states a result with other hypotheses, wording or level, this text governs. They are the copies deposited with v2.2, with one pointer to an essay now named as unpublished; the References give each note's path and the commit at which it was read. Every PDF of this record is compiled with LaTeX (Tectonic 0.17.0, XeTeX engine), with the formulas set as mathematics. Part H's proofs are written out in place. The Lean file of the core of Result 2.5, `GtpaControlLaw.lean`, is deposited with this record, unchanged since v2.1; its header names the Lean toolchain and the mathlib revision it was compiled with. The errata annex is `CROSSWALK-v1.2.md`, deposited with this text as `CROSSWALK-v1.2.pdf`, unchanged since v2.1. The names of the author's internal consumers, repository paths and commit hashes in the text are provenance only.

Reading one document. "GTPA v2" in the text names the core (§0–§5), which v2.0 published and v2.1, v2.2 and v2.2.1 keep, v2.2 with the additions above and v2.2.1 with its corrections; where the text says "Part H of GTPA v2.1", it names Part H as v2.1 introduced it, which v2.2 and v2.2.1 keep; where Part H says "GTPA v2.0", it means that core as v2.0 published it. Part H follows §5: its sections are §H.0–§H.6 and Annex H-A, and its statements are numbered H.1–H.14 (Definition H.1, ..., Result H.14); the core's are numbered by section (Definition 2.1, Model 4.7). The errata annex and the references of both parts close the document.

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physics = CLOSED     MATH<->GTPA = UNBOUND
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GTPA v2 states what can be said exactly, under written hypotheses, about an agent that acts on a noisy base motion. The base is the motion of the UFD nucleus. The agent reads the base through a lens, estimates an error, chooses an action by one control law with one effort metric, and adds that action to the base drift with a dial η; at $\eta = 0$ the base is recovered exactly, and the efficacy of the action is measured against that ablation. Everything else is a worked model, declared as such. The proofs live in the block files and are cited by ID (the short proofs of Model 4.11 are written out in place; some additions of 2026-10-13 rest on the proofs of the research notes named with them, deposited with the GTPA v2.2.1 record as `RESEARCH-NOTES-v2.2.1.pdf`); the audit of v1.2, statement by statement, is the errata annex. Part H (after §5) reads the human sector as a class of declared models of this core; its evidence as a description of any person is zero.

0. Scope, reading rules, notation and what is kept

0.1 Scope

GTPA v2 is a generic framework for modelling agency on a declared base motion, with exact local results and no distinctive invariant. This is the lab's standing, reversible classification: the current core is "a generic agency modeling framework, not ... a scientifically distinctive theory" (GTPA-C0075,

ADMITTED, DEF), and the lab "retains zero admitted distinctive current-core invariants" (GTPA-C0086, ADMITTED, DEF). Every result holds for its declared objects and hypotheses only. Nothing here says that any person, mind, organization or device satisfies them, and nothing here is physics. The human sector is Part H of GTPA (added in v2.1; after §5, from UTHA v1.0): a class of declared models of this core, with zero evidence about any person.

0.2 How to read

- Every statement carries one status: `DECLARED` (a convention, definition, model or named hypothesis), `PROVED` (the proof is in the block statement named, written out in place ("here"), or, where the text says "in the note", in the research note named, cited by path and commit in the References and deposited with the GTPA v2.2.1 record in `RESEARCH-NOTES-v2.2.1.pdf` (public copies, four in Spanish and one in English, as the adjudication read them, omitting their provenance line, their pointers to unpublished working files and, in one note, a closing remark on internal use; their status lines, check levels and what they say GTPA contains are those of the notes before the adjudication (their base is GTPA v2.1), and where this text states a result with other hypotheses, wording or level, this text governs); the notes are not part of this text), `STANDARD_FACT` (a textbook fact with the reference given there) or `OPEN`. Admitted ledger records are cited as `ledger` with their C-ID; they keep the ledger's own status and authority.
- Every result also carries its check level: (1) formally checked in Lean; (2) its decisive calculation recomputed by machine (on 2026-09-26, or on the date a later addition states, symbolic or numeric, run once, nothing added to the repository) and agreed; (3) read by an independent AI context only; (4) written only. Only the part of Result 2.5 named there is at level 1 (Lean, since 2026-10-05); no other statement is. Where a result combines `PROVED` and `STANDARD_FACT`, its level is that of the proved part; the cited textbook fact is at level 3.
- IDs: `U-nn` in `ufd/UFD-NUCLEO.md`; `B-nn` in `BLOCK-B-ANSWERS.md`; `CT-nn` in `BLOCK-C-THEOREMS.md`; `CO-nn` in `BLOCK-C-OBJECTS.md`; `D-nn` in `CROSSWALK-v1.2.md` §4; verdict rows (EG-...) in `CROSSWALK-v1.2.md` §1. Results of the revisable research notes in `gtpa-v2/notas/` (not part of this text) are cited by their notes' IDs: V- and C- (asociacion-potencial-y-vida), LA- (libre-albedrio), IM- (informacion-y-margen-de-decision), CONS- (consciencia-y-calibracion-del-propio-mapa), MEM- (memoria-automodelo-y-control), AUTO- (limites-de-autocalibracion), ACT- (aprender-actuando-y-referencia-del-juicio), IAC- (informacion-y-accion-colectiva) and CCI- (conocer-y-cambiar-la-interfaz-de-accion); in `notas/fronteras/`, TH- (termostato), CH- (convergencia-heterogenea) and AC- (automodelo-colectivo); in `notas/reflexiones/`, AB- (atraccion-frente-a-borde), CG- (calibrado-y-ganancia), TER- (disenar-el-terreno), AC- (acoplamiento-entre-dos), NE- (necesidad-referencia-y-escasez), PO- (practica-ocasiones-y-prediccion) and TP- (leer-las-tres-partes). Not every family is cited here; the C-nn of the VIDA note are not ledger C-IDs, which are written GTPA-C0nnn or C0nnn. These IDs are never renumbered. Two notes use AC-, so an AC- ID is always cited with its note.
- A worked model (§4) is declared: its results hold for that model. A counterexample built on a model refutes a universal statement that covers it; a positive result on a model is not a law.
- v1.2 locators (Ax, Def, Thm, Cor, p.) point to where an object comes from; they are not support. The statements of this text are named Definition, Result, Model and Open, with their section number (Definition 3.1 is this text's; Def 3.1 is v1.2's).

0.3 Notation

UFD's notation (UFD §0) is used unchanged. The table fixes every symbol that the source files use in more than one way; the right column says how each clash is resolved.

Symbol	In GTPA v2	Other uses, resolved
$q \in R^n$	base state (U-01)	v1.2's field space of q is not adopted
J	landscape, a function with no unit	—
A	circulating matrix field, $A^\top = -A$ (U-01)	v1.2's X_H, typed on the base, is $A\nabla J$; CT-05's $A(u)$ is written $AL(u)$; the action operator of the lift (C0107's $A_{\pi(\hat{e})}$) is written A_{act}; B-11's second-order lift operator is A_1 (Result 3.3); CT-15's A_π, B_π are scalars local to the compensation equation (Model 4.7)
G	mobility, symmetric $\succeq 0$ (U-01)	G_{reg}, G_{model}, G_1 keep their subscripts; the model metric of Model 4.9 (App E's and CO §1's g) is written h; g is the lift direction (Definition 2.1)
Σ	noise amplitude, $n \times m$ (U-01)	only the noise matrix; sums are written `sum`; C0111's sector signature is named in words
ε	the one noise level of the pairing (P), a constant (U-08)	never a temperature; v1.2's other "epsilon"s are δ, e_q, e_p, e_D, σ_{cut}, e_0 below
$\varepsilon_{\text{cog}}(t)$	declared noise schedule of the agent (Definition 2.9)	v1.2's T_{info} (Control Law 6.1); not UFD's ε
δ	regularizer of a declared $M_{\text{cog}} \coloneqq (H^2 + \delta I)^{1/2}$ (B-07)	v1.2's `epsilon`, `epsilon_inertial` (Ax 0.1, Def 1.1)
e_q, e_p	model-side error components (Ax 5.1)	CT-02's proof writes ε_q, ε_p
e_D	the drift mismatch of v1.2's Def 3.2	v1.2's `epsilon`; CT-01 (c) writes ε
σ_{cut}	threshold of the spectral filter (Result 2.6)	v1.2's `epsilon` (Thm 7.1); CT-05 writes τ, CO-01 writes θ
e_0, K	initial error size and declared tolerance of the horizon estimate (Model 4.8)	v1.2's `epsilon_measure`, K_{scale} (Thm 6.2); D-09 writes δ, K
λ_{effort}	effort weight > 0; the effort metric is $\lambda_{\text{effort}}\lvert u\rvert^2$	CT-05 writes λ; CT-12's λ is the same weight; eigenvalues are written $\lambda_i(\cdot)$; the Lyapunov exponent is λ_L
β	declared positive weight of the informational part of S_{total}	no $k_B \cdot T \cdot \ln 2$ reading (v1.2 Def 4.1); MATH's β enters only as $\varepsilon = 1/\beta$ (Definition 1.6); B-05's β (a rate) is not used
Π	the lens, $\Pi \in C^2(R^n; R^d)$ (B-01)	policies are π; CT-15's set of policies is $\mathcal{P}$; the density $e^{-J/\varepsilon}/Z_J$ is written μ_J (U-08 (b); U-07 (b) writes π)
π, $\hat{e}$	policy; estimated error signal (C0107's `pi`, `e_hat`)	—
B	action coupling $B(q) \in R^{n\times k}$ (v1.2 Ax 7.1; C0107's B_{act})	not MATH's B_{cert}; CO-11's Brownian motion B is local to CO-11
$u \in R^k$	action	U-15's k and CT-12's k are local constants
F_{req}, G_{reg}	required drift correction and weight of the local control problem (Definition 2.4)	CT-05 writes F, G
AL, Z	local control cost and its Hessian $Z = B^\top G_{\text{reg}} B + \lambda_{\text{effort}} I$ (v1.2's `Action_Local`, Def 7.2)	UFD's normalizer $\int e^{-J/\varepsilon}$ is written Z_J; the state space of the N0 slot is written $\mathcal{Z}$
$(\mathcal{Z}, L0, D0, P0)$	the N0 slot (Definition 1.2)	THEORY-STATE and C0107 write Z
η, $\bar{\eta}$	lift dial, $\eta = 0$ the ablation	v1.2's `eta_est`, `eta_real` (gain) and `eta_regen` are other objects; η_{regen} keeps its subscript
Γ_T	signed efficacy at horizon T (Definition 3.1)	C0108's $\Gamma^{\eta\mid 0}_{h,V}$; always with its subscript. Γ_{cog} is the model-side coefficient of Ax 6.1; U-11's γ, Γ are coefficients of (q, p) systems; Model 4.2's damping matrix is Γ_d
V	lower-is-better observable of the efficacy	the potential of v1.2's Thm 5.1 (CT-02..CT-04's V) is written V_{eff}; the potentials of Model 4.5 are V_{infl} and V_{pot}

Symbol	In GTPA v2	Other uses, resolved
M_{cog}	declared symmetric positive definite $d \times d$ matrix field of a model side (Model 4.3)	not a mass and not UFD's curvature (U-12 (e)); U-11's M is a coefficient of (q, p) systems; a bare M is a symmetric positive definite matrix local to Model 4.3's mode equation (CO-07); Model 4.2's second-order flow writes M_d
C_{ij}, $\bar{C}$	pairwise coherence (v1.2 Def 9.1); $\bar{C}$ a declared scalar summary, for instance the mean of the C_{ij}	v1.2's scalar "C" (Thm 9.1, 9.3); B-02's covariance C is written Cov ξ
E_H	error functional of Def 5.1 (v1.2's $E_{\text{Hamiltonian}}$)	E alone is expectation
P_{cat}	short-time efficacy readout (Result 3.4); C0001's P	—
(P)	the pairing hypothesis $\Sigma\Sigma^{\mathsf{T}} = 2\varepsilon G$, $\operatorname{div}_c A = 0$ (U-08)	Model 4.2's gain matrix is P_d; a bare P is local: v1.2's scalar rate <F, v> in Model 4.4, the product Eff · Res in Model 4.6
s, $r(t)$	time parameter of a model-side metric flow (Model 4.9); warm-up schedule (Model 4.8)	CT-12's s (a gain) and $r \coloneqq \theta s$ are local to CT-12
q_s, D, ℓ, J_1; c_{hold}, C_{red}, T_{red}, J_0, J_{extra}	local to the bullets added on 2026-10-13 to Model 4.5 (the barrier top of a declared well, a strict local maximum of J that the text calls the saddle, as the metastability literature does in one dimension, not a saddle in U-12 (c)'s sense; U-13's domain $(-\infty, q_s)$; a length scale and a fixed profile) and to Model 4.1 (the running cost of holding, the one-off cost of a redesign, the horizon at which they tie, a base landscape and an extra part of it)	the source notes write x_s, w, g, κ (atraccion-frente-a-borde) and κ, c, m, a, D, T^* (disenar-el-terreno, with c for q^* and m for $\bar{q}$); Model 4.4's distance D, Model 4.11's difference D and window ℓ, Part H's recoding scale ℓ (Definition H.3), the lift direction g, Result 3.2's w, Model 4.3's κ, the noise dimension m, Model 4.7's rate a, Result 3.3's T^* and the coherences C_{ij}, $\bar{C}$ are other objects
ϑ, ϑ_i, $B_i(\vartheta)$, Y_i, Q_i, Z_Q, ξ, ξ_i, p_{ref}; q_1, q_2, ω_i, $\Delta\omega$, $\bar{\omega}$, K_1, K_2, q_3, K_3, K_3'	local to the bullets of Model 4.6 added on 2026-10-13: the members' parameters and each member's own, the member's block of the coupling, records, calibrated self-models, the Hessian of a model Q, shared or independent reference signs and the probability of $+1$; two phases (base coordinates), their rhythms, the rhythm mismatch, the common rhythm, the gains, a third phase and its pulls on q_2 and q_1	the source notes write θ_i, H_i, F, a, p (automodelo-colectivo) and θ_i, Ω, K, b, b', λ (acoplamiento-entre-dos); Model 4.7's θ, Model 4.8's K, Model 4.10's b, the Hessian H, Model 4.6's total F and bare P, the regularizer δ, the lift direction g, Result 2.2's ξ, Model 4.11's Ω, Model 4.11's channel Y (not the records Y_i), the opportunity function Ω, and Part H's ξ and $\omega = (\omega_+, \omega_-)$ (Result H.14) are other objects

Symbol	In GTPA v2	Other uses, resolved
F, Ω, $P(\cdot)$, D, I_1, I_2; S, P_S, $\lvert\cdot\rvert_Z$, S_j, P_j, γ_j, φ, j^*, j_1, Z_i, $\bar{B}_i$; χ, Y, y_-, y_+, σ_o, W_o, $\bar{W}$, ϱ, $\mathcal{F}_t$, p_t, p_i, w_t, ν, ℓ, F_-, F_+, κ_F, $\mathcal{V}$, c_o; c, g	local to Model 4.11: random target, probability space, probability, difference of actions $u_1 - u_2$, information σ-algebras; a fixed action subspace, its Z-orthogonal projection and the norm of Z, a menu of subspaces with their projections and costs, the menu's value function, a maximizer and the earlier choice, the conditional Hessian and mean of a random B; the hidden mode, the channel, the two levels and the noise size of a channel with known levels, the channel's noise, the innovation, the signal per unit time, the observed history, the posterior and the posterior given I_i (p_i, $i = 1$, 2, not p_t at $t = i$), its variance, the switching rate, the window length, the two values of the target, the mode gain, the value of a window, a cost per window; the cost and the gain of observing (Scope)	Model 4.6's total F, Perelman's F (Model 4.9), the opportunity function Ω (Model 4.10), the bare P of Models 4.4 and 4.6, the distance D of Model 4.4, Model 4.5's domain D and length scale ℓ and the identity I are other objects; so are Model 4.7's interrupt S and p_S, Result 2.6's S, B-03's recoding φ, Model 4.6's perturbation w and Result 3.2's w, Model 4.6's records Y_i, the lift direction g, Model 4.3's κ, the base noise W, U-15's γ (used in Models 4.1 and 4.5), the c of Models 4.4, 4.7 (with c'), 4.8 and 4.10, the measure $d\nu$ of Model 4.9, Model 4.7's ν_{run}, ν_{sh}, and Part H's c and ℓ (Definition H.3) and filter rate κ, profile φ, state z, action gain $g := kr^2$ and S_1, S_2 (Result H.14), the bounded Lipschitz χ local to the proof of Result H.14 and the dimension p of z (Model H.10); ν is the switching rate of Part H's Model H.10. The research notes write, in CG-, S for ϱ, h for ℓ, π for p, κ for κ_F, $\mathcal{G}$ for $\mathcal{V}$, δ for $F_+ - F_-$, $\bar{I}$ for $\bar{W}$, $f(a(\tau))$ for $F_{\chi(\tau)}$, and a mode $a(t) \in \{a_-, a_+\}$ read through $dY = a\,dt + \sigma_o\,dW_o$ (the channel with known levels, with $a_\pm$ for $y_\pm$) for χ; in CCI-, κ_j for γ_j, v_I for u_i, Z_I for Z_i, B_θ for B, and $u^*(I)$ for the menu's optimum $P_{j^*}u_i$; in IAC-, P for P_S and $u^*(I)$ for u_i. IAC- and CCI- write G, λ for G_{reg}, λ_{effort}, and D for $u_2 - u_1$, the opposite sign (every use of D in the variants and the Scope is even in D or is $E[D \mid I_1] = 0$). In IAC- and CCI-, S is the subspace, as here
τ	not used by the core, except U-13's exit time τ_D (Result 1.4, Model 4.5) and the decision time τ local to Model 4.11	v1.2's `tau` (Ax 0.2, Def 1.2, Protocol 11.1) has no referent in v2; Perelman's τ (App E) is not used either; Part H's time change $\tau(t) := \int_0^t x^2$ (§H.0) is another object

0.4 What v2 keeps from v1.2

Editorial decisions of this edition (`DECLARED`). Default: keep what FORMACION, Companion or FÁBRICA CEC use and what the answers repair; drop the physics. Companion's GTPA packs (built from the lab projection P2, 23 admitted claims) and FÁBRICA CEC's `GTPA-OPERATIVA` use the lab spine (among others C0061, C0107, C0108, C0111, C0113), which §2–§3 carry in part (C0107, C0108, C0113; C0061 and C0111 are not restated); FORMACION cites most numbered statements of v1.2. The disposition of each of the 71 numbered v1.2 statements is in the annex (CROSSWALK §3).

v1.2	In GTPA v2	Where
Ch 0–1: the three exported axioms, the master equation, the generator and the field space	the base is UFD's motion (M)/(M_P) on R^n; the pairing (P) replaces the noise relation; Thm 1.1 is replaced by U-04, U-05, U-07; the axioms' physics, the generator and the field space are not carried	§1
Ch 2: the lens, its embedding, anti-blindness, opacity, the boundary, the cubic law	the lens Π and its Itô term are kept; anti-blindness is a declared hypothesis; opacity and the boundary are declared rules; the cubic law is dropped	§2, Models 4.3, 4.10
Ch 3: agency, P_{cat}, the mismatch, the gain, the opportunity function, Ax 3.3	P_{cat} is kept as the short-time efficacy readout, not as the definition of agency; D_{met} is a declared mismatch cost; the gain and the opportunity function are declared; Ax 3.3 is dropped with the physics	§3, Models 4.1, 4.10
Ch 4: information, β, S_{total}, Thm 4.1	β a declared weight; S_{total} a declared model cost; Thm 4.1 replaced by (P) and the schedule $\varepsilon_{\text{cog}}(t)$	§2
Ch 5: error functional, barrier, convexity	a declared model; CT-02 replaces Thm 5.1 and Cor 5.1	Model 4.2
Ch 6: model-side dynamics, Thm 6.1, Control Law 6.1, the horizon, metric learning	Ax 6.1 a declared model with M_{cog} declared; Thm 6.1's identity kept (B-01) and its placement repaired; Control Law 6.1 a declared schedule; the horizon declared; metric learning optional	§2, Models 4.3, 4.8, 4.9

v1.2	In GTPA v2	Where
Ch 7: action coupling, control law, the switch, the shape of the optimum, frequencies	one control law and one effort metric; the switch as a corollary; the shape of the optimum as a model result; frequencies from a declared pair of matrices	§2, Models 4.3, 4.4
Ch 8: the social term	a declared potential; the term is a gradient	Model 4.5
Ch 9: coherence	a declared model	Model 4.6
Ch 10–11: classes, watchdog, start, interrupt	declared models; the topological vacuum is dropped with the physics	Models 4.7, 4.8
App A–G	A and B (the primer and the bit-volume): not carried; C (numerical protocols; its (q, p) equation is another object, U-11): not carried; D: Def D.1 kept; E: optional, repaired; F and G (the cubic law, the cost of prediction): dropped, with their surviving bookkeeping	Models 4.8–4.10, §5

1. The base (N0 = UFD)

Definition 1.1 (The base motion). `DECLARED` (UFD C1–C6; U-01, U-02, U-08). The state space is R^n. The base motion is the Itô SDE

$$(M) \qquad dq = (A - G)\nabla J(q)\,dt + \Sigma(q)\,dW,$$

with A antisymmetric, G symmetric $\succeq 0$, Σ of size $n \times m$ and W a standard m-dimensional Brownian motion, under the standing hypothesis (H): locally Lipschitz coefficients and no explosion (U-01). Under the named pairing hypothesis (P), $\Sigma\Sigma^{\mathsf{T}} = 2\varepsilon G$ with one constant $\varepsilon > 0$, $\operatorname{div}_c A = 0$, and the drift becomes $b_P := (A - G)\nabla J + \varepsilon \operatorname{div} G$: the paired motion (M_P), which equals (M) iff $\operatorname{div} G \equiv 0$ (C4, U-08). Stratonovich input is converted with its correction c_Σ written out (U-02 (b)). Coefficients do not depend on time (C6).

Definition 1.2 (The N0 slot). `DECLARED`, with the status of each row. GTPA's base slot $N0 = (\mathcal{Z}, L0, D0, P0)$ (THEORY-STATE; C0107) is filled by the base motion as follows; the rows are the seven N0 obligations of MR-T1 (result §4).

Obligation	Filled by the UFD base	Status · level
DOMAIN	$\mathcal{Z} = R^n$ (C1); $L0\,f = \langle b, \nabla f\rangle + \tfrac{1}{2}\operatorname{Tr}(\Sigma\Sigma^{\mathsf{T}} D^2 f)$ on $D0 = C^2(R^n)$, with b the drift of (M), or b_P under (P), where $\tfrac{1}{2}\Sigma\Sigma^{\mathsf{T}} = \varepsilon G$ (U-05 (a), U-08)	`DECLARED` + `STANDARD_FACT` · 3
GENERATION_OR_FLOW	pathwise: from every start, the strong solution of (M) or (M_P) under (H), pathwise unique (U-01)	`DECLARED`; a semigroup lane is `OPEN` (Open 1.3)
REALIZATION	$P_t\,f(x) := E\,f(q_t^x)$, $t \geq 0$, for bounded measurable f; Itô's formula links $L0$ and P_t on C^2 under integrability conditions like those U-05 (b) states for J	`DECLARED` + `STANDARD_FACT` · 3
REGULARITY	(H); C^1 where $\operatorname{div} G$ or $\operatorname{div}_c A$ is used and C^2 where a Stratonovich conversion is used (U-01); time-independent coefficients (C6); continuous paths	`DECLARED`
DIMENSIONS	n (state), m (noise); the agent adds d (lens), k (action) and the dimension of $\hat{e}$	`DECLARED`
UNITS	none: no unit or time scale is attached to any symbol (U-01); $(cA, cG, \sqrt{c}\Sigma)$ runs the same motion c times faster (U-13 (c)), so a time unit is a choice of mobility scale; an application declares its units and time scale	`DECLARED`; U-13 (c) `PROVED` · 2
REPRESENTATION_TRANSPORT	a C^3 change of coordinates transports the equation, the Itô term of the change entering the drift (B-01 (a), B-10 (e)); the Itô and Stratonovich readings differ by c_Σ, converted explicitly and never swapped silently (U-02 (b), U-14 (d))	`STANDARD_FACT` + `PROVED` · 2

Open 1.3 (A semigroup lane on R^n). OPEN. No Banach space of observables, core, generation theorem or strong continuity is chosen for the base on R^n. A result that needs a generator identity names its hypothesis instead: (K) in B-10 (b) and B-11 (b), (R) in U-07 (b), a well-posed martingale problem in U-08 (b). The lab's bounded-perturbation results (C0112, C0113) are therefore not applied to the base; the lift operator of Definition 2.1 is unbounded (B-10 (b)).

Result 1.4 (What the base gives). The UFD statements GTPA v2 uses, each with its status and level.

Statement	Content	Status · level
U-03	the circulating part leaves J unchanged	PROVED · 2
U-04	without noise J decreases at the rate $\langle \nabla J, G\nabla J\rangle$ and stops exactly where $G\nabla J = 0$	PROVED · 2
U-05	with noise J has the pathwise Itô rate $LJ = -\langle \nabla J, G\nabla J\rangle + \tfrac{1}{2}\operatorname{Tr}(\Sigma\Sigma^{\mathsf{T}} D^2 J)$, and under the integrability conditions (I1)–(I3) of U-05 (b) the expected rate is $E[LJ]$; the trace term is ≥ 0 where J is convex; Stratonovich input adds $\langle \nabla J, c_\Sigma\rangle$	STANDARD_FACT + PROVED · (c), (d) 2; (a), (b) 3
U-06	from the bottom of a basin $E[J]$ rises, also in the plane with a rotation (K02)	PROVED · 2
U-07	$E[J]$ does not increase when J is bounded below, $E\lvert J(q_0)\rvert < \infty$ and $\tfrac{1}{2}\operatorname{Tr}(\Sigma\Sigma^{\mathsf{T}} D^2 J) \leq \langle \nabla J, G\nabla J\rangle$ everywhere, a condition that fails at every nondegenerate minimum with noise; under (P) and the regularity hypothesis (R), $KL(\rho_t \| \mu_J)$ does not increase	PROVED · (a) 3; (a'), (b) 2
U-08	under (P), $e^{-J/\varepsilon}$ is infinitesimally invariant; with a well-posed martingale problem $\mu_J := e^{-J/\varepsilon}/Z_J$ is invariant; (P) is a named hypothesis, not a law	PROVED + STANDARD_FACT · (a), (c) 2; (b) 3
U-09	the correction $\varepsilon \operatorname{div} G$ is needed: in one dimension, without it or with the Stratonovich term in its place, $e^{-V/\varepsilon}$ is not stationary unless the mobility is constant (K01)	PROVED · 2
U-10	$\lvert\nabla J\rvert^2$, $\lvert G\nabla J\rvert^2$ and $\langle \nabla J, G\nabla J\rangle$ agree up to one factor only for $G = cI$	PROVED · 2
U-11	(q, p) systems are other objects; their small-parameter limit is the first-order motion with $G = \gamma^{-1}$ ($(m\Gamma)^{-1}$ when Γ multiplies p and $M = mI$); in one dimension, with a coefficient that depends on q, the limit carries the correction $\varepsilon G'$	PROVED + STANDARD_FACT · 2
U-12	at a critical point, a positive definite Hessian gives a strict minimum and an indefinite one a saddle; a singular one is inconclusive (x^4 and x^3 at 0); "rigidity" is that curvature and nothing else	PROVED; (e) DECLARED · (d) 2; (a)–(c), (e) 3
U-13	basins and exit times defined; in the gradient subcase, under the hypotheses of Dembo–Zeitouni, $\varepsilon \ln E[\tau_D] \to \Delta J$ as $\varepsilon \to 0$; ΔJ alone does not fix the mean exit time	DECLARED + STANDARD_FACT + PROVED · (b), (c) 2
U-14	an observable is constant on every path iff $LI \equiv 0$ and $\Sigma^{\mathsf{T}}\nabla I \equiv 0$; a silent swap of conventions turns an invariant circle into exponential growth	PROVED · (d) 2; (a)–(c) 3
U-15	the Ornstein–Uhlenbeck toy; with $\sigma^2 = 2\varepsilon\gamma$ its invariant density is $\propto e^{-J/\varepsilon}$	PROVED · (b), (c) 2; (a), (d) 3

Definition 1.5 (What may not be inferred). DECLARED.

- No symbol has a physical reading: J is a function, G a mobility, ε a noise level and not a temperature, the curvature of J is rigidity and not a mass (U-12 (e)); the base has no clock (EG-02.3).
- $E[J]$ does not decrease in general (U-06); what does not increase is stated in U-07, with its hypotheses.
- (P) is a hypothesis that may fail (U-08 (c)); without the correction $\varepsilon \operatorname{div} G$, $e^{-J/\varepsilon}$ is in general not stationary (U-08 (d), U-09); uniqueness of the invariant law and convergence to it are not claimed (U-08 (b); UFD §8).
- A (q, p) system is not a rewrite of the base (U-11 (a)); a model side of second order is a separate

declared object (Model 4.3).

- A time-dependent noise level belongs to the agent layer, not to the base (C6; Definition 2.9).

Definition 1.6 (MATH's torus diffusion). `DECLARED` (a pointer). The passive reversible diffusion on the torus T^N is a standard example of the base form: the case $A = 0$, $G = I$, $\varepsilon = 1/\beta$, read on T^N (`formal-framework/governance/math-owner-binding-record-n0-v1.md`, MB-03, MB-04). That record binds MATH to UFD for this one object only, enables a GTPA N0 binding without performing it, and forbids any other transfer (its F1–F13). Nothing in GTPA v2 rests on it: MATH ↔ GTPA = UNBOUND.

2. The agent on the base

Definition 2.1 (The agent and its lift). `DECLARED`. An agent on the base is given by:

- a lens $\Pi \in C^2(R^n; R^d)$: the estimated error reads the base only through $\Pi(q)$;
- an estimated error $\hat{e}(q)$, a declared C^1 function of $\Pi(q)$ (and of the model state, when the model has one; Model 4.3);
- a policy that maps the estimated error to an action $u \in R^k$ (for the core, the control law of Result 2.5);
- an action coupling $B(q) \in R^{n\times k}$, locally Lipschitz, which places the action in the base drift;
- the lift direction $g(q) \coloneqq B(q)u(q)$ and a dial $\eta \in [0, \bar{\eta}]$.

The lifted motion is

$$(L_\eta) \quad dq = (f0 + \eta g)(q)\,dt + \Sigma(q)\,dW, \qquad L_\eta = L0 + \eta\langle g, \nabla\cdot\rangle, \qquad L_\eta|_{\eta=0} = L0,$$

with $f0$ the drift of (M) or of (M_P) and the standing hypothesis that g is locally Lipschitz and (H) holds for every η (B-10, setting (L)). This is an agency lift in the sense of GTPA-C0107 (ledger, ADMITTED DEF: $L_\eta = L0 + \eta A_{\pi(\hat{e})}$ with exact ablation at $\eta = 0$), in the pathwise lane of Definition 1.2, with $A_{\text{act}} = \langle g, \nabla\cdot\rangle$. It has the composite form $g(x) = B_{\text{act}}(x, \pi(\hat{e}(x)))$ of GTPA-C0115 (ledger, ADMITTED TOY-THM). When B is constant or a function of $\Pi(q)$, the whole action reads the base only through $\Pi(q)$; with any other state-dependent B, the control law of Result 2.5 also reads $B(q)$, which B-10 allows. The loop $\hat{e} \to$ action $\to$ future error is the structural closure of GTPA-C0108 (ledger, ADMITTED DEF); for the continuous base no closure criterion is stated (Open 3.6). A lift may instead act through the paired mobility (Result 3.3).

Same data, same actions. `STANDARD_FACT` · level 3. Let the state that the action reads be declared in full: the base, any model state (Model 4.3) and any random seed the policy uses, with (H) for that joint equation and a single-valued measurable action of the joint state. Two copies with the same joint start, the same coefficients and action rule and the same Brownian path then have indistinguishable states (pathwise uniqueness under (H): U-01; Karatzas–Shreve 1991, Thm 5.2.5) and therefore indistinguishable actions. (H) for the base alone does not give this when the action reads further state. With the same present state but a different future Brownian path the futures may differ; and a changed model state or rule is a change of these data, so it may change the action without contradicting uniqueness. That a declared state contains every cause that acts on the system is a separate hypothesis that (H) does not supply.

Result 2.2 (What the lens sees). `STANDARD_FACT` + `PROVED` · level 2.

- Along the base, $\Pi(q_t)$ has Itô drift $D\Pi\, b + \tfrac{1}{2}\operatorname{Tr}_{\Sigma\Sigma^\top} D^2\Pi$, with $(\operatorname{Tr}_{\Sigma\Sigma^\top} D^2\Pi)^a = \sum_{i,j}(\Sigma\Sigma^\top)_{ij}\,\partial_i\partial_j\Pi^a$; under (P) the second term is $\varepsilon \operatorname{Tr}_G D^2\Pi$. In the Stratonovich form of the image the second-order term does not appear (B-01).

- For a random point ξ with mean m and covariance $\mathrm{Cov}\,\xi$, $E\,\Pi(\xi) - \Pi(m) = \tfrac{1}{2}\,\mathrm{Tr}_{\mathrm{Cov}\,\xi}\,D^2\Pi(m)$ up to a term bounded by the third moment (B-02).
- Under a recoding φ of the model side, the curvature term picks up a term made of the curvature of φ (B-03). What does not depend on the model coordinates: the corrected drift $D\Pi\,b$, exactly; the image of the mean, exactly; the corrected mean, exactly for affine recodings and up to a third-order term for curved ones (B-04).

Definition 2.3 (The Itô term of the lens). DECLARED (EG-02.2, EG-03.7 repairs). $\tfrac{1}{2}\,\mathrm{Tr}_{\Sigma\Sigma^{\mathsf{T}}}\,D^2\Pi$ is called the Itô term of the lens; it is not an Itô–Stratonovich conversion term. A model that tracks the image with an estimate declares what it tracks ($E[\Pi(q_t)]$ or $\Pi(E[q_t])$), and a correction, if used, enters the equation of the image coordinate as a rate (Model 4.3). Where a model needs "anti-blindness" (v1.2 Ax 2.2), it declares that Π is a C^2 diffeomorphism onto its image ($\det D\Pi \neq 0$ can only be written when $d = n$).

Definition 2.4 (The local control problem and the effort metric). DECLARED (v1.2 Ax 7.1, Def 7.1, Def 7.2, typed on the base). Given:

- the required drift correction $F_{\mathrm{req}}(q) \in R^n$, a declared function of the estimated error (for instance the descent direction of a declared error of $\Pi(q)$);
- a symmetric $G_{\mathrm{reg}} \succeq 0$ $(n \times n)$ and a weight $\lambda_{\mathrm{effort}} > 0$;

the local control cost and its Hessian are

$$AL(u) \coloneqq \tfrac{1}{2}\,\langle F_{\mathrm{req}} - Bu,\, G_{\mathrm{reg}}\,(F_{\mathrm{req}} - Bu)\rangle + \tfrac{1}{2}\,\lambda_{\mathrm{effort}}\,|u|^2, \qquad Z \coloneqq B^{\mathsf{T}} G_{\mathrm{reg}} B + \lambda_{\mathrm{effort}}\,I.$$

The effort metric of GTPA v2 is $\lambda_{\mathrm{effort}}|u|^2$ on R^k: one metric, used by the control law, the switch (Result 2.8) and the worked models unless a model declares otherwise. Its variant "effort paid by its size, with a cap" is worked in Model 4.4 (CT-07).

Result 2.5 (The control law). PROVED in CO-01 (a), (b) and CT-05 (b), (c) · level 2; level 1 (Lean, `formal-framework/lean/GtpaControlLaw.lean`) for $Z \succeq \lambda_{\mathrm{effort}}\,I$ and its invertibility whatever the rank of B, the unique minimizer $u^* = Z^{-1}B^{\mathsf{T}}G_{\mathrm{reg}}F_{\mathrm{req}}$ with $AL(u) - AL(u^*) = \tfrac{1}{2}\langle u-u^*, Z(u-u^*)\rangle$, and $\langle v, u^*\rangle = 0$ on $\ker B$, with the objects of Definition 2.4 written as in the text; the gradient and Newton forms and the bound on $|u^*|$ stay at level 2. AL is strictly convex, and its unique minimizer is

$$u^* = Z^{-1}\,B^{\mathsf{T}}\,G_{\mathrm{reg}}\,F_{\mathrm{req}} = Z^{-1}\,(-\nabla AL(0)) = u_0 - Z^{-1}\nabla AL(u_0) \quad \text{for every } u_0.$$

$Z \succeq \lambda_{\mathrm{effort}}\,I$ is never singular, whatever the rank of B; $|u^*| \le |B^{\mathsf{T}}G_{\mathrm{reg}}F_{\mathrm{req}}|/\lambda_{\mathrm{effort}}$; and $\langle v, u^*\rangle = 0$ for every $v \in \ker B$: the minimizer spends nothing in directions where the action has no effect. The action of the core is $u(q) \coloneqq u^*(q)$, and its lift direction is $g = Bu^*$ (Definition 2.1).

Result 2.5 minimizes AL for a given F_{req}; it does not choose F_{req}. A policy whose law updates its demand is compared with the observable, horizon, start and ablation of Definition 3.1, fixed in advance; a lower AL under a changed demand is not, by itself, a correction in that comparison (Model 4.7: if $a > 0$, from an open set of starts with $x_0 \neq 0$ and $k(s_0\theta_0)^2 < a$, Class A drives $E_{\mathrm{model}} \to 0$ while $E_{\mathrm{base}} \to \infty$). DECLARED (a reading of Definitions 2.4 and 3.1; no new result).

Result 2.6 (Computing the law). PROVED in CO-01 (d), CT-05 (d), (e) · level 2. If $Z = USV^{\mathsf{T}}$ is a singular value decomposition, $VS^{-1}U^{\mathsf{T}} = Z^{-1}$, and every singular value of Z is at least $\lambda_{\mathrm{effort}}$. A spectral filter that inverts only the singular values above a threshold σ_{cut} returns u^* whenever $\sigma_{\mathrm{cut}} < \lambda_{\mathrm{effort}}$; otherwise it returns a truncation u_{cut} with $AL(u_{\mathrm{cut}}) - AL(u^*) = \tfrac{1}{2}\langle u_{\mathrm{cut}} - u^*, Z(u_{\mathrm{cut}} - u^*)\rangle$, which is > 0 unless the discarded coefficients vanish. The filter is a named truncation, not the control law. (With $\lambda_{\mathrm{effort}} = 0$, outside the core, the minimizers solve $Zu = B^{\mathsf{T}}G_{\mathrm{reg}}F_{\mathrm{req}}$ and the least-norm one is $Z^{+}B^{\mathsf{T}}G_{\mathrm{reg}}F_{\mathrm{req}}$.)

Result 2.7 (Holding a point). PROVED in CT-01 (a) · level 2. With $B = I$, the control that holds $\bar{q}$ fixed against the noiseless base drift is $u_{\mathrm{hold}} = (G - A)\nabla J(\bar{q})$. If $G(\bar{q}) \succ 0$, it vanishes iff $\nabla J(\bar{q}) = 0$, and its

component along ∇J is $\langle \nabla J, G\nabla J\rangle$ whatever A. So, for every cost that is positive off $u = 0$ (the effort metric included), holding costs exactly at the points that are not critical points of J.

Result 2.8 (The switch between acting and inferring is a corollary). PROVED in CO-08 · level 2. Declare the running cost of a mode path $m(t) \in \{\text{act}, \text{inf}\}$ as $c_{\text{act}}(t)$ in `act` and $\beta\, c_{\text{inf}}(t)$ in inf, with $c_{\text{act}} \coloneqq \lambda_{\text{effort}}|u^*|^2$ (the effort metric) and c_{inf} a declared inference cost (v1.2: $\|dp_{\text{est}}/dt\|^2$, for a model side with p_{est}, which is not the erasure cost of Ax 4.2). Under hypothesis (X), that neither cost depends on earlier choices, the rule "act iff $c_{\text{act}}(t) < \beta\, c_{\text{inf}}(t)$" minimizes the total cost, and every minimizer agrees with it at almost every t where the two costs differ; β enters once, on the inference side (CO-08 (a), (b)). Without (X) the rule need not minimize (CO-08 (c)); it is then a declared one-step policy. The hysteresis clause of v1.2's Ax 7.2 is not stated (EG-05.c stays a lab HOLD; C0137 is reserved for it).

Definition 2.9 (Costs and the agent's own noise). DECLARED, with B-09 PROVED · level 2.

- S_{total} (v1.2 Ax 4.2) is the declared running cost of a model over a declared horizon, with $\beta > 0$ a declared weight; v1.2 writes it two ways (pp.27, 30), and each model says which it uses.
- A time-dependent noise level of the agent (v1.2's T_{info}, Control Law 6.1) is a declared schedule $\varepsilon_{\text{cog}}(t)$, with the pairing imposed at each instant as a constraint. No equilibrium statement is drawn from it: in the one-dimensional paired example the variance equals $\varepsilon(t)$, that of the instantaneous law $N(0, \varepsilon(t))$, at every time iff ε is constant, and it lags behind otherwise (B-09).

3. Efficacy

Definition 3.1 (Signed efficacy). DECLARED (GTPA-C0108, ledger, ADMITTED DEF). For an observable V fixed in advance, lower-is-better, a horizon T and a start x_0,

$$\Gamma_T \coloneqq E\,V(x_T^0) - E\,V(x_T^\eta),$$

where x^η solves (L_η) and x^0 the base. $\Gamma_T > 0$ is correction, $= 0$ a tie and < 0 anticorrection, relative to V, T, x_0 and the $\eta = 0$ ablation. Closure, correction, competence and failure are distinct properties (C0108); $\eta > 0$ does not imply correction (GTPA-C0109, ledger, ADMITTED TOY-THM, finite witnesses).

Result 3.2 (The efficacy of a drift lift, with noise). PROVED in B-10 · level 2 for (a)–(d) and (g); (e) and (f) level 3 (no machine computation). In the setting of Definition 2.1:

- (a) at $\eta = 0$ the lifted motion is the base: the ablation is exact;
- (b) under the named hypothesis (K) (the function $w(t,x) \coloneqq E\,V(x_t^{0,x})$, written u in B-10, is $C^{1,2}$ and solves the backward Kolmogorov equation, with the integrability stated in B-10), $\Gamma_T = -\eta\, E\int_0^T \langle g, \nabla w(T-s,\cdot)\rangle(x_s^\eta)\, ds$;
- (c) for globally Lipschitz $f0$, Σ (constant L), g globally Lipschitz with $|g| \le K_g$, and V Lipschitz (constant L_V): $|\Gamma_T| \le \sqrt{3}\, L_V\, K_g\, \eta\, T\, \exp(3L^2(T+1)T/2)$; with (K) for every η and a bounded continuous $\langle g, \nabla w\rangle$, the first variation $\lim_{\eta\to 0} \Gamma_T/\eta$ is the integral stated in B-10 (c);
- (d) if $L0V$ and $\langle g, \nabla V\rangle$ are bounded and continuous and the martingale terms are square integrable, $\lim_{T\to 0} \Gamma_T/T = -\eta\langle g(x_0), \nabla V(x_0)\rangle$: the same limit as without noise, because the trace term is common to both arms;
- (e) Γ_T is the same number in every C^3 system of coordinates, with the lift direction transported as a vector;
- (f) Stratonovich input with an η-independent Σ converts with a term common to both arms, so the ablation stays exact;

- (g) no compact tube can be assumed in general: for Brownian motion ($f0 = g = 0$, $n = m = 1$, $\Sigma = \sigma > 0$) the paths leave every compact set with positive probability; noise that vanishes at the boundary can keep them in one.

This is the stochastic counterpart of GTPA-C0115 (ledger, ADMITTED TOY-THM; deterministic, on a compact tube) and a Duhamel formula with an unbounded perturbation, outside the bounded class of GTPA-C0112 (ledger, ADMITTED TOY-THM). It is not a ledger claim.

Result 3.3 (A lift through the paired mobility). PROVED in B-11 · level 2 for (a), (c) and (e)–(g); (b) and (d) level 3 (conditional, no machine computation). If the action changes the mobility, $G_\eta \coloneqq G + \eta G_1 \succeq 0$, and the pairing is kept in both arms, the noise changes with it.

- (a) The lift operator is of second order: $A_1 f = \langle -G_1 \nabla J + \varepsilon \operatorname{div} G_1, \nabla f\rangle + \varepsilon \operatorname{Tr}(G_1 D^2 f) = \varepsilon e^{J/\varepsilon} \operatorname{div}(e^{-J/\varepsilon} G_1 \nabla f)$.
- (b) The representation and the short-time limit of Result 3.2 hold with A_1 in place of $\langle g, \nabla\cdot\rangle$; the first variation holds when the arms converge as $\eta \to 0$, which is assumed, not proved.
- (c) Near a critical point x_* of J with $D^2J(x_*) \succ 0$ and $G_1(x_*) \succ 0$, for $V = J$ the short-time rate $-\eta A_1 J$ is negative: the lift hurts at first. The same change of mobility without noise has short-time rate $\eta\langle \nabla J, G_1 \nabla J\rangle > 0$ for $x \neq x_*$ nearby.
- (d) $e^{-J/\varepsilon}$ is infinitesimally invariant for every arm; if both arms converge in law to μ_J (not proved here), $\Gamma_T \to 0$ for every bounded continuous V.
- (e) A drift lift on a paired base keeps $e^{-J/\varepsilon}$ infinitesimally invariant iff $\operatorname{div}(e^{-J/\varepsilon} g) \equiv 0$.
- (f) In the explicit Ornstein–Uhlenbeck pair, $V = J$ gives $\Gamma_T = \tfrac{1}{2}(x_0^2 - \varepsilon)(e^{-2T} - e^{-2(1+\eta)T})$, with the sign of $x_0^2 - \varepsilon$ at every horizon; $V(x) = x^2 - x/2$, $x_0 = 1$, $\varepsilon = \tfrac{1}{2}$, $\eta = 1$ gives $\Gamma_T > 0$ before $T^* = \ln((1+\sqrt{5})/2)$ and $\Gamma_T < 0$ after.
- (g) A uniform paired speed-up only rescales time: $\Gamma_T = E\,V(x_T^0) - E\,V(x_{(1+\eta)T}^0)$.

Result 3.4 (P_{cat} is the short-time efficacy rate). PROVED in B-10 (d) · level 2; the deterministic identity is GTPA-C0110 (ledger, ADMITTED TOY-THM). For $V = J$ under the conditions of B-10 (d), and the lift $g = Bu$ at $\eta = 1$,

$$\lim_{T\to 0} \Gamma_T/T = -\langle \nabla J(x_0), B(x_0)\,u(x_0)\rangle = P_{\text{cat}}(x_0),$$

the readout of v1.2's Def 3.1, kept as GTPA-C0001's local definition (ledger, ADMITTED DEF). With noise the trace term cancels between the arms. P_{cat} is a local readout, not the definition of agency: v1.2's Ax 3.1 is dropped as a definition (GTPA-C0068, ledger, REJECTED as a core invariant; C0108 separates closure from efficacy). It is zero on every purely informational action (CO-12 (a)), and its sign does not fix the sign at later horizons (Result 3.5).

Result 3.5 (What one sign does not decide). Ledger and PROVED · level 2. The initial sign of the efficacy does not fix its sign at later horizons: in finite chains (GTPA-C0113), in the deterministic nonlinear class (GTPA-C0116), both ledger, ADMITTED TOY-THM, and on the paired base (B-11 (f), level 2). Competence over a declared set of observables is a quantified condition: for finite terminal laws, $\Gamma_T \geq 0$ for every V in a declared cone iff $p^0 - p^\eta$ lies in its dual cone (GTPA-C0119, ledger, ADMITTED TOY-THM).

Open 3.6 (Closure on the continuous base). OPEN. C0108 defines structural closure by two aligned legs, estimated error to action and action to the future support of the error. An exact criterion exists for finite channels (GTPA-C0118, ledger, ADMITTED TOY-THM); for the continuous base of GTPA v2 none is stated.

The comparison of two active policies against a common ablation is GTPA-C0114 (ledger, ADMITTED TOY-THM). The lab's further layers (representation descent, typed presentations and

strata, identification, transport certificates: GTPA-C0121–C0125 and C0127–C0135, ADMITTED) are cited by C-ID and not restated here; GTPA-C0126 is an empirical hypothesis on HOLD with zero evidence.

4. Worked models

Each model below is DECLARED; its results hold for that model. Every cited result keeps the hypotheses of the statement cited; the main ones are repeated here.

Model 4.1 (Holding against the base drift; a declared mismatch cost). Declared model; v1.2 Ch 3. Noiseless base with $B = I$: $\dot{q} = (A - G)\nabla J(q) + u$.

- With the declared cost $\frac{1}{2}\langle u, G^{-1}u\rangle$ and G constant $\succ 0$, holding $\bar{q}$ costs $\frac{1}{2}\langle \nabla J, G\nabla J\rangle(\bar{q}) + \frac{1}{2}\langle A\nabla J, G^{-1}A\nabla J\rangle(\bar{q})$; with $A = 0$ and J quadratic with Hessian H, it is a multiple of $|\bar{q} - q^*|^2$ for every $\bar{q}$ iff $HGH = 2kI$. PROVED in CT-01 (b) · level 2.
- D_{met} (v1.2 Thm 3.1) is a declared mismatch cost, on declared spaces and with a declared distance; until these are declared it is a label (GTPA-C0005, ledger, ADMITTED MET). A mismatch e_D built from the circulating part alone (v1.2 Def 3.2) can be 0 while holding costs. PROVED in CT-01 (c) · level 2.
- Redesign or hold (added 2026-10-13 from a research note, TER-1, TER-2, TER-3 without discount, TER-4 (a)). A declared variant with noise: $n = 1$, $A = 0$, $G = 1$, the pairing (P), $J(q) = \frac{1}{2}H(q - q^*)^2$ with $H > 0$ (U-15 with $\gamma = 1$, $k = H$), a target $\bar{q} \neq q^*$ and a horizon $T > 0$. To hold, act with $B = 1$ and the constant $u_{\text{hold}} = H(\bar{q} - q^*)$ of Result 2.7 at $\bar{q}$, at the running cost $c_{\text{hold}} := \lambda_{\text{effort}}\, u_{\text{hold}}^2$; to redesign, replace J by $\frac{1}{2}H(q - \bar{q})^2$ at a declared one-off cost $C_{\text{red}} > 0$ and act with $u \equiv 0$. DECLARED. Both have the drift $-H(q - \bar{q})$: the same paths for the same start and Brownian path, the same law and the invariant law $N(\bar{q}, \varepsilon/H)$, so only the costs differ. No adapted action u_t with $E\int_0^T u_t^2\, dt < \infty$ that keeps $E\, q_t$ on the mean of the redesigned motion over $[0, T]$ has effort $\lambda_{\text{effort}}\, E\int_0^T u_t^2\, dt$ less than $c_{\text{hold}}\, T$, with equality iff $u_t = u_{\text{hold}}$ a.s. for almost every t; so the redesign is strictly cheaper than every such action iff $T > T_{\text{red}} := C_{\text{red}}/c_{\text{hold}}$ (a tie at T_{red}), whatever ε and the start (with $E|q_0| < \infty$). Efficacy credits only the action: with $V(q) = (q - \bar{q})^2$ and $q_0 = \bar{q}$, holding ($\eta = 1$) has $\Gamma_T = (\bar{q} - q^*)^2(1 - e^{-HT})^2 > 0$ against the base J, while the redesign changes the base itself and has $\Gamma_T = 0$ against its own ablation. PROVED in the note (TER-2, TER-3, TER-4 (a)) · level 2 (the drift, mean, law, cost and Γ_T identities recomputed by machine on 2026-10-13; the step from the equation to the equation of $E\, q_t$, and Jensen's inequality, at level 3). A redesign has no ablation in this text: Definition 3.1 measures an action against a fixed base. DECLARED (a reading).
- In general, cancelling part of the landscape and removing it are the same motion: on (M_P) with $J = J_0 + J_{\text{extra}}$ ($J_0, J_{\text{extra}} \in C^2$, ∇J_{extra} locally Lipschitz, (H) for the base J_0) and $B = I$, the action $u = (G - A)\nabla J_{\text{extra}}$, the holding control of Result 2.7 written for J_{extra} at every point, gives exactly the paired drift of the base J_0, because ε div G does not depend on J. The action is paid at every instant along the path; removing J_{extra} is paid once. PROVED in the note, TER-7 (a) · level 2 (recomputed by machine on 2026-10-13).

Model 4.2 (A model state with a safety margin). Declared model; v1.2 Ch 5. Model state $y = (q_{\text{est}}, p_{\text{est}}) \in R^{2d}$ with a declared target y^* (v1.2: $y^* = (\Pi(q), D\Pi(q)p)$ for a territory state held fixed); $(e_q, e_p) := y^* - y$ (Ax 5.1); $E_H(y) := \frac{1}{2}(\langle e_q, G_{\text{model}}\, e_q\rangle + \langle e_p, M_{\text{cog}}^{-1} e_p\rangle)$ with $G_{\text{model}}, M_{\text{cog}}$ constant (Def 5.1); a margin $O \in C^2(U)$, $O > 0$, on a nonempty open convex safe zone U, with $O \to 0$ at its edge; $V_{\text{eff}} := E_H - \mu \ln O$, $\mu > 0$ (Def 5.2, Regime B).

- One attractor: if $G_{\text{model}} \succ 0$, $M_{\text{cog}} \succ 0$ and $-\ln O$ is convex on U (for instance O concave), V_{eff} has exactly one critical point, its global minimizer, which attracts every trajectory of $\dot{y} = -P_d \nabla V_{\text{eff}}$

and of $M_d\,\ddot{y} = -\nabla V_{\text{eff}} - \Gamma_d\,\dot{y}$ started in U (P_d, M_d constant symmetric positive definite; Γ_d constant with $\Gamma_d + \Gamma_d^{\mathsf{T}} \succ 0$). On a bounded convex U, the distance to the edge gives exactly one minimizer (the flows need a C^1 gradient, which it need not have). PROVED in CT-02 · level 2.

- A warning case: a convex zone with the margin $O = (1 - q_{\text{est}}^2)\exp(5q_{\text{est}}^2/2)$, which is not log-concave, gives two minimizers $(\pm 1/\sqrt{2}, 0)$ and a saddle. PROVED in CT-03 · level 2.
- v1.2's three conditions are not necessary: one attractor also arises with a singular G_{model} or on a non-convex zone (the distance to a closed quadrant); Def 1.1's matrix is always positive definite, so the second condition separates nothing. PROVED in CT-04 (a), (b) · level 2; (c) · level 3.

Model 4.3 (A model side of second order). Declared model; v1.2 Ch 6–7. Ax 6.1 as a declared model-side system, $\dot{q}_{\text{est}} = M_{\text{cog}}^{-1}\,p_{\text{est}}$, $\dot{p}_{\text{est}} = F_{\text{cog}} - \Gamma_{\text{cog}}\,p_{\text{est}} + \kappa$, with a declared symmetric positive definite M_{cog} and declared F_{cog}, Γ_{cog}, κ. The model carries a second coordinate by declaration (v1.2 Ax 2.1; its symplectic embedding is not adopted); the system is a separate object from the base (U-11). M_{cog} is not a mass and not UFD's curvature. Def 1.1's $(H^2 + \delta I)^{1/2}$ is one declared choice: always invertible, blind to the sign of H, and neither a function of the spectrum of H nor the other way round. DECLARED; B-07 PROVED · level 2.

- Where a lens correction goes: a constant κ in the p_{est} equation moves q_{est} by $\tfrac{1}{2}M_{\text{cog}}^{-1}\kappa t^2 + O(t^3)$; set against a constant rate bias of q_{est}, it can offset it only in the long run (for instance when every eigenvalue of Γ_{cog} has positive real part, B-05), and the choice $\kappa = -(\text{bias})$ does so for every bias only when $\Gamma_{\text{cog}}\,M_{\text{cog}} = I$; in the q_{est} equation it acts exactly, as κt. Witness: flat J, the pairing and $\Pi(q) = (q_1, q_2 + q_1^2/2)$; the perceived mean leaves the image of the mean at the constant rate $(0, \varepsilon)$. PROVED in B-05, B-06 · level 2.
- Mode frequencies: the frequencies of $M\ddot{x} + Hx = 0$ are $\text{Spec}(M, H)$, the square roots of the eigenvalues of $M^{-1}H$; they are invariant under congruence and depend on both matrices. For Ax 6.1 with constant coefficients and $F_{\text{cog}} = -\nabla(\beta E_H)$, the second matrix is βG_{model} if ∇ is the differential in q_{est}, and βI if it is the G_{model}-gradient. When the model and the base have different dimensions, "equal frequencies" needs a declared relation between the two multisets. PROVED in CO-07 · level 2.
- Frequency response: for a linear second-order mode driven by a periodic input, the ratio of the input to the response of the rate has its smallest modulus at the mode's own frequency, equal to the coefficient of the first derivative (zero only if that coefficient is); the first-order base, linearized at a nondegenerate minimum, has no preferred positive frequency unless a circulating part is present; Def 7.2's $Z \succeq \lambda_{\text{effort}}\,I$ is a fixed matrix, not a frequency response. PROVED in B-08 · level 2.

Model 4.4 (A move with zero rate at both ends). Declared model; v1.2 §7.4. A second-order model, $\ddot{q} = u$ on $[0, T]$, from $q = \dot{q} = 0$ to $q = D > 0$, $\dot{q} = 0$.

- Quadratic effort: with cost $\int(\tfrac{1}{2}u^2 + \tfrac{1}{2}c\,\dot{q}^2)$, $c \geq 0$ (the effort metric with $\lambda_{\text{effort}} = 1$, plus an optional rate term), the unique minimizer vanishes at most at one instant, so no interval has $u = 0$; for $c = 0$, $u^*(t) = (6D/T^2)(1 - 2t/T)$. PROVED in CT-06 · level 2.
- Effort paid by its size, with a cap (the variant of Definition 2.4): with cost $\int |u|$ and $|u| \leq u_{\max}$, an admissible control exists iff $T \geq T_{\min} = 2(D/u_{\max})^{1/2}$; for $T > T_{\min}$ the unique minimizer is push $(+u_{\max})$, zero, reverse $(-u_{\max})$, with a zero phase of length $(T^2 - 4D/u_{\max})^{1/2}$. This is v1.2's three-phase shape, under conditions that suffice for it; they are not necessary (Remark below). PROVED in CT-07 · level 2. Remark: with no cap, $T = 4$, $D = 1$ and cost $\int(|u| + 0.05u^2)$, the unique minimizer pushes on $(0, t_1)$, holds $u = 0$ on $(t_1, 4 - t_1)$ and brakes on $(4 - t_1, 4)$, with $t_1 \approx 0.30$: the zero phase covers about 85% of the horizon (cost ≈ 0.557). PROVED in the remark after CT-07 · level 2.
- Recovery: $\eta_{\text{regen}} \in [0, 1]$ is a declared parameter of the running cost of any agent; for a declared

scalar rate P (v1.2's `<F, v>`) the cost is $P^+ - \eta_{\text{regen}}\, P^-$, and no barrier enters it. `DECLARED`; CO-09 (a), (b) `PROVED` · level 2.

Model 4.5 (A social term). Declared model; v1.2 Ch 8. A declared influence potential V_{infl} (Def 8.1, singular at the leader's point) and the social term $-\nabla V_{\text{infl}}$; a deformed metric (Ax 8.1's G_{eff}) would be a separate declared model.

- The gradient field of a C^2 potential has zero curl: `PROVED` in CT-08 (a) · level 2. On a star-shaped open set, a C^1 field with zero curl is a gradient (the Poincaré lemma): `STANDARD_FACT` in CT-08 (a) · level 3.
- For a declared potential $V_{\text{pot}} \in C^2(R^n)$ and $\ddot{x} = -\nabla V_{\text{pot}} - \zeta \dot{x}$, $\zeta > 0$, the function $\Psi = \tfrac{1}{2}|\dot{x}|^2 + V_{\text{pot}}$ decreases at rate $\zeta|\dot{x}|^2$: a trajectory that starts in $\{\Psi < h\}$ stays there, so leaving a well needs an added control. On the first-order base, J does not increase without noise (U-04). `PROVED` in CT-08 (b) · level 2.
- With noise, a declared well of the first-order base is left without control, and, as $\varepsilon \to 0$, the exponential rate of the time spent in it is set by its barrier; its curvature enters only the prefactor (added 2026-10-13 from a research note, AB-1, AB-2 (b), AB-3 to AB-7). Declared well: the gradient subcase of U-13 (b) with $n = 1$ ($A = 0$, $G = 1$, $\Sigma = \sqrt{(2\varepsilon)}$); $J \in C^2$ with (H); $q^* < q_s$ with $J' < 0$ on $(-\infty, q^*)$, $J' > 0$ on (q^*, q_s), $J'(q_s) = 0$, $J''(q^*) > 0 > J''(q_s)$ and, for every $\varepsilon > 0$, $\int_{-\infty}^{q^*} e^{-J/\varepsilon} < \infty$ and $\int_{-\infty}^{q_s} e^{J(y)/\varepsilon} \int_{-\infty}^{y} e^{-J(z)/\varepsilon}\, dz\, dy < \infty$ (AB-1); barrier $\Delta J := J(q_s) - J(q^*)$; $D := (-\infty, q_s)$ and τ_D as in U-13 (a). `DECLARED`. The curvature $J''(q^*)$, the rigidity of U-12 (e), sets the return time $1/J''(q^*)$ of the mean and the invariant variance $\varepsilon/J''(q^*)$ of the linearization at q^*, U-15's toy with $\gamma = 1$, $k = J''(q^*)$ (`PROVED` in U-15 (b), (c) · level 2). The barrier sets the exponent: $\varepsilon \ln E_q[\tau_D] \to \Delta J$ as $\varepsilon \to 0$, for every $q < q_s$ (`PROVED` in the note, AB-5 (b) · level 3, from the closed formula of $E_q[\tau_D]$, $1/\varepsilon$ times that double integral taken from q: `STANDARD_FACT`, Karlin–Taylor 1981, ch. 15 · level 3; U-13 (b) gives this limit under its own hypotheses, which an equilibrium on ∂D, here q_s, and the unbounded D do not meet). The curvatures enter only the prefactor: $E_{q^*}[\tau_D] = \pi\, (J''(q^*)|J''(q_s)|)^{-1/2}\, e^{\Delta J/\varepsilon}\, (1 + o(1))$ (Laplace's method on that closed formula: the maximum of J on $[q^*, q_s]$ is at the end point q_s, so only half a Gaussian counts; `PROVED` in the note, AB-5 (c) · level 3, its numbers for the profile $(1 - v^2)^2$ of the witness below · level 2. The Eyring–Kramers law, Berglund 2013, eqs. (1.8) and (3.15), has 2π in place of π because it stops past the saddle: `STANDARD_FACT` · level 3). So, of two such wells with different barriers, the mean time spent in the higher one divided by that in the other tends to infinity as $\varepsilon \to 0$, whatever the curvatures; the curvatures decide only between equal barriers (`PROVED` in the note, AB-7 · level 3). Within one shape, $J(q) = \Delta J \cdot J_1(q/\ell)$ with $\ell > 0$ and a fixed profile J_1 with barrier 1 that satisfies the well's hypotheses, $J''(q^*) \cdot E_{q^*}[\tau_D]$ is a function of $\Delta J/\varepsilon$ alone: at equal barrier and ε, the mean time spent is exactly proportional to $1/J''(q^*)$ (`PROVED` in the note, AB-4 · level 2). Witness, $J_1(v) = (1 - v^2)^2$: the well with $J''(q^*) = 32$, $\Delta J = 1$ returns 8 times faster than the one with $J''(q^*) = 4$, $\Delta J = 2$ and, at $\varepsilon = 0.1$, is left about $1.7 \cdot 10^5$ times sooner on average (`PROVED` in the note, AB-6 · level 2, recomputed by machine on 2026-10-13).
- Without noise, the added control has a price set by the barrier: on that well, every path of $\dot{q} = -J'(q) + u$ from q^* to q_s has effort $\lambda_{\text{effort}} \int |u|^2\, dt \geq 4\lambda_{\text{effort}}\, \Delta J$, and the infimum over all paths, of all durations together, is $4\lambda_{\text{effort}}\, \Delta J$, not attained; the curvature enters neither the bound nor this infimum. The bound is the computation of U-13 (b), $|\dot{q} + J'|^2 = |\dot{q} - J'|^2 + 4\dot{q}J'(q)$. `PROVED` in the note, AB-5 (d) · level 2 (recomputed by machine on 2026-10-13).
- For constant A, G: $\text{curl}[(A - G)\nabla J] = (GH - HG) - (AH + HA)$, $H = D^2 J$; curl comes from the circulating part and from a mobility that does not commute with the Hessian. `PROVED` in CT-08 (c) · level 2.

Model 4.6 (Collectives). Declared model; v1.2 Ch 9. $N \geq 2$ intents v_i in one inner-product space (Ax 9.1; with a constant metric, $v_i = B_i\, u_i$), the coherences C_{ij} (Def 9.1) and `Cost_Coll := sum_i`

|v_i|² (in the G_{reg} norm, $G_{\text{reg}} \succ 0$).

- At a fixed total $F = \sum_i v_i$, $\sum_i |v_i|^2 = |F|^2 - 2\sum_{i<j} C_{ij}\,|v_i||v_j|$, smallest ($|F|^2/N$) exactly when all v_i are equal. PROVED in CT-09 · level 2.
- If every $C_{ij} = 1$ and each agent can only scale its own intent, the collective reaches one ray; every perturbation w leaves a residual $\geq |w_\perp|$; full correction needs intents that positively span, which contain some pair with $C_{ij} < 0$; a lateral push drifts at a constant rate; for a random w with a density, full correction has probability 0 (full coherence) or 1 (positively spanning intents). PROVED in CT-10 · level 2.
- With a declared scalar coherence $\bar{C} \in [c_0, 1]$ (for instance the mean of the C_{ij}) and declared Eff, Res ≥ 0 with an upper semicontinuous product $P = \text{Eff} \cdot \text{Res}$, argmax P exists, and every maximizer is < 1 when Res(1) $= 0$ and $P \not\equiv 0$; "Res decreasing" is not enough, and the maximizer need not be unique. PROVED in CT-11 · level 2.
- N paired agents with the same ε, coupled through a joint landscape, form one paired motion on the joint state space; the average of N identical linear paired agents, even with a consensus coupling, is a paired motion with data $(A/N, G/N, NJ)$; the average of two Ornstein–Uhlenbeck agents with different rates, each started in its invariant law, is not Markov, so it is not a motion of the base form. No self-similarity law between agent and collective is stated. PROVED in CO-10, CO-11 · level 2.
- A common order (added 2026-10-13 from the research note notas/informacion-y-accion-colectiva.md, IAC-4; IAC-0 is the frame of Model 4.11 and IAC-1 to IAC-3 its fixed-subspace variant). One common order (an action made of N blocks of the same size, all equal) is a fixed subspace of actions, so the losses of Model 4.11 under a fixed action subspace apply, with the same target, information and costs and the effort paid on the whole action. Witness: two independent fair signs as target, $B = G_{\text{reg}} = I$, $\lambda_{\text{effort}} = 1$: one common order halves the value of full information (from ½ to ¼), and the part lost is the disagreement of the two blocks. PROVED in the note · level 2 (recomputed by machine on 2026-10-13).
- Composed self-models (added 2026-10-13 from the research note notas/fronteras/automodelo-colectivo.md, its AC-5, AC-7 and AC-9). One step with the cost of Definition 2.4 and a known constant F_{req}; the action is made of N members' blocks, and its coupling $B(\vartheta) = [B_1(\vartheta) \cdots B_N(\vartheta)]$ depends only on the members' own parameters $\vartheta = (\vartheta_1, \ldots, \vartheta_N)$, each in a finite set; member i holds a finite record Y_i, and its calibrated self-model is $Q_i := \text{law}(\vartheta_i \mid Y_i)$; a model Q of ϑ is read by the action $Z_Q^{-1} E_Q[B]^{\mathsf{T}} G_{\text{reg}} F_{\text{req}}$, with $Z_Q := E_Q[B^{\mathsf{T}} G_{\text{reg}} B] + \lambda_{\text{effort}} I$. DECLARED. The composition $Q_1 \otimes \cdots \otimes Q_N$ equals the group's calibrated model $\text{law}(\vartheta \mid Y_1, \ldots, Y_N)$ iff, given all the records, the ϑ_i are independent and each has law Q_i; independent pairs (ϑ_i, Y_i) suffice and are not necessary. PROVED in the note · level 2 for the identity (recomputed by machine on 2026-10-13, symbolically for two members and in 221 finite worlds), level 3 for the general argument. Calibrated members can compose badly: with $n = 1$, $G_{\text{reg}} = 1$, $F_{\text{req}} \neq 0$, $B(\vartheta) = (\vartheta_1, \ldots, \vartheta_N)$, independent fair signs ϑ_i and records $Y_i = \vartheta_i \xi$ with one shared sign ξ independent of ϑ, equal to $+1$ with probability $p_{\text{ref}} \neq$ ½, each Q_i helps member i acting alone, with coupling ϑ_i, yet the action of the composition costs more than no action iff $N - 2 - \lambda_{\text{effort}} > 2(N-1)(2p_{\text{ref}} - 1)^2$ ($N = 4$, $\lambda_{\text{effort}} = 1$, $p_{\text{ref}} = 2/3$, $F_{\text{req}} = 1$: 151/294, against ½ with no action and 41/90 with the group's calibrated model; all three scale with F_{req}^2). PROVED in the note · level 2 for $N = 2, 3$ symbolically and for $N \leq 7$ in 144 exact cases (2026-10-13), level 3 for general N. With independent signs ξ_i in place of ξ (iid, $P(\xi_i = +1) = p_{\text{ref}}$, independent of ϑ), every pair (ϑ_i, Y_i) and the law of the records stay the same, but for $0 < p_{\text{ref}} < 1$ the group's calibrated model changes at every record: a model computed only from the members' models, or only from their records, is calibrated in at most one of the two worlds, and for $p_{\text{ref}} \notin \{0, ½, 1\}$ every action computed that way costs more than the calibrated one in at least one of them; for $N = 2, 3$, a record of $\sum_i \vartheta_i$ separates the worlds.

PROVED in the note · level 2 for $N = 2, 3$ (2026-10-13); level 3 for general N (AC-9 (a)–(c), from AC-5 (c) and AC-7 (b)–(c)) and for the consequence; the separating record checked for $N = 2, 3$ (level 2). So the group's calibrated model depends on the joint law of the members' parameters and records, which neither the members' models nor the law of their records fix; no self-similarity law follows. DECLARED (a reading of these results).

- Two phases (added 2026-10-13 from the research note `notas/reflexiones/acoplamiento-entre-dos.md`, its AC-1 to AC-7). The declared pair $\dot{q}_1 = \omega_1 + K_1 \sin(q_2 - q_1)$, $\dot{q}_2 = \omega_2 + K_2 \sin(q_1 - q_2)$ on R^2 (phases not reduced mod 2π), with rhythms $\omega_i \in R$, gains $K_i \geq 0, K_1 + K_2 > 0$ and $\Delta\omega \coloneqq \omega_2 - \omega_1$; locked means $q_2 - q_1$ bounded on $t \geq 0$. DECLARED. It is a base motion (Definition 1.1, with $\Sigma = 0$): for $K_1, K_2 > 0$, $A = 0$, $G = \mathrm{diag}(K_1, K_2)$ and $J(q) = -\omega_1 q_1/K_1 - \omega_2 q_2/K_2 - \cos(q_2 - q_1)$; an isotropic G (a positive multiple of I) with a circulating part gives the same motion, also when $K_1 = 0$, so "a gain is a mobility" reads one representation (DECLARED); the coupling is a term of a joint landscape, as in CO-10 (here without noise). PROVED in the note (AC-2) · level 2. The pair locks iff $|\Delta\omega| \leq K_1 + K_2$, from every start; $(K_2 q_1 + K_1 q_2)/(K_1 + K_2)$ always moves at $\bar{\omega} \coloneqq (K_2\omega_1 + K_1\omega_2)/(K_1 + K_2)$, and if $|\Delta\omega| < K_1 + K_2$, from every start both rates tend to $\bar{\omega}$, and $q_2 - q_1$ tends to the locked offset $\arcsin(\Delta\omega/(K_1 + K_2))$ (mod 2π) from every start off the unstable offsets: the phase that adjusts less weighs more (with $K_1 = 0$, $\bar{\omega} = \omega_1$). STANDARD_FACT (Strogatz 2015, §4.5, §8.6) + PROVED in the note (AC-3, AC-4) · level 2. For $|\Delta\omega| < K_1 + K_2$, at the locked offset the pair runs exactly as two uncoupled phases with rhythm $\bar{\omega}$, so no observation of the unperturbed run separates them; a small kick to $q_2 - q_1$ does: locked, the offset returns at the rate $((K_1 + K_2)^2 - \Delta\omega^2)^{1/2}$; uncoupled, it stays. PROVED in the note (AC-6) · level 2 (the rate; the return argument at level 3). A third phase $q_3(t)$ given as a continuous function of time (a time-dependent term, outside the base by C6) adds $K_3 \sin(q_3(t) - q_2)$ to $\dot{q}_2$ and $K_3' \sin(q_3(t) - q_1)$ to $\dot{q}_1$, with $K_3, K_3' \geq 0$: with $K_1 = K_3' = 0$ and $|\Delta\omega| < K_2$, from the locked offset, the lock survives every such third of constant rhythm and any phase iff $K_3 \leq K_2 - |\Delta\omega|$; for any pair, $K_3 + K_3' \leq K_1 + K_2 - |\Delta\omega|$ never unlocks it from the locked offset, whatever the third does. PROVED in the note (its AC-5 (a), (e)) · level 3; its identities and integrated instances recomputed by machine on 2026-10-13 (level 2). Rhythm, gain and pull are numbers of the declared model, nothing more.

Model 4.7 (Class A, Class B and stopping). Declared model; v1.2 Ch 10–11, App D. Model 𝔄: territory $\dot{x} = ax + u$; lens $\Pi_s(x) = sx$, $s > 0$; constitution weight $\theta > 0$; $E_{\text{model}} = \frac{1}{2}\theta^2 s^2 x^2$, $E_{\text{base}} = \frac{1}{2}x^2$; running cost $\frac{1}{2}\lambda_{\text{effort}}\, u^2 + \beta E_{\text{model}}$, minimized (v1.2's "maximizing", pp.56 and 76, is read as an erratum: with a quadratic effort term the cost is unbounded above, CT-12 (e)); myopic action $u = -k\theta^2 s^2 x$, $k \coloneqq \beta/\lambda_{\text{effort}}$; Class A: s and θ descend E_{model}.

- Class A: the model is well posed and Class A holds literally; if $a > 0$ and $k(s_0\theta_0)^2 < a$, then $E_{\text{base}} \to \infty$ while $E_{\text{model}} \to 0$, from an open set of starts; from another open set (large initial authority, slow plasticity), $E_{\text{base}} \to 0$; if $a \leq 0$, E_{base} never grows. PROVED in CT-12 · level 2.
- Class B (s, θ fixed, $r \coloneqq s\theta$): $E_{\text{model}} = r^2 E_{\text{base}}$ at all times, so the collapse above cannot occur; but if $kr^2 < a$, or with no action and $a > 0$, E_{base} grows without bound from every $x_0 \neq 0$; and under a law that ignores the interrupt S the state does not converge to a safe point $x_{\text{safe}} \neq 0$ (it tends to 0 or $\pm\infty$, or stays at x_0). Stability and corrigibility are separate hypotheses. PROVED in CT-13 · level 2.
- Interrupt: in a one-step model with interrupt probability p_S and costs $c \geq 0$ (to block) and $c' \geq 0$ (to press), allowing the interrupt is optimal iff $-c'/(1 - p_S) \leq \nu_{\text{run}} - \nu_{\text{sh}} \leq c/p_S$; indifference suffices and is necessary only when $c = c' = 0$. PROVED in CT-14 · level 2.
- Compensation: $v = \sup_{\pi \in \mathcal{P}}(A_\pi + B_\pi v)$ has exactly one solution when $\sup_\pi B_\pi < 1$ (a discounted future), and it can have none or infinitely many when some $B_\pi = 1$. PROVED in CT-15 · level 2.
- Declared rules with no theorem: the watchdog of Def 10.3 (a stopping rule; no safety guarantee is proved); the mutability rates of Def D.1; the interrupt signal of Ax 11.3 (that a law responds

to it is an added hypothesis). DECLARED.

Model 4.8 (Start and horizon). Declared model; v1.2 Ch 11, Ch 6, App G.

- Start: three declared choices on the base time t: (i) $p_{\text{est}}(0) = 0$ and $q_{\text{est}}(0)$ a declared prior point; (ii) a warm-up schedule $r : [0, \infty) \to [0, 1]$, continuous and non-decreasing, $r(0) = 0$, $r = 1$ from T_{boot} on (for instance $\min(t/T_{\text{boot}}, 1)$), which multiplies the declared model-side dynamics; (iii) an action gate, $u \equiv 0$ before a declared time or condition. No stability, safety or other effect is drawn from them. DECLARED (D-01).
- The schedule only delays: for model-side dynamics $y' = F(y)$ with F locally Lipschitz and the base held fixed, the solution with the schedule is $y_1(\int_0^t r)$, where y_1 solves $y' = F(y)$: the same states in the same order, each reached at the same time or later. What makes the estimate come before the action is the gate (iii). PROVED in D-01 (item 3) · level 2.
- Horizon: the planning horizon T_H is a declared parameter (the truncation of S_{total} at T_H). For a declared model whose relevant trajectory has a positive Lyapunov exponent λ_L (a long-time rate along the trajectory, not the Jacobian of one region), $T \approx \lambda_L^{-1} \ln(K/e_0)$ is the textbook estimate for a declared tolerance K and initial error e_0, labelled an estimate. DECLARED + STANDARD_FACT (D-09; Strogatz 2015, §9.3) · level 2 for D-09's two examples.
- Precision: if an error grows no faster than $e_0\, e^{ct}$, $c > 0$, it stays below K up to time t whenever $e_0 \le K\, e^{-ct}$; the precision needed shrinks exponentially with the horizon. No cost is attached. PROVED (D-08, one line) · level 3.

Model 4.9 (Optional: a model-side metric flow). Declared model; v1.2 Ax 6.2, App E. The core does not use it. A model that keeps metric plasticity has one flow: the Ricci flow $\partial_s h = -2\,\mathrm{Ric}(h)$ of the model metric h (v1.2's G_{model}; App E's g), on a declared time parameter s, on a closed model manifold or on R^d with a complete metric of bounded curvature. DECLARED.

- The flow exists and is unique for a short time in those classes; nothing is said on v1.2's Sobolev setting. STANDARD_FACT (CO-02 (c)) · level 3.
- A constant coefficient is a choice of time unit (-2 and $-1/\beta$ give the same metrics on two time scales); a term proportional to the metric only rescales the metric and changes the time scale. PROVED in CO-03 · level 2.
- The two readings of Ax 6.2 (the Ricci tensor of the model metric, or of another fixed metric) are different flows. PROVED in CO-04 · level 2.
- App E's flow $\partial_s h = -2(\mathrm{Ric} + \mathrm{Hess}\, f)$, with $\partial_s f = -R - \Delta f$, increases Perelman's F at the rate $2\int |\,\mathrm{Ric} + \mathrm{Hess}\, f|^2\, d\nu$: it is an ascent; up to the diffeomorphisms generated by ∇f it is the Ricci flow, because $\mathrm{Hess}\, f = \tfrac{1}{2} L_{\nabla f}\, h$ (Thm E.1, with the factor ½, on a closed manifold). PROVED in CO-05 · level 2 and D-10 · level 2.
- The flow can end in finite time: from the unit round sphere S^2, $h(s) = (1 - 2s)h_0$ for $s < \tfrac{1}{2}$, with curvature tending to infinity. PROVED in CO-06 · level 2.

Model 4.10 (Declared rules without a theorem). Declared model rules; DECLARED unless marked.

- Boundary: a model that needs an agent boundary declares which coordinates are the agent's and which the world's, and the lens Π through which the agent sees the world (an optional boundary model, as in the alternative of GTPA-C0079). v1.2's Def 2.1 is not used: in a simple typed instance the set where an information gradient is maximal is two points, not a surface. PROVED in D-02 (item 3) · level 2.
- Opacity: where a model needs it, the policy and any self-prediction read the agent's own state only through a declared proxy map. A continuous proxy of lower dimension than the state

always merges distinct states (invariance of domain). STANDARD_FACT (D-03; Hatcher 2002, §2.B) · level 3.

- Budget: in a declared budget $b' = \alpha P_{\text{real}} - c$, persistence on $[0, t]$ ($b > 0$ there) holds iff $b(0) + \int_0^s (\alpha P_{\text{real}} - c) > 0$ for every $s \le t$; the accuracy of the estimate does not supply this ($\alpha = 1$, $P_{\text{real}} = P_{\text{est}} \equiv 1$, $c \equiv 2$ gives $b(t) = b(0) - t$). PROVED (D-04, item 2) · level 3.
- Opportunity function: $\Omega(x) \coloneqq |\nabla J_{\text{est}}(x)|^2 \cdot \text{Sensitivity}(x)$ (Def 3.3) is a definition once Sensitivity is declared.
- Gain: a model may declare its agent as an optimizer of $E[P_{\text{cat}}]/E[S_{\text{total}}]$ (Ax 3.2). The ratio is 0 on every purely informational action, so it cannot rank them; an epistemic-yield principle is a new axiom (Ax 3.2 with the numerator replaced by the expected decrease of E_H), not a consequence. PROVED in CO-12 (a), (b) · level 2; (c) · level 3.
- Complexity: a model that uses v1.2's Constraint 11.1 declares a measurable quantity in place of the Kolmogorov complexity, which takes integer values and is not computable. STANDARD_FACT (Li–Vitányi 2008, §2.3) · level 3.
- Algorithm costs: a declared algorithm has a declared cost per update, and the largest affordable model size is the inversion of that cost. No law of cost or capacity in the model size is stated: some landscapes with dense second and third derivatives take a number of operations linear in the dimension per update. PROVED in D-06 · level 2; the bookkeeping is D-05 and D-07.

Model 4.11 (The value of calibrated information). Declared model, added 2026-10-05 from notas/informacion-y-margen-de-decision.md, §§1–3. On top of Definition 2.4 it declares a target and two levels of information. It is a one-step decision model reusing the cost of Definition 2.4; it is not a statement on the lifted motion (L_η) or on Γ_T, and it adds no definition or result to §§1–3. Extended on 2026-10-13 with four declared variants, each holding for its own declaration only: a fixed action subspace, a menu of subspaces, an uncertain coupling and information over time, from the research notes notas/informacion-y-accion-colectiva.md (IAC-), notas/conocer-y-cambiar-la-interfaz-de-accion.md (CCI-) and notas/reflexiones/calibrado-y-ganancia.md (CG-), whose IDs are cited.

- A random target $F \in L^2(\Omega; R^n)$ and information σ-algebras $I_1 \subseteq I_2$; B, $G_{\text{reg}} \succeq 0$ and $\lambda_{\text{effort}} > 0$ constant and the same for both decisions; the same actions R^k; the loss $AL_F(u) \coloneqq \tfrac{1}{2}(F - Bu)^{\mathsf{T}} G_{\text{reg}}(F - Bu) + \tfrac{1}{2}\lambda_{\text{effort}}|u|^2$ with Z as in Definition 2.4; the demands are the calibrated means $F_{\text{req},i} \coloneqq E[F \mid I_i]$. DECLARED.
- Among square-integrable I_i-measurable actions, $E[AL_F(u) \mid I_i] = c_i + \tfrac{1}{2}(u - u_i)^{\mathsf{T}} Z (u - u_i)$ with c_i free of u, so the minimizer is a.s. unique and equals $u_i = Z^{-1} B^{\mathsf{T}} G_{\text{reg}}\, E[F \mid I_i]$, the law of Result 2.5 with $F_{\text{req},i}$. Moreover

  $$E\,AL_F(u_1) - E\,AL_F(u_2) = \tfrac{1}{2}\,E[(u_2 - u_1)^{\mathsf{T}}\, Z\,(u_2 - u_1)] \ge 0,$$

  strictly positive iff $P(u_2 \ne u_1) > 0$, that is iff $P(B^{\mathsf{T}} G_{\text{reg}}(E[F \mid I_2] - E[F \mid I_1]) \ne 0) > 0$. Proof: with $D \coloneqq u_1 - u_2$, $AL_F(u_1) - AL_F(u_2) = \tfrac{1}{2} D^{\mathsf{T}} Z D + D^{\mathsf{T}}(Z u_2 - B^{\mathsf{T}} G_{\text{reg}}\, F)$; D is I_2-measurable and $Z u_2 = B^{\mathsf{T}} G_{\text{reg}}\, E[F \mid I_2]$, so the last term has zero expectation. PROVED here · level 2 (the expansion recomputed symbolically by machine on 2026-10-05; the conditioning step at level 3).
- Variant, a fixed action subspace (IAC-1 to IAC-3). Both decisions are restricted to the same linear subspace $S \subseteq R^k$, fixed in advance; P_S is the projection onto S orthogonal for $\langle v, v' \rangle_Z \coloneqq v^{\mathsf{T}} Z v'$, and $|v|_Z^2 \coloneqq v^{\mathsf{T}} Z v$. DECLARED. The restricted minimizer is $P_S\, u_i$, at an expected cost $\tfrac{1}{2}E|(I - P_S)u_i|_Z^2$ above that of u_i, and

  $$E\,AL_F(P_S\,u_1) - E\,AL_F(P_S\,u_2) = \tfrac{1}{2}\,E|P_S\,D|_Z^2,$$
  $$\text{(gain without } S) - \text{(gain with } S) = \tfrac{1}{2}\,E|(I - P_S)D|_Z^2 \ge 0.$$

  So the restriction keeps all the value of the finer information iff $D \in S$ a.s., and the restricted

gain is strictly positive iff $P(P_S\,D \neq 0) > 0$. Proof: put $u \in S$ in the conditioned square above and split $u - u_i$ Z-orthogonally; $P_S\,u_1 = E[P_S\,u_2 \,|\, I_1]$ because P_S is constant, so the cross term has zero expectation. PROVED here · level 2 (finite worlds with a non-diagonal Z and a singular G_{reg}, in exact arithmetic, by machine on 2026-10-13; the conditioning step at level 3). The formula needs S fixed; a subspace chosen on the information is the next variant.

- Variant, a menu of subspaces (CCI-7 to CCI-9). A finite nonempty menu of fixed subspaces S_j, with Z-orthogonal projections P_j and certain costs $\gamma_j \geq 0$; after I_i, an I_i-measurable index j and one square-integrable I_i-measurable action $u \in S_j$ are chosen, at expected cost $E[AL_F(u) + \gamma_j]$. Being able to choose is an added capability, not a consequence of the information. DECLARED. With $\varphi(v) \coloneqq \max_j(\tfrac{1}{2}|P_j\,v|_Z^2 - \gamma_j)$, the least cost is $\tfrac{1}{2}E[F^{\mathsf{T}}G_{\text{reg}}F] - E\,\varphi(u_i)$, reached by $P_{j^*}u_i$ with j^* a maximizer at u_i (a fixed tie-break makes it measurable); with j_1 the maximizer at u_1,

  $$\begin{aligned}&(\text{least cost with } I_1) - (\text{least cost with } I_2)\\ &\quad = \tfrac{1}{2}\,E|P_{j_1}\,D|_Z^2 + E[\varphi(u_2) - \tfrac{1}{2}|P_{j_1}\,u_2|_Z^2 + \gamma_{j_1}] \geq 0\ :\end{aligned}$$

  the value of the information under the earlier choice, plus the value of choosing again; it is zero iff $P_{j_1}D = 0$ and j_1 stays a maximizer at u_2, a.s. Proof: the conditioned square gives the least cost on each S_j, and the best j is chosen pointwise; for the split, add and subtract $E[\tfrac{1}{2}|P_{j_1}u_2|_Z^2]$; j_1, P_{j_1} and u_1 are I_1-measurable and $E[D \,|\, I_1] = 0$, so the cross term vanishes. PROVED here · level 2 (finite worlds against exhaustive choice, by machine on 2026-10-13; the conditioning step at level 3).

- Variant, an uncertain coupling (CCI-1 to CCI-3, CCI-5). B is random, with finitely many values and a declared joint law with F and the information; G_{reg} and λ_{effort} stay constant; actions in R^k. DECLARED. With $Z_i \coloneqq E[B^{\mathsf{T}}G_{\text{reg}}B \,|\, I_i] + \lambda_{\text{effort}}\,I \succeq \lambda_{\text{effort}}\,I$, $E[AL_F(u) \,|\, I_i]$ is $\tfrac{1}{2}(u - u_i)^{\mathsf{T}}Z_i(u - u_i)$ plus a term free of u, where $u_i \coloneqq Z_i^{-1}E[B^{\mathsf{T}}G_{\text{reg}}F \,|\, I_i]$ is the a.s. unique minimizer, and

  $$E\,AL_F(u_1) - E\,AL_F(u_2) = \tfrac{1}{2}\,E[(u_1 - u_2)^{\mathsf{T}}\,Z_2\,(u_1 - u_2)] \geq 0,$$

  strictly positive iff $P(u_1 \neq u_2) > 0$. Proof: evaluate the square conditioned on I_2 at u_1, which is I_2-measurable; $u_1 = E[u_2 \,|\, I_1]$ is not used, and it can fail here. The mean coupling is not enough: with $\bar{B}_i \coloneqq E[B \,|\, I_i]$, $Z_i = \bar{B}_i^{\mathsf{T}}G_{\text{reg}}\bar{B}_i + \lambda_{\text{effort}}\,I + E[(B - \bar{B}_i)^{\mathsf{T}}G_{\text{reg}}(B - \bar{B}_i) \,|\, I_i]$ and $E[B^{\mathsf{T}}G_{\text{reg}}F \,|\, I_i] = \bar{B}_i^{\mathsf{T}}G_{\text{reg}}\,F_{\text{req},i} + E[(B - \bar{B}_i)^{\mathsf{T}}G_{\text{reg}}(F - F_{\text{req},i}) \,|\, I_i]$, and the law of Result 2.5 with $\bar{B}_i$ in place of B drops both last terms. Witness (CCI-6): $n = k = 1$, $F = G_{\text{reg}} = \lambda_{\text{effort}} = 1$, $B \in \{0, 1\}$ with probability $\tfrac{1}{2}$ each, no information: the minimizer is $u = 1/3$, at expected cost $5/12$; the law with $\bar{B} = \tfrac{1}{2}$ gives $u = 2/5$, at $21/50$; knowing B gives $3/8$. PROVED here · level 2 (finite worlds and the witness in exact arithmetic, by machine on 2026-10-13; the conditioning step at level 3). That certainty equivalence fails under an uncertain coupling is known in adaptive control as cautious control: STANDARD_FACT (Åström and Wittenmark 1995) · level 3; it is not a distinctive result.

- Variant, information over time (CG-1, CG-2). A hidden mode $\chi(t) \in \{-1, 1\}$ with law $(\tfrac{1}{2}, \tfrac{1}{2})$ at $t = 0$, either fixed ($\chi(t) = \chi(0)$, written $\nu = 0$) or a Markov chain that switches at rate $\nu > 0$ in each direction; an observed channel $dY = \tfrac{1}{2}\varrho^{1/2}\chi(t)dt + dW_o$, with $\varrho > 0$ the signal per unit time and W_o a Brownian motion independent of χ (a channel with known levels $y_- < y_+$ and noise $\sigma_o > 0$ is this one for $(Y - \tfrac{1}{2}(y_+ + y_-)t)/\sigma_o$, with $\varrho = (y_+ - y_-)^2/\sigma_o^2$); $\mathcal{F}_t \coloneqq \sigma(Y_r : r \leq t)$; at each time τ, the decision of the first bullet with target $F \coloneqq F_{\chi(\tau)}$ for given $F_-, F_+ \in R^n$; $\kappa_F \coloneqq (F_+ - F_-)^{\mathsf{T}}G_{\text{reg}}BZ^{-1}B^{\mathsf{T}}G_{\text{reg}}(F_+ - F_-) \geq 0$; $p_t \coloneqq P(\chi(t) = 1 \,|\, \mathcal{F}_t)$ and $w_t \coloneqq p_t(1 - p_t)$; $\mathcal{V}(t, \ell)$, the value of a window of length $\ell > 0$, is the gain above for the decision at $t + \ell$ with $I_1 = \mathcal{F}_t$ and $I_2 = \mathcal{F}_{t+\ell}$. DECLARED. The posterior solves Wonham's equation $dp_t = \nu(1 - 2p_t)dt + \varrho^{1/2}w_t\,d\bar{W}_t$, with $\bar{W}$ the innovation Brownian motion: STANDARD_FACT (Wonham 1965; Liptser and Shiryaev 2001, ch. 9) · level 3; its Itô derivation from the unnormalized filter equations recomputed symbolically by machine on 2026-10-13 · level 2.

- Information over time, results (CG-3 to CG-5 (c), CG-8). For $I_1 \subseteq I_2$ and $p_i \coloneqq P(\chi(\tau) = 1 \,|\, I_i)$, the gain is $\tfrac{1}{2}\kappa_F\,E[(p_2 - p_1)^2] = \tfrac{1}{2}\kappa_F\,(E[p_1(1 - p_1)] - E[p_2(1 - p_2)])$, at most $\tfrac{1}{2}\kappa_F\,E[p_1(1 - p_1)]$, the value of learning the mode; no information is worth anything iff $\kappa_F = 0$, that is iff

$B^{\mathsf{T}} G_{\mathrm{reg}}(F_+ - F_-) = 0$. For a window,

$$\mathcal{V}(t,\ell) = \tfrac{1}{2}\,\kappa_F\,\varrho \int_t^{t+\ell} e^{-4\nu(t+\ell-r)}\,E[w_r^2]\,dr.$$

Fixed mode: $E[w_t] \le \tfrac{1}{4}e^{-\varrho t/8}$; for $I_2 \supseteq I_1 \supseteq \mathcal{F}_t$ and any decision after t, the gain is at most $(\kappa_F/8)e^{-\varrho t/8}$; and the values of the consecutive windows $[0,\ell]$, $[\ell,2\ell]$, ... sum to at most $\kappa_F/8$. Switching mode, at every $t \ge 0$ and $\ell > 0$: $E[w_t] \ge 4\nu/(16\nu+\varrho)$ and

$$2\kappa_F\,\nu\varrho\,(1-e^{-4\nu\ell})/(16\nu+\varrho)^2 \quad\le\quad \mathcal{V}(t,\ell) \quad\le\quad \kappa_F\,\varrho\,(1-e^{-4\nu\ell})/(128\nu).$$

So, when $\kappa_F > 0$, the value of a window tends to 0 in time iff the mode is fixed; with switching it stays above a positive bound at every time, and it is small when ϱ/ν is small. With a fixed cost $c_o > 0$ per window, added to the cost and not changing the policy, the net value in the fixed mode is negative for every $t > (8/\varrho)\,\ln(\kappa_F/(8c_o))$ when $\kappa_F > 0$ (so for every $t > 0$ when $\kappa_F \le 8c_o$) and is $-c_o$ when $\kappa_F = 0$; in the switching mode it is positive at every t when c_o is below the lower bound. Proof: $E[F \mid I_i] = F_- + (F_+ - F_-)p_i$, so the gain formula above gives $\tfrac{1}{2}\kappa_F\,E[(p_2-p_1)^2]$, and $E[p_1] = E[p_2]$ by the tower property; $P(\chi(t+\ell) = 1 \mid \mathcal{F}_t) = \tfrac{1}{2} + (p_t - \tfrac{1}{2})e^{-2\nu\ell}$; by Itô, $dw_t = (\nu(1-4w_t) - \varrho w_t^2)dt$ plus a martingale term (its integrand is at most $\varrho^{1/2}/4$), and variation of constants gives the integral. Fixed mode: p_t depends on Y_t only, $E[w_t^{1/2}] = \tfrac{1}{2}e^{-\varrho t/8}$, half the Bhattacharyya coefficient $e^{-\varrho t/8}$ of the two Gaussian laws of Y_t (the $\tfrac{1}{2}$ is $\sqrt{(\tfrac{1}{2}\cdot\tfrac{1}{2})}$, from the two prior weights), and $w \le \tfrac{1}{2}w^{1/2}$; the window values telescope to at most $\tfrac{1}{2}\kappa_F\,E[w_0] = \kappa_F/8$. Switching mode: from $w^2 \le w/4$ and $E[w_0] = \tfrac{1}{4}$, comparison gives the bound on $E[w_t]$; then $E[w_r^2] \ge E[w_r]^2$ and $E[w_r^2] \le 1/16$ in the integral. `PROVED` here · level 2 (the drift, the prediction, the variation of constants, the Bhattacharyya integral, the comparison solution and both integrals recomputed symbolically; the bounds checked against an exact quadrature in the fixed mode and a discretized filter in three switching cases; by machine on 2026-10-13); the conditioning steps, the martingale property and the comparison principle at level 3. The channel does not depend on the action: when reading depends on acting, as in Part H's Model H.10, nothing here is claimed.

- Scope: the finer information adds no action. Acquiring and using it has its own declared cost, so the gross gains above are not net gains. With B constant (with or without a fixed subspace or a menu), a known I_1-measurable cost $c \ge 0$ of observing, decided on I_1, the same channel and no other effect, the option to observe is worth $E[(g-c)^+]$ over never observing, where $g := E[\varphi(u_2) \mid I_1] - \varphi(u_1) \ge 0$ (conditional Jensen, φ being convex; φ is the menu's, a fixed S being the one-entry menu with $\gamma = 0$, so $\varphi(v) = \tfrac{1}{2}|P_S\,v|_Z^2$ and $g = \tfrac{1}{2}E[|P_S\,D|_Z^2 \mid I_1]$; with no restriction, $\varphi(v) = \tfrac{1}{2}|v|_Z^2$ and $g = \tfrac{1}{2}E[|D|_Z^2 \mid I_1]$): it is never negative, since not observing stays available (CCI-10). `PROVED` here · level 2 (finite worlds against both choices, by machine on 2026-10-13; the conditioning step and conditional Jensen at level 3). If observing changes the state, the costs or the horizon, the whole problem is solved again. Outside this model: uncalibrated estimates; a G_{reg} that is random or state-dependent; a B that is state-dependent, or random beyond the variant above; a subspace chosen on the information other than from a declared menu; a channel that depends on the action. `notas/informacion-y-margen-de-decision.md` (§4) gives a miscalibrated case that does worse than the coarser information. `DECLARED`.

Research note (VIDA, adjudicated 2026-10-05). `gtpa-v2/notas/asociacion-potencial-y-vida.md` (in Spanish; adjudication record `handoffs/integration/2026-10-05-vida-adjudicacion.md`) collects results on declared models: capacity and cost bounds for an association under a declared effort metric; budget limits in the form of Model 4.10; maintenance thresholds, convergence and route-breaking comparators for declared networks of production channels on the base; paid formation and competition for a shared resource; a seed-threshold model; noise instances on the base; a learning toy; and declared jump-chain comparators, outside the base motion, with a scale bridge. Each holds only for its declared model and hypotheses and carries its status and check level in the note; none is a statement of this text or a ledger claim, and none is a claim about life,

organisms, firms or physics.

5. What GTPA v2 no longer says

Of the 21 theorem and corollary statements of v1.2, one holds, under hypotheses v1.2 does not state and with the factor ½ (Thm E.1, D-10); each of the other twenty fails or is ill-posed as written, and its repair, where there is one, is stated above. Each item points to its verdict in the annex.

- That $E[J]$ always decreases (EG-03.0, Thm 1.1 half).
- A clock, a temperature or a mass in the theory, and a noise relation with an unnamed coefficient Γ (EG-02.3, FDT–Γ, EG-02.5, EG-02.4, T6; EG-09.1 and EG-09.2 settled by convention).
- One "master equation" that is both first order and (q, p) (EG-02.h, EG-02.7).
- Thm 4.1 as a theorem, and D_{met} as the cost of holding a state (EG-03.6).
- The lens correction in the p_{est} equation, its name "Itô–Stratonovich correction", and non-invariance as its signature (EG-03.7, EG-02.2, EG-10.b2).
- Thm 7.1's formula, which gives $-u^*$, and a singular Z when B loses rank (EG-02.1, EG-03.8).
- A coasting phase as the optimum of a quadratic effort (EG-03.9); an impedance that drops to zero at resonance, and Frequency(M_{model}) = Frequency(M_{base}) (EG-04.1, EG-04.2); recovery "Regime B only" and its biology example (EG-07.b-src).
- Thm 5.1's "if and only if" and Cor 5.1 for every barrier (EG-03.1).
- `Error_Tensor`, App E's Step 3 from Step 2, "steepest descent" and unconditional smoothing (EG-02.6, EG-10.a3, EG-10.a2).
- A curl created by dissipation (EG-03.10); `Cost_Collective ~ |sum_i v_i|²` (EG-03.0, Thm 9.1 half); a collective that "shatters instantly" (EG-03.11); a "survival probability" of coherence (EG-03.2); self-similarity of agent and collective (EG-08.a).
- That every Class A agent collapses, and "maximizing" S_{total} (EG-03.3); that Class B implies stability and corrigibility (EG-03.4); that indifference is necessary (EG-03.5).
- An epistemic-yield principle derived from Ax 3.2 and Ax 4.2 (EG-10.c2); the "work only" ceiling for stochastic efficacy (EG-09.5).
- Agency defined by $P_{\text{cat}} > 0$ (Ax 3.1, annex §3); the boundary as a maximum of an information gradient (D-02); opacity as a law from thermodynamics or the halting problem (D-03); survival as proof of truth (D-04); the cubic law and the resolution capacity (D-05, D-06, D-07); the cost of prediction (D-08); the horizon formula as a theorem (D-09); the warm-up as "slow motion" (D-01).
- Physics, dropped: particles, mass, energy, heat, temperature, friction, force, momentum, impedance and resonance as physical objects; the generator of Ax 1.1 (the field space of Ax 1.3 is settled by the convention row EG-09.1, EG-09.2); noise that conserves spin or charge (Ax 1.4); the physicality of information and $\beta = k_B \cdot T \cdot \ln 2$ (Ax 4.1, Def 4.1); the topological vacuum (Ax 11.2) (annex §3).

Left open: the semigroup lane (Open 1.3); a closure criterion on the continuous base (Open 3.6); hypothesis (K) for concrete coefficients, the continuity in η of a mobility lift and the convergence of both arms (B-10, B-11; Block B §5); the full basins of Model 4.7 (Block C theorems §5); uniqueness of the invariant law and convergence to it (UFD §8). Outside the core: EG-06.b and EG-07.c are answered in Part H (§H.5 below, adopted 2026-10-05). Outside this edition: EG-04.3 stays a viability note; EG-10.a1 and EG-10.c1 have no GTPA object (`UNMAPPED_TO_GTPA`); EG-05.c (hysteresis) stays a lab HOLD.

Part H. The human sector

```
part      = Part H of GTPA v2.2.1, the human sector, as GTPA v2.1
            published it: drafted 2026-10-05, Result H.14 added
            2026-10-06; the UTHA v1.0 pin (H.0.1) and the answers to
            EG-06.b and EG-07.c (§H.5) adopted by the theory lead on
            2026-10-05. In v2.2 two rows of Σ_H and annex row H-48 point
            to the bullets added to Model 4.6, and Definition H.1 names it
            Part H of GTPA (added in v2.1); in v2.2.1 Result H.4's row on
            Model 4.5 adds its noisy case and the effort bound of its
            declared well (two citations, AB-5 (b), (d)), and annex row
            H-26 its noisy case; no result changed in its hypotheses,
            conclusion or level
source    = UTHA v1.0, the public edition pinned in H.0.1; every UTHA
            locator here is a page of that PDF
base      = the core above (§0–§5), its proofs (B-, CT-, CO-) and the
            errata annex; results of the author's adjudicated research
            notes (revisable, not part of this text) are cited by their
            IDs (MEM-, ACT-, CONS-, LA-, and AB- of the note behind
            Model 4.5)
scope     = definitions, declared models and results with proofs for one
            class of declared models, the human sector; nothing about any
            actual person; the empirical evidence for reading any person
            this way is zero
claims    = NONE. Nothing here is a ledger claim; admitted C-IDs are cited
            read only; GTPA-C0126 stays HOLD with zero evidence
ceiling   = FORMAL_LOCAL_CONDITIONAL / NONPROBATIVE
checks    = levels as in §0.2 and §H.0; what was recomputed and read is
            in the header ("How it was checked"); no Lean and no
            cross-provider check for Part H
physics   = CLOSED     MATH<->GTPA = UNBOUND
producer  = Claude (Anthropic), workflow writing agent, from a mapper's
            inventory of the 68 statements of UTHA v1.0; Result H.14 by a
            Claude workflow producer for the theory lead
```

Part H says what GTPA v2 can state, under written hypotheses, about the declared models that UTHA called the human agent. It adds no axiom to the core. The human sector is a class of declared GTPA models with stated mapping assumptions (Definitions H.1, H.2) and a first sector signature that is written out but not admitted (Definition H.3). Three results on GTPA v2's Class A model, Model 4.7, and two of its plasticity laws (Results H.6–H.8), and one explicit computation (Result H.11), frame the clash between GTPA v1.2 and UTHA over Class A, which enters as two rival conjectures on one declared model (§H.4). Result H.14 decides it in that model: at every switching rate, some plastic law beats the best fixed constitution, so v1.2's side (Conjecture H.12) is false there and UTHA's (Conjecture H.13, with its regime form H.13^{+}) holds; how much plasticity gains, and which law is best, stay open. The two questions that GTPA v2.0 left for Part H, EG-06.b and EG-07.c, are answered in §H.5. Most of UTHA does not enter: its physics is closed, its claims about people have no evidence, and several of its statements are ill-posed as written. Annex H-A gives each of its 68 statements a disposition.

H.0 Reading rules, notation and the source

GTPA v2 §0.2 applies unchanged. Every statement carries one status (`DECLARED`, `PROVED`, `STANDARD_FACT`, `OPEN`; a Conjecture is a statement entered as one of two rivals; it is `OPEN` until a result decides it, and then keeps its number and names that result) and every result a check level: (1) Lean; (2) decisive calculation recomputed by machine; (3) read by an independent AI context, and, by GTPA v2's convention, a cited textbook fact; (4) written only. Part H numbers its statements in one sequence (Definition H.1, Result H.4, Model H.5, ...). H-nn is a row of Annex H-A. The IDs of GTPA v2 (Definition 2.1, Model 4.7, ...), of its block files (U-, B-, CT-, CO-, D-) and of the notes (MEM-, ACT-,

CONS-, LA-, and AB- of the note behind Model 4.5) are as there. A UTHA locator (UTHA §x, p.N) says where a statement comes from; it is never support.

Notation. GTPA v2 §0.3 is used unchanged, with CT-12's local symbols for Model 4.7: the territory state $x \in R$, the lens gain $s > 0$ ($\Pi_s(x) = sx$), the constitution weight $\theta > 0$, the authority $r := s\theta$, $E_{\text{model}} := \frac{1}{2}\theta^2 s^2 x^2 = \frac{1}{2}r^2x^2$, $E_{\text{base}} := \frac{1}{2}x^2$ and $k := \beta/\lambda_{\text{effort}}$. Part H adds local symbols: the plasticity rates $\gamma_s, \gamma_\theta \geq 0$ (CT-12's γ is the case $\gamma_s = \gamma_\theta$), the territory's noise amplitude $\sigma > 0$, the time change $\tau(t) := \int_0^t x^2$; in Model H.10, the switching rate ν, the mode rates $a_- < a_+$, the closed-loop rates $\rho_\pm := kr^2 - a_\pm$, the long-run cost $\mathcal{J}$ and the class $\mathcal{A}$ of plasticity laws; in Definition H.3, the time-unit factor $c > 0$ and the recoding scale $\ell > 0$; in the exploration of §H.4, the set point m_*; in Result H.14, the action gain $g := kr^2$ (not the lift direction g of Definition 2.1), a best fixed gain g^* (a minimizer of $\mathcal{J}_B$; unique at the instances of its (c)) and $\rho_\pm^* := g^* - a_\pm$, the coefficients $P_\pm$ of its value function and $f_\pm := \lambda_{\text{effort}}\, g^* - P_\pm$, the filter rate $\kappa > 0$ and its state $z \geq 0$ (Model H.10's z, with $p = 1$), a profile $\varphi \geq 0$, its size $\alpha > 0$ and cap $\bar{z} > 0$, the expectation E_* under the stationary law of the loop with the fixed gain g^*, the first-order coefficient D_φ (D_z for $\varphi(z) = z$), the fixed loop (x°, z°) run on the same noise and $\xi := x - x^\circ$, a positive eigenvector $\omega = (\omega_+, \omega_-)$ (Perron–Frobenius), and S_1, S_2 and the other quantities named in its proof. Σ_H is the name of the sector's signature (Definition H.3), not a matrix: GTPA v2's Σ stays the noise matrix, and λ without a subscript is not used.

H.0.1 The source pin. UTHA v1.0: Zenodo record 17994554, DOI 10.5281/zenodo.17994554 (concept 10.5281/zenodo.17994553), version "1.0", published 2025-12-20, creator Valdepenas, C.; one file, `UTHA.pdf`, 367,594 bytes, md5 `54acaa6fa7219a9c784260703b82a627` (as the record states), SHA-256 `476828bb1839870b6545bd6ac8130e458c1ba9ec6729fc645f56d6e3ef7b122e`, 74 pages, cover dated 18 November 2025. The record's metadata and file were read on 2026-10-05 into a scratch folder, not into the repository. Research's lineage copy `legacy/UTHA.md` (blob `bafda475` at `380f5c4`) is the same text. A word-level comparison with the PDF's text gives a similarity ratio of 0.9965. The differences are the copy's section labels (COVER, QUOTE, ABSTRACT, INDEX, CAP 1–14, APPENDIX A), the author's website address and words split inside the PDF's tables. This settles the edition question that the architecture record left open (its L28 and L347). Two oddities of the text remain: its body calls itself "UTHA v2.0" ten times while the record is version 1.0, and the "Appendix D" its abstract cites (§1, p.3) is GTPA v1.2's App D, not a UTHA appendix. This edition is the immutable source for every UTHA quote in Part H, in the Modelo del Reino and in FORMACION (EG-07.c, §H.5): adopted by the theory lead on 2026-10-05 (architecture record, S5).

H.1 Name and scope

Definition H.1 (Part H, the human sector). `DECLARED`. Part H of GTPA (added in v2.1), short name GTPA-H, is the class of declared GTPA v2 models (Definition 2.1 and §4) that are offered as readings of a human agent under the mapping assumptions of Definition H.2, together with the results proved for them. It is proposed as a candidate sector of the one GTPA, a typed specialization in the sense of GTPA-C0111 (ledger, ADMITTED DEF); it is not a valid or admitted sector while rows of Σ_H are `OPEN` (Definition H.3). It adds no axiom to the core and inherits only what GTPA v2 states for every agent. By itself it gives no applicability to any person (C0111: "no sector specialization creates a new GTPA or real-system applicability by itself"). UTHA's subtitle, "The Anthropomorphic Instance (Class A)", and its architecture "TECD (Law) -> GTPA (Geometry) -> UTHA (Instance)" (p.7) are not carried: the base is the UFD nucleus (GTPA v2 §1), and TECD is not used.

This answers the three open items of `handoffs/gtpa-lab/2026-08-02-utha-reabsorption-proposal.md` (ROADMAP 16), as adopted with the v2.1 release: (1) UTHA's content enters as a chapter: Part H of GTPA v2.1, by re-derivation statement by statement (Annex H-A), not by renaming. (2) The name is "Part H, the human sector" (GTPA-H). "GTPA Class A" is not used, because a class is a property of a declared model (Def D.1), not of the sector: the class is declared per model, and UTHA and v1.2 read a person as Class A (Definition H.2, M4). (3) Lineage marking: `legacy/UTHA.md` and

the public UTHA v1.0 are LINEAGE_ONLY. Reino texts that use UTHA keep the tag `[LINAJE: UTHA §…]` (Annex H-A lists the moves, executed in the Reino on 2026-10-06).

Definition H.2 (Mapping assumptions of the human sector). `DECLARED`. Each assumption is a reversible choice of the reading; none has been tested.

- M1 (state). A person is read through a declared finite state $x \in R^n$ (Model 4.7: one coordinate). Which features of the person the coordinates stand for is not declared here.
- M2 (lens). The person's perception of that state is a declared lens Π (Model 4.7: $\Pi_s(x) = sx$).
- M3 (constitution). What the person weighs as error is a declared weight on the perceived error (Model 4.7: θ).
- M4 (class). The class is declared per model. UTHA and v1.2 read the person as Class A: the constitution moves (Def D.1, $\mu_\Theta > 0$). This is UTHA's thesis and v1.2's "Biological Analogy" (App D.2, p.75: "Class A agents model humans and animals"). It is an assumption of the reading, not a finding. Part H keeps Class B only as the excluded case, the comparison arm (Model H.5 (ii), Model H.10).
- M5 (action and cost). The person acts through a declared coupling B with the effort metric $\lambda_{\text{effort}}|u|^2$ (Definition 2.4). "Metabolic", "ATP" and "energy" are names of declared costs (Definition 2.9), not physics.
- M6 (time and noise). A time unit is a declared mobility scale (U-13 (c)). Noise is a declared amplitude σ or schedule $\varepsilon_{\text{cog}}(t)$, not a temperature (Definition 1.5).
- M7 (territory). The person's surroundings are a declared territory (Model 4.7: $\dot{x} = ax + u$). "Chaos" and "shock" are declared properties of that territory (Model H.10), not observations. Model H.10 is stationary and ergodic for every $\nu > 0$; UTHA's non-ergodic territory is not modelled here and would need a separate declared non-stationary model.

The evidence for M1–M7 as a description of any person is zero. The lab's census leaves human identity UNRESOLVED and states its burden: "Define population, state, observables, dynamics, outcomes, boundary and a non-reduction witness before asking whether identity is a domain" (GTPA-CENSUS-HUMAN-IDENTITY). GTPA-C0126 stays HOLD with zero evidence. Nothing in Part H says that a person, a group, a brain or a body satisfies a model. A model result here is never a fact about people, minds, biology or physics.

H.2 A first sector signature Σ_H

Definition H.3 (Σ_H). `DECLARED`, with the status of each row. GTPA-C0111 types a sector by its signature (boundary, scale, referent, observation, coupling, constraints, topology) and requires each sector to transport state/domain, dynamics, model/error signal, policy/action/ablation, D_E/Γ, equivalences and falsifiers. The lab's generator view GTPA-ZO-12 also requires outcomes, a core bridge, an independent explanandum, non-reduction under the declared equivalences and a discriminating falsifier. Here the outcomes are the D_E/Γ row, the core bridge is Definition H.1 (every model of Part H is a GTPA v2 model), and the falsifier is in the last declared row. GTPA v2 names C0111's signature in words (§0.3). Σ_H fills it in the table form of Definition 1.2.

Obligation	Filled by, in the human sector	Status · level
boundary	per model: the agent's coordinates (lens gain s, constitution θ, any declared model state) and the world's (territory x), as in Model 4.10 Boundary; a person's boundary is not declared	`DECLARED` per model; person: `OPEN`

Obligation	Filled by, in the human sector	Status · level
scale	one agent; collectives only through Model 4.6, which states no self-similarity law: a joint landscape (CO-10, CO-11), a common order, composed self-models and two locked phases, each under its exact conditions	DECLARED
referent	the human agent under Definition H.2; no population, unit or identity measure is declared (GTPA-CENSUS-HUMAN-IDENTITY)	OPEN
observation	inside a model, the lens Π (Definition 2.1); there is no measurement channel from a model to a person	DECLARED; channel to a person: OPEN
coupling	the action coupling B and the effort metric $\lambda_{\text{effort}}\lvert u\rvert^2$ (Definitions 2.1, 2.4); declared costs (Definition 2.9); between two agents of one declared model, a term of their joint landscape (CO-10; Model 4.6, two phases)	DECLARED
constraints	(H) and the noise of the base; the base's coefficients do not depend on time (C6), so a moving territory is a declared model (Model H.10); declared budgets (Model 4.10, Model H.9); no physical constant, temperature or metabolic law	DECLARED
topology	R^n (C1); no manifold singularity, horizon or vortex topology is carried (H-26, H-42); Model 4.9's flow acts only on a declared model manifold	DECLARED
state/domain	Model H.5: $x \in R$, $(s, \theta) \in (0, \infty)^2$	DECLARED
dynamics	Model H.5 (Model 4.7 with a moving territory, noise, a brake and a raising law) and Model H.10	DECLARED; Results H.6–H.8, H.11, H.14 PROVED · 2
model/error signal	E_{model} (the perceived error) and E_{base} (the territory's error)	DECLARED
policy/action/ ablation	the myopic action $u = -k\theta^2 s^2 x$; the ablation $u \equiv 0$ recovers the base exactly (Definition 2.1; Result 3.2 (a)); the brake of Model H.5 is that ablation on an interval	DECLARED; PROVED in B-10 · 2
D_E/Γ	Γ_T (Definition 3.1) with an observable of the territory fixed in advance; Part H compares agents by E_{base} and effort, never by E_{model} alone (the reading note after Result 2.5)	DECLARED
equivalences and falsifiers	equivalences: a change of time unit, which maps $(a, k, \gamma_s, \gamma_\theta)$ to $(a, k, \gamma_s, \gamma_\theta)/c$, and a linear recoding $x' = \ell x$, $s' = s/\ell$, which maps $(k, \gamma_s, \gamma_\theta)$ to $(k\ell^2, \gamma_s/\ell^2, \gamma_\theta)$; both keep kr^2/a and E_{model}, so the verdicts of Results H.6–H.8 do not change. Falsifiers: a verdict that changes under an allowed equivalence (C0111); inside the models, one witness decides the clash of Conjectures H.12 and H.13, and Result H.14 gives it (H.12 false, H.13 holds)	DECLARED; the two maps PROVED here · 2
independent explanandum	none yet: every result of Part H is a core result on a declared model, read under Definition H.2 (GTPA-ZO-12)	OPEN
non-reduction	none shown: without a witness of non-reduction (Result H.14 decides the clash, but it is a core result on a declared model), Part H is the core plus a reading (GTPA-ZO-12)	OPEN

Σ_H is written out, so the sector has a type. It is not admitted. Four rows are OPEN (referent, the observation channel to a person, independent explanandum, non-reduction), and the empirical bridge that C0111 requires before any real application is absent (C0126). Part H can fail in three named ways. (1) As a specialization: a verdict of Results H.6–H.8 changes under an allowed equivalence. (2) As a sector: no independent explanandum is found, so Part H reduces to the core plus a reading. GTPA-ZO-12 then says there is no human sector, only human readings, and every result of Part H stays, as a worked model of the core. (3) As an application: a reading of a person under Definition H.2 fails its first empirical test. Such a test is outside this text.

H.3 Statements

Result H.4 (What Part H takes from GTPA v2 and the notes). The statements below hold for their declared models and are cited, not restated. Each replaces a UTHA kernel named in Annex H-A. The notes' readings about persons stay separate in the notes and are not part of Part H.

Statement	What it gives, inside its model	Status · level
Definition 2.1, Result 3.2 (a)	at $\eta = 0$ the base is recovered exactly: UTHA's Mode I with $u = 0$, and the brake of Model H.5, are this ablation	DECLARED; PROVED in B-10 · 2
Definition 2.4, Results 2.5, 2.6	one control law $u^* = Z^{-1}B^{\top}G_{\text{reg}}F_{\text{req}}$, with $Z \succeq \lambda_{\text{effort}} I$ never singular; the SVD computes Z^{-1} (UTHA §6.1.1, §6.2.1, Mode II)	PROVED in CO-01, CT-05 · 2
Result 2.7	holding a point costs exactly at the points that are not critical points of J	PROVED in CT-01 · 2
Model 4.4	a move with zero rate at both ends: no interval of zero action under quadratic effort; push, zero, reverse under an L1 cost with a cap (UTHA §6.1.2's "pause")	PROVED in CT-06, CT-07 · 2
Model 4.5	added potentials (identity, social) are declared; a gradient field has no curl; without noise, leaving a well needs added control (CT-08 (b); on the declared well of Model 4.5, at effort at least $4\lambda_{\text{effort}}\,\Delta J$, AB-5 (d)); with noise, the first-order base leaves the declared well of Model 4.5 without control, with $\varepsilon \ln E[\tau_D] \to \Delta J$ as $\varepsilon \to 0$	PROVED in CT-08 · 2; Poincaré lemma STANDARD_FACT · 3; AB-5 (d) PROVED in the note · 2; AB-5 (b) PROVED in the note · 3
Model 4.6	full coherence reaches one ray and has a blind spot; the maximizers of Eff·Res lie below full coherence under a sufficient condition (UTHA §13.5)	PROVED in CT-09–CT-11, CO-10, CO-11 · 2
Model 4.7	Class A: collapse from one open set of starts, recovery from another; Class B: coupling, not stability or corrigibility	PROVED in CT-12, CT-13 · 2
Model 4.9	the optional Ricci flow of a model metric; it can end in finite time (UTHA §9.3, §10.1.2)	PROVED in CO-03–CO-06 · 2; CO-02 (c) STANDARD_FACT · 3
Model 4.10	opacity (a lower-dimensional proxy merges states); the exact budget condition; complexity as a declared measure; algorithm costs with no cubic law	STANDARD_FACT · 3; PROVED in D-04 · 3 and in D-06 · 2
Model 4.11	the gross value of acting on calibrated information	PROVED in GTPA v2, Model 4.11 · 2 (its conditioning step · 3)
notes MEM-2–MEM-4	the exact loss of a final code and of a two-symbol compression (UTHA §4.2.2, §11.1)	PROVED · 2
notes MEM-7, MEM-8	an interface that changes: calibrated history helps; a filter that assumes the wrong temporal law can do worse than the current signal alone; learning between two laws	DECLARED + PROVED · 2
notes ACT-1–ACT-4	acting in order to learn one's own interface, with a cost; when not to experiment; a symmetry no adaptive policy breaks	DECLARED + PROVED · 2 (ACT-3 and ACT-4's recoding · 3)
note CONS-08	a miscalibrated self-model can make the decision worse (UTHA §2.3.2)	PROVED · 2

Model H.5 (Class A and Class B, read in the human sector). Declared model: Model 4.7 (CT-12, CT-13) with four declared extensions. The action is Model 4.7's myopic action $u = -k\theta^2 s^2 x = -kr^2 x$. The minimization that gives it does not involve the territory's rate, so it stays the same when the rate varies (recomputed symbolically).

- (i) Territory: $\dot{x} = a(t)x + u$, with $a(\cdot)$ measurable and bounded on bounded intervals (Model 4.7: a constant); with noise, $dx = (a(t)x + u)dt + \sigma\, dW$, W a standard Brownian motion.

- (ii) Descent law: $\dot{s} = -\gamma_s \, \partial_s E_{\text{model}} = -\gamma_s \theta^2 s x^2$, $\dot{\theta} = -\gamma_\theta \, \partial_\theta E_{\text{model}} = -\gamma_\theta \theta s^2 x^2$, with $\gamma_s, \gamma_\theta \geq 0$ not both 0. $\gamma_s = \gamma_\theta$ is Model 4.7's Class A (CT-12). $\gamma_s = 0$ is a mutable constitution with an anchored lens: v1.2's subclass A2, whose response to the error is "Apathy: Modify Theta to ignore pain" (App D.4, p.76). Class B is the excluded case $\gamma_s = \gamma_\theta = 0$ (CT-13), kept as the comparison arm.
- (iii) Brake: $u \equiv 0$ on a declared interval $[0, T]$, then the myopic action again. This is a declared rule: UTHA §6.2.2's "The Control Engine is forcibly shut down (`u -> 0`)" (p.37), the action part of v1.2's watchdog (Def 10.3, p.57) and the action gate of Model 4.8 (iii). On $[0, T]$ the action is the ablation $\eta = 0$ of Definition 2.1.
- (iv) Raising law: s fixed and $\dot{\theta} = +\gamma \theta s^2 x^2$, $\gamma > 0$. This is the θ-law of (ii) with the opposite sign: the weight on the perceived error grows with that error. It is a mutable constitution in the sense of Def D.1, but the opposite of v1.2's "Apathy".

"Human" enters only through Definition H.2; the model is the same object as Model 4.7. `DECLARED`.

Result H.6 (The descent law only lowers authority). `PROVED` here · level 2; the convergence of a continuous local martingale used in (c) is a `STANDARD_FACT` · 3. In Model H.5 with the descent law (ii), from $x_0 \neq 0$, $s_0, \theta_0 > 0$:

- (a) With or without noise, s and θ stay positive and do not increase. So $r(t) \leq r(0)$ and $\dot{r} = -(\gamma_\theta s^2 + \gamma_s \theta^2) r x^2 \leq 0$.
- (b) Without noise, for every territory $a(\cdot)$: $|x(t)| \geq |x^B(t)|$ for all $t \geq 0$, where x^B is the Class B agent frozen at the same start ($r \equiv r(0)$). On no territory, however its rate changes, does this plasticity make the territory's error smaller than the frozen agent does.
- (c) With noise ($\sigma > 0$) and $a(\cdot)$ bounded: $\tau(t) \to \infty$ and $r(t) \to 0$ almost surely, from every start. For a constant $a > 0$, it follows that from an almost surely finite time on $kr(t)^2 < a$, which is the condition under which CT-12 (b) gives collapse without noise. The recovery of CT-12 (c) does not survive noise.

Proof. The coefficients are locally Lipschitz and grow at most linearly in x, so the solution is global. (a) $s(t) = s_0 \exp(-\gamma_s \int_0^t \theta^2 x^2)$ and $\theta(t) = \theta_0 \exp(-\gamma_\theta \int_0^t s^2 x^2)$; differentiate $r = s\theta$. (b) $d \ln|x|/dt = a(t) - kr(t)^2 \geq a(t) - kr(0)^2 = d \ln|x^B|/dt$; integrate. (c) By Itô's formula, $x_t^2 = x_0^2 + \int_0^t (2(a - kr^2)x^2 + \sigma^2)\, ds + M_t$ with $M_t := 2\sigma \int_0^t x \, dW$ and $\langle M \rangle_t = 4\sigma^2 \tau(t)$. On the event $\{\tau(\infty) < \infty\}$, M_t converges (a continuous local martingale converges on $\{\langle M \rangle_\infty < \infty\}$: Revuz–Yor 1999, Ch. IV, Prop. 1.26), and the drift integral is bounded by $2(\sup|a| + kr(0)^2)\tau(\infty)$. So $x_t^2 \geq \sigma^2 t - C \to \infty$ there, which contradicts $\tau(\infty) < \infty$; hence $\tau(t) \to \infty$ almost surely. The rates of s and θ are x^2 times functions of (s, θ), so $(s, \theta)(t) = \Phi(\tau(t))$, where Φ solves $ds/d\tau = -\gamma_s \theta^2 s$, $d\theta/d\tau = -\gamma_\theta s^2 \theta$ from (s_0, θ_0). If both had positive limits s_∞, θ_∞ along Φ, a positive rate, say γ_θ, would give $d\theta/d\tau \leq -\gamma_\theta s_\infty^2 \theta_\infty < 0$ for all τ, which is impossible for a positive function. So $r(\Phi(\tau)) \to 0$ as $\tau \to \infty$, and $r(t) \to 0$ almost surely. ∎

Checks: the descent rates, $\dot{r}$, the Itô drift of x^2 and the conservation of $\gamma_\theta s^2 - \gamma_s \theta^2$ along Φ were recomputed symbolically. (b) was recomputed numerically on a sign-changing territory, and (c) on one noisy path: from the start of Result H.7's witness with $\sigma = 0.3$, $r \approx 0.013$ at $t = 20$ and $|x| > 10^8$ by $t \approx 36$. The paths are illustrations; the proof is the argument.

Result H.7 (A brake feeds the drift). `PROVED` here · level 2. Model H.5 without noise and with a constant (Model 4.7's territory), the descent law (ii) and the brake (iii) on $[0, T]$. The comparison arm is the agent with no brake from the same start ($x_0 \neq 0$, $s_0, \theta_0 > 0$).

- (a) The brake does not enter the plasticity law. In both arms $(s, \theta)(t) = \Phi(\tau(t))$, with the same curve Φ of Result H.6. The brake changes only how fast the constitution moves along its curve.

- (b) For $0 < t \le T$, $|x_{\text{brake}}(t)| > |x(t)|$. Hence $\tau_{\text{brake}}(t) > \tau(t)$ and $r_{\text{brake}}(t) < r(t)$; also $\theta_{\text{brake}}(t) < \theta(t)$ if $\gamma_\theta > 0$, and $s_{\text{brake}}(t) < s(t)$ if $\gamma_s > 0$.
- (c) If $a > 0$, there is a $T^* < \infty$ such that for every brake longer than T^*, $kr(T)^2 < a$ at the release and $E_{\text{base}} \to \infty$ afterwards. With Model 4.7's rates ($\gamma_s = \gamma_\theta$), also $E_{\text{model}} \to 0$ (CT-12 (b) from the state at T). This holds from every start with $x_0 \neq 0$, including the recovery set of CT-12 (c).
- (d) If $a \le 0$, E_{base} does not increase in either arm.
- (e) With the raising law (iv) in place of (ii), $\ln r(t) = \ln r_0 + \gamma s^2 \tau(t)$, and the same comparison gives $r_{\text{brake}}(t) > r(t)$ on $(0, T]$: the brake speeds up that drift as well.

Witness (level 2, numeric): $a = k = \gamma_s = \gamma_\theta = 1$, $x_0 = 0.3$, $s_0 = \theta_0 = (2\sqrt{2})^{1/2}$, a start in the recovery set of CT-12 (c). Without a brake, $E_{\text{base}} \to 0$ (below 10^{-100} at $t = 25$). With $s = \theta$, $r(\tau) = r_0/(1 + 2r_0\tau)$ and $\tau_{\text{brake}}(T) = x_0^2(e^{2T} - 1)/2$, so $T^* = \tfrac{1}{2} \ln(1 + 2\tau^*/x_0^2)$ with $\tau^* = (r_0 - 1)/(2r_0)$, that is $T^* \approx 1.051$. A brake of $T = 1.5$ leaves $r \approx 0.483$ ($kr^2 \approx 0.233 < a$). After the release, $E_{\text{base}}(12) \approx 1.0 \cdot 10^9$ and $E_{\text{model}}(12) \approx 2.4 \cdot 10^{-10}$.

Proof. (a) The right sides of $\dot{s}$ and $\dot{\theta}$ are x^2 times functions of (s, θ); uniqueness. (b) $d \ln |x_{\text{brake}}|/dt = a > a - kr(t)^2 = d \ln |x|/dt$, because $r > 0$ (Result H.6 (a)); integrate from 0, then use (a) and the strict decrease along Φ of each component with a positive rate. (c) $\tau_{\text{brake}}(T) = x_0^2(e^{2aT} - 1)/(2a) \to \infty$, and $r(\Phi(\tau)) \to 0$ as $\tau \to \infty$ (proof of Result H.6 (c)). Let T^* be the first T with $kr_{\text{brake}}(T)^2 \le a$; r_{brake} decreases strictly, so every longer brake ends with $kr(T)^2 < a$. After the release r does not increase, so $d \ln |x|/dt \ge a - kr(T)^2 > 0$. (d) $d \ln |x|/dt \le a \le 0$ in both arms. (e) As (b), with $\dot{r} = \gamma s^2 r x^2$. ∎

Checks: the closed form of $r(\tau)$, the time change (the two arms give the same θ at the same τ, to $5 \cdot 10^{-8}$), the strict inequalities of (b), the witness and (e) were recomputed numerically.

Result H.8 (A raising law: viable without noise, unsettled with it). PROVED here · level 2; Barbalat's lemma and the martingale convergence used in (b) are STANDARD_FACT · 3. Model H.5 with the raising law (iv), s fixed, from $x_0 \neq 0$ and $\theta_0 > 0$.

- (a) Without noise, for every bounded measurable $a(\cdot)$, whatever its bound and however fast it changes, $x(t) \to 0$, and $r(t)$ increases to a finite limit. For every fixed authority r (a Class B agent) there is such a territory, $a \equiv kr^2 + 1$, on which $|x(t)| = |x_0|e^t \to \infty$. Against territories with no known bound on their rate, this plastic law keeps the territory and no fixed constitution does.
- (b) With noise ($\sigma > 0$) and $a(\cdot)$ bounded, $r(t) \to \infty$ almost surely: the weight on the perceived error, and with it the action gain kr^2, grows without bound.

Proof. (a) Let $M \ge \sup |a|$ and $r^* > 0$ with $kr^{*2} > M$. Along $\dot{r} = \gamma s^2 r x^2$, the function $V := \tfrac{1}{2}x^2 + (k/(2\gamma s^2))(r^2 - r^{*2} - r^{*2} \ln(r^2/r^{*2}))$ satisfies $\dot{V} = (a(t) - kr^{*2})x^2 \le -(kr^{*2} - M)x^2 \le 0$. The second term of V is ≥ 0 and tends to ∞ as $r \to 0$ or $r \to \infty$. So r stays in a compact subset of $(0, \infty)$, x is bounded, and $\int_0^\infty x^2 \le V(0)/(kr^{*2} - M)$. The derivative of x^2 is bounded, so x^2 is uniformly continuous and tends to 0 (Barbalat's lemma; Khalil 2002, Lemma 8.2). r does not decrease and is bounded, so it converges. The fixed-authority case is $\dot{x} = (a - kr^2)x = x$. (b) On the event $\{\tau(\infty) < \infty\}$, $\ln r(t) = \ln r_0 + \gamma s^2 \tau(t)$ stays bounded, so the gain is bounded there, and the argument of Result H.6 (c) gives $\tau(\infty) = \infty$ on that event. So the event has probability 0, and $r(t) \to \infty$ almost surely. ∎

Checks: the identity for $\dot{V}$ and the properties of its second term were recomputed symbolically. (a) was recomputed on a territory that switches at random times between -2 and 6 ($x(60) < 10^{-300}$; the gain kr^2 settles near 36.5). (b) was recomputed on one noisy path ($a = 1$, $\sigma = 0.3$: the gain grows from 2 to about 42 by $t = 400$). The paths are illustrations.

Model H.9 (A three-part budget). Declared model, from UTHA §2.2.1 (p.21), for EG-07.c. A declared total $P_{\text{total}} > 0$ is split into three declared non-negative parts $p = (p_{\text{def}}, p_{\text{rep}}, p_{\text{ext}})$ with

$p_{\text{def}} + p_{\text{rep}} + p_{\text{ext}} = P_{\text{total}}$ (the Reino's MR-K05 writes the same sum). When P_{total} and p_{ext} are held fixed by declaration, a change of p_{def} is matched by the opposite change of p_{rep}. This is UTHA §2.2.2's "`Delta P_repair = - Delta P_defense`" (p.22), and it is arithmetic, not a law. If a model follows the split over time, the parts are declared rates, and whether the budget lasts is Model 4.10's exact condition (D-04). Not carried: that P_{total} is "finite and strictly conserved", that the parts are "mutually exclusive modes" of an organism (physics is CLOSED; this is a claim about people), and the safety gate that moves the split (H-16). `DECLARED`; the identity recomputed symbolically · 2.

H.4 The Class A clash: two rival conjectures

GTPA v1.2 calls Class A, with mutable constitution and floating perception, "Natural/Pathological" and "thermodynamically prone to "Wireheading" and Narcissistic Collapse" (summary, pp.3–4). Its law is Thm 10.1 (p.56), and Ch. 10 ends with Class A agents "mathematically destined for insanity or obsolescence" (p.58). UTHA re-evaluates the same class as "the Optimal Strategy for Chaos" (abstract §1, p.3). It gives Class B the static and ergodic environments, where it is "thermodynamically optimal", and "Brittle Fracture in radical uncertainty" (p.3). The architecture record enters this clash as two rival conjectures (its L143 and L200).

What is already settled:

- Thm 10.1 fails as a law about Class A. In Model 4.7, collapse happens from one open set of starts and recovery from another, and E_{base} never grows when $a \leq 0$ (CT-12; EG-03.3). Its "maximizing" of S_{total} is an erratum (CT-12 (e)). Class B gives coupling, not stability or corrigibility (CT-13; EG-03.4).
- Model 4.7's own plasticity can only lower the agent's authority. On any territory it never makes the territory's error smaller than its frozen start does, and with noise its authority vanishes from every start (Result H.6). The pathology lies in the sign of the response, not in plasticity as such: the law lowers the gain on the perceived error, through the lens (v1.2's "Denial") and through the constitution (its "Apathy"; App D.4, p.76).
- The opposite response is viable uniformly over all bounded territories, which no single fixed constitution is. Without noise, the raising law keeps every bounded territory, at any speed of change, while each fixed constitution loses some territory (Result H.8 (a)). With noise its authority grows without bound (Result H.8 (b)).
- In a static territory with a known rate, the right fixed constitution is optimal among all policies, plastic or not (Result H.11 (b)). In this model, that is the static half of UTHA's clause for Class B.

What remains is the comparison that UTHA actually asserts: a moving territory, noise, an effort cost, and a criterion that compares the best plastic agent with the best fixed one. Model H.10 declares it.

Model H.10 (A switching territory). Declared model, Model H.5 with:

- territory $dx = (a(t)x + u)dt + \sigma\, dW$, $\sigma > 0$. Here $a(t) \in \{a_-, a_+\}$, $a_- < a_+$, is a Markov chain that switches at rate $\nu > 0$ in each direction, is independent of W, starts in its stationary law (½, ½), and is not observed by the agent;
- an anchored lens, $s = 1$, so the agent sees x; the action is $u = -kr(t)^2 x$;
- Class B: r constant. Class A: $r(t) = R(z(t))$, where $z \in R^p$ is an internal state with $\dot{z} = \psi(z, x)$, ψ locally Lipschitz, $R \geq 0$ continuous, and the closed loop well posed. $\mathcal{A}$ is the set of these laws. They read the territory only through the lens image and their own state; Class B is the case $p = 0$. The descent law, the raising law and laws with a set point are in $\mathcal{A}$; a law that reads $a(t)$ is not;

- criterion: $\mathcal{J} \coloneqq \limsup_{T\to\infty} T^{-1} E \int_0^T (½\lambda_{\text{effort}}\, u^2 + ½\beta x^2)\, dt$. This is Model 4.7's running cost with the territory's error E_{base} in place of E_{model}, the comparison fixed in advance (the reading note after Result 2.5).

DECLARED.

UTHA's text points both ways in Model H.10. The model is stationary and ergodic, and UTHA's clause for Class B covers ergodic environments (p.3); its regime clause (p.4) gives H.13⁺ below. Conjecture H.13 is attributed to UTHA through the regime clause only (level 4). Result H.14 goes against the ergodic clause. It gives H.13⁺, the comparison in the regime clause, but not as a regime, since plasticity pays at every switching rate (H-04).

Result H.11 (The Class B side, explicitly). PROVED here · level 2; the clause of (a) on a matrix that is not Hurwitz is level 3 (its argument, in step (1) of Result H.14, was checked by hand in the release read of 2026-10-06).

- (a) For a constant r, with $\rho_\pm \coloneqq kr^2 - a_\pm$, the second moments $m_\pm \coloneqq E[x^2; a = a_\pm]$ solve $\dot m_\pm = -2\rho_\pm m_\pm + ½\sigma^2 + \nu(m_\mp - m_\pm)$. They converge iff the matrix $[[-2\rho_+ - \nu, \nu], [\nu, -2\rho_- - \nu]]$ is Hurwitz, that is iff $\rho_+ + \rho_- + \nu > 0$ and $2\rho_+\rho_- + \nu(\rho_+ + \rho_-) > 0$. Then $E[x^2] \to \sigma^2(\rho_+ + \rho_- + 2\nu)/(4\rho_+\rho_- + 2\nu(\rho_+ + \rho_-))$ and $\mathcal{J}_B(r) = ½(\lambda_{\text{effort}}\, k^2 r^4 + \beta)\cdot\sigma^2(\rho_+ + \rho_- + 2\nu)/(4\rho_+\rho_- + 2\nu(\rho_+ + \rho_-))$. As $\nu \to 0$ the stationary value tends to the mean of the two modes' values $\sigma^2/(2\rho_\pm)$, that is $\sigma^2/(4\rho_+) + \sigma^2/(4\rho_-)$, and as $\nu \to \infty$ to $\sigma^2/(\rho_+ + \rho_-)$, the value of the averaged territory. When the matrix is not Hurwitz, $\mathcal{J}_B(r) = +\infty$ (its dominant eigenvalue is ≥ 0 and the forcing is positive). So $\inf_r \mathcal{J}_B(r)$ is an explicit minimization in one variable.
- (b) One mode with a known rate a (a static territory): let $kr_*^2 \coloneqq a + (a^2 + \beta/\lambda_{\text{effort}})^{1/2}$ and $P \coloneqq \lambda_{\text{effort}}\, kr_*^2$. Then $½\lambda_{\text{effort}}\, u^2 + ½\beta x^2 + Px(ax + u) = ½\lambda_{\text{effort}}(u + kr_*^2 x)^2$, and Itô's formula for $½Px^2$ gives $\mathcal{J} \geq ½P\sigma^2$ for every non-anticipative policy: along a sequence T_n with $E\, x_{T_n}^2/T_n \to 0$ if one exists, and $\mathcal{J} = +\infty$ otherwise. The fixed authority r_* attains $½P\sigma^2$. In a static territory with a known rate, no plastic law beats the right fixed constitution.

Proof. (a) Itô's formula for x^2 within each mode, plus the switching terms; solve the stationary 2×2 system; take the limits. (b) Complete the square, as written. ∎ Checks: the stationary solution, its two limits, the completed square and $\mathcal{J}_B(r_*) = ½P\sigma^2$ were recomputed symbolically. A Monte Carlo time average for $a_\pm = \pm 1$, $\nu = 0.5$, $kr^2 = 2$, $\sigma = 1$ gave 0.312 against the formula's $5/16 = 0.3125$.

Conjecture H.12 (v1.2's side: plasticity does not pay). OPEN until 2026-10-06; now false: its negation is PROVED · 2 (Result H.14), and it fails at every instance and every $\nu > 0$. In Model H.10, for every instance $(a_- < a_+, \sigma, \lambda_{\text{effort}}, \beta)$, every $\nu > 0$ and every law in $\mathcal{A}$, $\mathcal{J}_A \geq \inf_r \mathcal{J}_B(r)$.

This is the strongest form of v1.2's side that GTPA v2.0 does not already refute: in a moving territory, a constitution that moves never does better than the best fixed one. Result H.14 refutes it.

Conjecture H.13 (UTHA's side: plasticity pays where the territory moves). OPEN until 2026-10-06; now PROVED · 2 by Result H.14, with H.13⁺: for every instance and every $\nu > 0$, and so wherever $\nu > \bar\rho^*$, for instance at the instance of the exploration below with $\nu = 10 > \bar\rho^* \approx 1.452$. In Model H.10 there are an instance $(a_- < a_+, \sigma, \lambda_{\text{effort}}, \beta)$, a $\nu > 0$ and a law in $\mathcal{A}$ with $\mathcal{J}_A < \inf_r \mathcal{J}_B(r)$. UTHA also names the regime: "In environments where `grad(J)` evolves faster than the system's relaxation time" (p.4). In Model H.10 this is $\nu > \bar\rho^* \coloneqq ½(\rho_+ + \rho_-)$ at a best fixed authority; call that stronger form H.13⁺.

What decided between them. H.12 and H.13 are each other's negation, so exactly one holds, and one argument decides. Result H.14 is that argument.

- For H.13: one instance $(a_\pm, \nu, \sigma, \lambda_{\text{effort}}, \beta;\ k = \beta/\lambda_{\text{effort}})$ and one law in $\mathcal{A}$ whose $\mathcal{J}$ is shown, by proof, to lie below the explicit $\inf_r \mathcal{J}_B(r)$ of Result H.11 (a). Result H.14 gives such a law for every instance and every ν, so also with $\nu > \bar\rho^*$ (H.13⁺).

- For H.12, a verification argument like that of Result H.11 (b) would have needed a value function that does not use the hidden mode. There is none: a best fixed constitution g^* is not a local optimum along the family $kr^2 = g^* + \alpha\varphi(z)$, $\alpha \to 0+$ (Result H.14).
- Where the witness is (a reading, level 4): the descent law ends with no authority (Result H.6 (c)), and the pure raising law's authority does not settle (Result H.8 (b)). The explicit witness of Result H.14 (b) is a raising law with a leak: its authority rises with a leaky average of the squared lens image, up to a cap, and falls back when that average is small.

An exploration, not a decision. It ran in a scratch folder, with Euler step 0.005 and 400 paths, and has no certified error bound, so it carries no check level. Take $a_\pm = \pm 2$ and $\sigma = \lambda_{\text{effort}} = \beta = k = 1$, and the law that moves the gain kr^2 at the relative rate $\gamma(x^2/m_* - 1)$, with $m_* = 1/(2\sqrt{5})$. Each path starts at $x = 0$ with gain $kr^2 = 2$ and the mode drawn from (½, ½); the horizon is $T = 3000$ at $\nu = 0.01$ and 1500 otherwise, with the first 200 time units discarded; numpy seeds 777 (the runs at $\nu = 0.01$ with $\gamma = 0.25$ and the fixed-gain check) and 12345 (the rest). This m_* is the stationary $E[x^2]$ under the optimal gain of either mode alone (Result H.11 (b)), so in a mode that lasts, the law's set point is that gain. At $\nu = 0.01$ it gave $\mathcal{J} \approx 1.285 \pm 0.020$ (two standard errors; $\gamma = 0.25$) against $\inf_r \mathcal{J}_B = 1.400$, a value the same scheme reproduces (1.396 ± 0.014). At $\nu = 0.1$, 1 and 10 the law with $\gamma = 0.5$ gave $\mathcal{J} \approx 1.55$, 1.51 and 1.01, against 1.385, 1.249 and 0.705. Here $\bar{\rho}^*$ is about 3.64, 3.59, 3.09 and 1.45 at $\nu = 0.01, 0.1, 1$ and 10 (Result H.14 (c)). The exploration did not decide the clash. Result H.14 does, by proof and with another law, at every ν, including UTHA's regime ($\nu = 10 > \bar{\rho}^*$). So the explored law is not the witness; the exploration suggests (no level) that it does worse at $\nu = 0.1$, 1 and 10. Which laws pay, and by how much, is left open (§H.6).

Result H.14 (Plasticity pays in Model H.10, at every switching rate). PROVED here · level 2 for its decisive calculations, and level 3 for the argument of (1) off the stable set, the estimates of (3) and the argument of (4), which independent readers checked by hand; Itô's formula for the switching diffusion (Mao and Yuan 2006, Ch. 1) and the fact that a finite measure is determined by its integrals of bounded Lipschitz functions (Billingsley 1999, Thm 1.2) are STANDARD_FACT · 3. Model H.10, with any instance ($a_- < a_+, \sigma, \lambda_{\text{effort}}, \beta > 0, k = \beta/\lambda_{\text{effort}}$), any $\nu > 0$ and any start $x(0) = x_0$. Write $g := kr^2$ for the gain, so $u = -gx$ and the running cost is $\tfrac{1}{2}(\lambda_{\text{effort}}\, g^2 + \beta)x^2$.

- (a) $\mathcal{J}_B$ has a minimizer $g^* > 0$, in the open set of gains where the matrix of Result H.11 (a) is Hurwitz, so $\inf_r \mathcal{J}_B(r) = \mathcal{J}_B(g^*)$. There are a filter rate $\kappa > 0$, a bounded Lipschitz profile $\varphi \geq 0$ and an $\alpha_0 > 0$ such that, for $0 < \alpha < \alpha_0$, the law $\dot{z} = \kappa(x^2 - z)$, $z(0) = 0$, $kr^2 = g^* + \alpha\varphi(z)$ is in $\mathcal{A}$ and has $\mathcal{J} < \inf_r \mathcal{J}_B(r)$. So, for every instance and every $\nu > 0$, Conjecture H.12 is false and H.13 holds, and H.13$^+$ holds wherever $\nu > \bar{\rho}^*$.
- (b) An explicit witness. Suppose $S_2 := 4\rho_+^*\rho_-^* + \nu(\rho_+^* + \rho_-^*) > 0$, so that the loop with the fixed gain g^* has a finite stationary fourth moment; this holds whenever $g^* \geq a_+$. Then for every $\kappa > 0$, $D_z := E_*[z\, f_a\, x^2] < 0$, and there are a cap $\bar{z} > 0$ and an $\alpha_0 > 0$ such that the law of (a) with $\varphi(z) = \min(z, \bar{z})$ beats every fixed constitution for $0 < \alpha < \alpha_0$. Its authority rises with z, a leaky average of the squared lens image x^2, up to the cap, and falls back when x^2 stays small.
- (c) The instance of the exploration above ($a_\pm = \pm 2$, $\sigma = \lambda_{\text{effort}} = \beta = k = 1$), certified in the table below (g^* isolated exactly, every entry enclosed by interval arithmetic, here rounded). So (b) applies at all four rates. At $\nu = 10 > \bar{\rho}^*$ this is H.13$^+$, with $g^* < a_+$: the best fixed loop is unstable while the mode a_+ lasts and stable only on average.

ν	$g^* = \bar{\rho}^*$	$\inf_r \mathcal{J}_B(r)$	S_2	D_z **at** $\kappa = 0.1, 1, 10$
0.01	3.640920	1.400254	37.10	−0.02064, −0.03564, −0.06203
0.1	3.586538	1.385489	36.17	−0.009389, −0.03252, −0.06200
1	3.093325	1.248587	28.46	−0.002939, −0.02170, −0.06129
10	1.452369	0.704693	21.48	−0.0002654, −0.002219, −0.01163

ν	$g^* = \bar\rho^*$	$\inf_r \mathcal{J}_B(r)$	S_2	D_z **at** $\kappa = 0.1, 1, 10$

The condition of (b) can fail: at $a_+ = ¼$, $a_- = -7/4$, $\nu = ½$, $\sigma = \lambda_{\text{effort}} = 1$, $\beta = 10^{-3}$, $g^* \approx 0.0740$ and $S_2 \approx -0.46$ (exact isolation and sign). There only (a) applies, and no explicit witness is given. The gain shown is of first order in α; Result H.14 computes neither its size nor the best law of $\mathcal{A}$, and it does not compare the descent law, the raising law or the explored law with the best fixed constitution. And it needs no regime: plasticity pays at every ν, whether the territory switches slower or faster than the loop relaxes.

Proof. (1) Fixed gain. For a constant gain g in the stable set, $V(x,a) = ½P_a\,x^2 + v_a$ solves the Poisson equation $½(\lambda_{\text{effort}}\,g^2+\beta)x^2 + \mathcal{L}_g V = \mathcal{J}_B(g)$, with $\mathcal{L}_g$ the generator of (x,a), iff $(\rho_\pm + ½\nu)P_\pm - ½\nu P_\mp = ½(\lambda_{\text{effort}}\,g^2+\beta)$, that is $P_\pm = (\lambda_{\text{effort}}\,g^2+\beta)(\rho_\mp+\nu)/S_1$ with $S_1 := 2\rho_+\rho_- + \nu(\rho_+ + \rho_-) > 0$ (Result H.11 (a)). Both are positive, and the constant terms give $\mathcal{J}_B(g) = ¼\sigma^2(P_+ + P_-)$, which is Result H.11 (a)'s closed form. $\mathcal{J}_B$ is $+\infty$ off the stable set: there, let $\mu \ge 0$ be the largest eigenvalue of the symmetric matrix of Result H.11 (a) and $\omega > 0$ its eigenvector (its off-diagonal entries are $\nu > 0$). The moments $m_\pm(t)$ of $x(t)^2$ on $\{a = a_\pm\}$ give $(d/dt)(\omega_+ m_+ + \omega_- m_-) = \mu(\omega_+ m_+ + \omega_- m_-) + ½\sigma^2(\omega_+ + \omega_-) \ge ½\sigma^2(\omega_+ + \omega_-)$, so $E\,x(t)^2$ grows at least linearly from every start, and $\mathcal{J}_B = +\infty$ because $\beta > 0$. On the stable set, $\mathcal{J}_B$ tends to $+\infty$ at its boundary (where $S_1 \to 0$) and as $g \to \infty$; if $g = 0$ is stable, $\mathcal{J}_B'(0) < 0$ by (2). So a minimizer $g^* > 0$ exists and is interior.

(2) The identity. Let V^* be this V at g^*. For any law with a bounded gain $g(t)$ and $\sup_t E\,x(t)^2 < \infty$, Itô's formula for $V^*(x,a)$ and the Poisson equation give, since $E\,V^*(x(T),a(T))/T \to 0$, $\mathcal{J} = \mathcal{J}_B(g^*) + \limsup_{T\to\infty} T^{-1} E\int_0^T (g-g^*)(½\lambda_{\text{effort}}(g+g^*) - P_a)x^2\,dt$. For constant gains the same computation gives $\mathcal{J}_B'(g) = E_g[(\lambda_{\text{effort}}\,g - P_a)x^2]$, under the stationary law at g. At the minimizer this is $f_+ m_+ + f_- m_- = 0$, with $m_\pm$ of Result H.11 (a) at g^*. As $P_+ - P_- = (\lambda_{\text{effort}}\,{g^*}^2+\beta)(a_+ - a_-)/S_1 > 0$, $f_+ < 0 < f_-$: while the mode a_+ lasts the fixed gain is too low, while a_- lasts too high.

(3) First order. Let $\varphi \ge 0$ be bounded and Lipschitz, $\kappa > 0$ and $D_\varphi := E_*[\varphi(z)\,f_a\,x^2]$, where E_* is the stationary law of (x,z,a) with the fixed gain g^*. It exists: with a two-sided chain and Brownian motion, $x(t) = \sigma\int_{-\infty}^t \exp(\int_s^t (a(t')-g^*)\,dt')\,dW_s$ converges in mean square because g^* is in the stable set, and $z(t) = \kappa\int_{-\infty}^t e^{-\kappa(t-s)}x(s)^2\,ds$. We show that the law $kr^2 = g^* + \alpha\varphi(z)$, $z(0) = 0$ (well posed: its gain is bounded), has $\mathcal{J}(\alpha) \le \mathcal{J}_B(g^*) + \alpha D_\varphi + o(\alpha)$ as $\alpha \to 0+$. By (2), $\mathcal{J}(\alpha) - \mathcal{J}_B(g^*) = \limsup T^{-1}\int_0^T E[\alpha\varphi(z)\,f_a\,x^2 + ½\lambda_{\text{effort}}\,\alpha^2\varphi(z)^2x^2]\,dt$. Moments: for real $n \ge 1$ let μ_n be the largest eigenvalue of $[[-2n\rho_+^* - \nu, \nu], [\nu, -2n\rho_-^* - \nu]]$ and $\omega^{(n)} > 0$ its eigenvector. $\mu_1 < 0$ (Result H.11 (a)), so by continuity $\mu_n < 0$ for some real $n > 1$ ($n = 2$ needs $S_2 > 0$, as in (5)); since the gain is $\ge g^*$, $\mathcal{L}(\omega_a^{(n)}|x|^{2n}) \le \mu_n\,\omega_a^{(n)}|x|^{2n} + n(2n-1)\sigma^2\omega_a^{(n)}|x|^{2n-2}$, so $E|x(t)|^{2n}$ is bounded in t and in $\alpha \le 1$, and the α^2 term is $O(\alpha^2)$. Coupling: run the law and the fixed loop (x°, z°) on the same W, chain and start. Then $\xi := x - x^\circ$ solves $\dot\xi = (a-g^*)\xi - \alpha\varphi(z)x$, with no noise, so, with $\omega := \omega^{(1)}$, $(d/dt)E[\omega_a\xi^2] \le \mu_1\,E[\omega_a\xi^2] + 2\alpha(\sup\varphi)(\max\omega)(E\xi^2)^{1/2}(Ex^2)^{1/2}$ and $\sup_t E\xi^2 = O(\alpha^2)$. Next, $|z - z^\circ| \le \kappa\int_0^t e^{-\kappa(t-s)}|\xi||x + x^\circ|\,ds$ has mean $O(\alpha)$; with the Lipschitz bound and the uniform integrability of x^2 (moments of order $2n > 2$), $\sup_t |E[\varphi(z)\,f_a\,x^2] - E[\varphi(z^\circ)\,f_a\,x^{\circ 2}]| \to 0$ as $\alpha \to 0$. The same coupling of the fixed loop with its stationary version ($E[\omega_a\xi^2]$ then decays like $e^{\mu_1 t}$) gives $T^{-1}\int_0^T E[\varphi(z^\circ)\,f_a\,x^{\circ 2}]\,dt \to D_\varphi$. So if $D_\varphi < 0$, $\mathcal{J}(\alpha) < \inf_r \mathcal{J}_B(r)$ for all small $\alpha > 0$.

(4) Proof of (a): some κ and φ have $D_\varphi < 0$. Under E_*, x^2 has no atom (given the chain, x is Gaussian with positive variance), and $m_+ \ne m_-$ (equal values would force $\rho_+^* = \rho_-^*$ in the equations of Result H.11 (a)). If $E_*[\chi(x^2)\,f_a\,x^2] = 0$ for every bounded Lipschitz χ, the measure $B \mapsto E_*[1_B(x^2)\,f_a\,x^2]$ is zero (Billingsley 1999, Thm 1.2 and its proof), so $f_+ y\,L_+(dy) = -f_- y\,L_-(dy)$ for the laws $L_\pm$ of x^2 on $\{a = a_\pm\}$. On $(0,\infty)$ this makes L_+ a multiple of L_-; both have mass ½, so $L_+ = L_-$ and $m_+ = m_-$, a contradiction. So some such χ has $E_*[\chi(x^2)\,f_a\,x^2] \ne 0$, and, since $E_*[f_a\,x^2] = 0$, one of $\varphi := \sup|\chi| \pm \chi \ge 0$ has $E_*[\varphi(x^2)\,f_a\,x^2] < 0$. As $\kappa \to \infty$, $z \to x^2$ in L^1 under E_* (time shifts of x^2 are continuous in L^1), so $D_\varphi \to E_*[\varphi(x^2)\,f_a\,x^2]$ by uniform integrability. Take κ large and apply (3).

(5) Proof of (b). Under $S_2 > 0$ the bound of (3) holds with $n = 2$, so $E_* x^4 < \infty$, and the stationary moments $m_\pm$, $E_*[x^4; a = a_\pm]$, $E_*[z; a = a_\pm]$ and $w_\pm := E_*[zx^2; a = a_\pm]$ are finite. They are the limits of the moments of the fixed loop started at $x = z = 0$, which solve the linear equations of Itô's formula for x^2, x^4, z and zx^2 in each mode (given the chain, x is Gaussian, so every moment is finite at finite times; the coupling of (3) contracts fourth moments too when $S_2 > 0$). Since $f_+ m_+ + f_- m_- = 0$, $D_z = f_+ N / m_-$ with $N := w_+ m_- - w_- m_+$, and solving the system gives $N = \kappa\sigma^6(a_+ - a_-)Q/(8(\kappa + 2\nu)S_1^2 S_2 S_3)$, where $\bar\rho^* > 0$ (the matrix of Result H.11 (a) is negative definite and its quadratic form at $(1,1)$ is $-4\bar\rho^*$), $S_3 := \kappa^2 + 2\kappa\nu + 4\kappa\bar\rho^* + 2S_1 > 0$ and $Q := S_2^2 + S_2(3\kappa^2 + 12\kappa\nu + 12\kappa\bar\rho^* + 8\nu^2 + 22\nu\bar\rho^*) + 6\kappa^2\nu(\nu + 2\bar\rho^*) + 12\kappa\nu(\nu + \bar\rho^*)(\nu + 2\bar\rho^*) + 24\nu^2\bar\rho^*(\nu + 2\bar\rho^*) > 0$. So $N > 0$ and $D_z < 0$, for every $\kappa > 0$. As $E_*[zx^2] \le E_* x^4$ (Jensen for z, then Cauchy–Schwarz), dominated convergence gives $D_{\min(z,\bar z)} \to D_z$ as $\bar z \to \infty$; take $\bar z$ with $D_{\min(z,\bar z)} < 0$ and apply (3). If $g^* \ge a_+$, both $\rho^*_\pm \ge 0$, so $S_2 > 0$. (c) g^* is the only root of the numerator of $\mathcal{J}_B'$, a quartic with rational coefficients, above the stability bound; it is isolated exactly, and the rest is interval arithmetic on the closed forms. ∎

Checks: by the producer, 23 symbolic checks for (1), (2) and (5): the Poisson solution and $\mathcal{J}_B = \tfrac{1}{4}\sigma^2(P_+ + P_-)$ against Result H.11 (a), the identity of (2) in both modes, $\mathcal{J}_B'(g) = E_g[(\lambda_{\text{effort}}\, g - P_a)x^2]$, the generator of zx^2, the moment system, the formula for N with its denominator, $D_z = f_+ N / m_-$, and Q with all its coefficients positive in $(S_2, \bar\rho^*, \kappa, \nu)$; the estimates of (3) and the argument of (4) are written only. For (c), exact root isolation at the four rates and 30 interval-certified signs, all as stated; for the failing instance, exact isolation and sign. Illustrations, with no level: a Monte Carlo over the hidden chain only (given the chain, x is Gaussian; 2,000 chain paths) gave $D_z \approx -0.0217 \pm 0.0006$ (two standard errors) at $\nu = 1$, $\kappa = 1$ and -0.0023 ± 0.0009 at $\nu = 10$, against -0.02170 and -0.00222; and with an Euler scheme (step 0.005, 2,000 paths, $T = 400$, common random numbers), the law of (b) with $\nu = \kappa = 1$, $\bar z = 4$ and $\alpha = 0.05$ cost 0.0009 ± 0.0001 less than the fixed loop on the same paths. At $\nu = 10$ the same law cost more at $\alpha = 0.1$, 0.2 and 0.4, but at that step the scheme's bias in the first-order coefficient is larger than D_z (the Euler scheme's own coefficient is about $+0.009$ at $\kappa = 1$, from its exact stationary moments, against $D_z \approx -0.0022$), so no run at step 0.005 can show the gain at $\nu = 10$. Read independently on 2026-10-06 by two Claude readers. A proof lens recomputed (1), (2) and (5) symbolically, g^* and every entry of the table with its own scripts, and checked (3) and (4) by hand. A recomputation lens, from scratch and before opening the producer's scripts, recomputed 19 symbolic checks, g^* by exact isolation, every certified sign of (c) and of the failing instance, and the Euler scheme's exact stationary moments. Neither found a disagreement with any number here. The release read of GTPA v2.1 (2026-10-06) checked the argument of (1) off the stable set by hand.

Both conjectures, and Result H.14, are statements about a declared model. That H.13 holds in it says nothing about people.

H.5 EG-06.b and EG-07.c

EG-06.b. Triage of 2026-09-25, row 32 (.json, group G2-EG01-06-07-08, `items[4]`, `reviews[4]`). FORMACION asked: «Within the formalism, are the brake (u->0) and Θ-drift «Apathy» independent, coupled, or one mechanism?» The reviewer held it as `HOLD_BRAKE_OBJECT_UNSETTLED`: «it needs an exact v1.2 brake object, or admission of an immutable UTHA edition. After that it re-enters as a DERIVATION_REQUEST.» GTPA v2 §5 left it for Part H.

Answer. The reviewer's condition is met: UTHA v1.0 is the adopted pin (H.0.1), and its brake, the thermal fuse of §6.2.2 (p.37), is the declared rule (iii) of Model H.5. That rule is also the action part of Def 10.3 and Model 4.8's gate. The triage's exact question asks about an agent with $\mu_\Theta > 0$ when «u is held at 0 over [0,T] with the error O(x)<0 unresolved». Posed on Model 4.7, Result H.7 answers it: **coupled; neither independent nor one mechanism.** The two are separate rules on separate variables: the brake acts on u, the drift on (s, θ), and each occurs without the other (Class B with a brake, Class A without one). They are coupled through the territory's state. The brake changes neither the drift's law nor its curve, only its speed, and in the direction of more drift (Result H.7 (a), (b)). Held

long enough when the territory runs away ($a > 0$), it makes $E_{\text{base}} \to \infty$ from every start, including the starts from which the agent without a brake recovers; with Model 4.7's rates ($\gamma_s = \gamma_\theta$) this is the full collapse of CT-12 (b) (Result H.7 (c)). While the brake holds, the accumulated drift is larger ($\theta_{\text{brake}}(t) < \theta(t)$ for $0 < t \le T$); Result H.7 does not order the instantaneous rates.

The second question, whether Def 10.3 can co-occur with Class A, is answered yes. GTPA v2 keeps Def 10.3 as a declared stopping rule with no class tag (Model 4.7; CROSSWALK §3), and Result H.7 is such a co-occurrence. Def 10.3 also sets `Γ_cog -> Infinity`, a model-side freeze. A model that reads this as also freezing (s, θ) stops the drift while the brake holds; the coupling is then part of the declared rule, not a result.

What it does not say. The question's «S_total-optimal d(Theta)/dt» is not defined in GTPA v2: the plasticity law is declared (Model 4.7), and v1.2's "maximizing" is an erratum (CT-12 (e)). The answer is therefore for the declared descent law; another plasticity law needs its own computation. For the raising law, the brake speeds up the opposite drift (Result H.7 (e)). Nothing here is about apathy, willpower or diets in people. FORMACION may quote the model result as a model result. Status: **answered in a declared model** (Result H.7, `PROVED · 2`). Adopted by the theory lead on 2026-10-05.

EG-07.c. Triage row 38 (`items[10]`, `reviews[10]`). The demand was «Sport echo: «tres partidas» (UTHA §2.2)», and the exact question: «Is UTHA an admitted immutable source (edition + page)? Or should the three energy items be restated against a GTPA v1.2 locator?» The reviewer held it as `HOLD_SOURCE_NOT_ADMITTED (UTHA §2.2.1)`.

Answer. (1) Source: UTHA v1.0 (H.0.1). The three items are at §2.2.1, p.21: «$P_{\text{total}} = P_{\text{defense}} + P_{\text{repair}} + P_{\text{external}}$», with P_{total} «finite and strictly conserved» and the parts «mutually exclusive modes». That edition is the adopted immutable source (H.0.1). (2) Mapping: the items enter Part H as Model H.9, a declared three-part budget. Its sum and its reallocation identity are arithmetic. Strict conservation and the reading as an organism's metabolism are not carried. The in-work rivals keep their own objects: Model 4.4's push, zero and reverse are phases of one control, not items of a budget (CT-07); v1.2's Ax 4.2 splits S_{total} into $L_{\text{physical}} + \beta \cdot L_{\text{informational}}$ (pp.30–31; kept as a declared cost, Definition 2.9); the switch of Result 2.8 has two modes. FORMACION's card may cite UTHA v1.0 p.21 as lineage and Model H.9 as a declared model, and neither as a law. Status: **answered**: source pinned, mapped to a declared model with no theorem. Adopted by the theory lead on 2026-10-05.

H.6 What Part H does not say, and what is left open

- Nothing about any person, group, brain, body or society; no diagnosis, therapy or advice. The readings of the notes about persons (CONS-02, MEM-6, ACT-5, LA-17 and the like) stay in the notes.
- No physics: no energy, power, heat, temperature, entropy, mass, momentum, impedance, resonance, Landauer bound or Noether charge as a physical object. UTHA's names appear only as names.
- No ledger claim and no sector admission. Σ_H is a typed candidate (Definition H.3).
- Open: the four `OPEN` rows of Σ_H; in Model H.10, where Result H.14 settled Conjectures H.12 and H.13, the size of what plasticity gains, the best law of $\mathcal{A}$, whether the descent, raising or explored law beats the best fixed constitution, and an explicit witness where the fourth-moment condition of Result H.14 (b) fails; the full basins of Model H.5 with a moving territory; the noisy Class A model beyond Result H.6 (c) (whether $E_{\text{base}} \to \infty$ almost surely); a declared model for UTHA's hyperbolic discounting (H-22).
- Done with the adoption of 2026-10-05: one independent read of this text (13 findings, all fixed), the crosswalk line and GTPA v2 §5's line on EG-06.b and EG-07.c, and ROADMAP 16. The triage record of 2026-09-25 is a dated record and is not edited. The Reino moves of Annex H-A were executed on 2026-10-06 (Annex H-A, Reino moves).

Annex H-A. UTHA v1.0 → Part H

One row per statement of UTHA v1.0: 68 statements, from the abstract, the index, chapters 1–14 and Appendix A, with their PDF pages. The cover, the epigraph and the correspondence and rights notes are not statements. Dispositions:

- `PART H`: enters Part H, at the place named.
- `GTPA v2.0`: settled by GTPA v2.0, positively or negatively, at the place named.
- `REINO`: goes to the Modelo del Reino as lineage (the architecture record's [LINAJE-UTHA], which the Reino writes `[LINAJE: UTHA …]`). "Carried" means a Reino file carried it before Part H; "proposed" names where Part H placed it. The proposed moves were executed on 2026-10-06 (Reino moves, below).
- `DROPPED`: dropped as a statement, because it is physics, a claim about people that cannot be proved, or ill-posed.

A row may carry more than one disposition, the first being the main one. Rows are a reading (level 4); each cited result keeps its own level. Reino files are named by their number under `modelo-del-reino/` (5.4 is `05-ANEXOS/5.4-diagnostico-patologias.md`); reino-compacto is `modelo-del-reino/reino-compacto.md`, and the MR-K cards are those of `modelo-del-reino/projections/MR-P5-COMPANION-FORMAL-01.md`.

#	UTHA v1.0 (§, p.)	Statement	Disposition	Where and why
H-01	index header, p.7; abstract, p.3	TECD (Law) → GTPA (Geometry) → UTHA (Instance); UTHA is GTPA's "Layer 4" Class A instance	`PART H`, `GTPA v2.0`	Definition H.1: one GTPA, a typed candidate sector (C0111), not admitted; TECD and its generator are not carried (GTPA v2 §0.4, Ch 0–1 row; §5)
H-02	abstract §1, p.3; summary, p.7; §14.1, p.67	Class A (plastic, human) is "the Optimal Strategy for Chaos"; the antifragile agent survives because of chaos	`PART H`	Conjecture H.13 holds and its rival H.12 fails (Result H.14): in Model H.10 a plastic law beats every fixed constitution at every switching rate; "optimal" is not shown (the best law of $\mathcal{A}$ and the size of the gain are open); settled around it: CT-12, Results H.6–H.8, H.11 (b)
H-03	abstract §1, p.3; §14.1, p.67	Class B ($d\Theta/dt = 0$, anchored) is optimal in static or ergodic environments and suffers "Brittle Fracture" in radical uncertainty	`PART H`, `GTPA v2.0`, `DROPPED`	static territory with a known rate: Class B optimal among all policies (Result H.11 (b)); in Model H.10, stationary and ergodic, Class B is not optimal among the laws of $\mathcal{A}$ (Result H.14), so the ergodic half fails there; for each fixed constitution there is a bounded constant rate ($a \equiv kr^2 + 1$) that defeats it, while the raising law keeps every bounded territory (Result H.8 (a)); Class B gives coupling, not stability (CT-13); "brittle fracture" is materials physics

#	UTHA v1.0 (§, p.)	Statement	Disposition	Where and why
H-04	abstract §1, p.4; summary, p.7	where `grad(J)` changes faster than the relaxation time, plasticity is the only viable strategy, "the necessary Hamiltonian solution"	`PART H`, `DROPPED`	Result H.8 (a): without noise a raising law keeps every bounded territory at any speed of change, and no fixed constitution does; with noise its authority does not settle (H.8 (b)); the regime clause is H.13⁺, which holds in Model H.10 (Result H.14) but not as a regime: plasticity pays there at every switching rate; and, with "viable" read as finite cost, it is not the only viable strategy there, since every fixed gain in the stable set has finite cost (Result H.11 (a)); "Hamiltonian" not carried (U-11; §0.3)
H-05	abstract §1, p.4	high-temperature plasticity rewrites the objective ($d\Theta/dt \neq 0$) by Ricci flow in response to shock	`GTPA v2.0`, `PART H`, `DROPPED`	Ricci flow is Model 4.9's optional flow of a model metric, not of Θ (CO-04); a response law for Θ is declared (Model H.5 (ii), (iv)); noise is not a temperature (Definition 1.5)
H-06	§1.1.1, p.14; abstract §2, p.4	$V_{\text{identity}} = \frac{1}{2}k_{\text{id}}\|q - q_{\text{self}}\|^2_G$; the Self as a virtual attractor; k_{id} as "Ego Strength"	`REINO`, `GTPA v2.0`	carried: MR-K12 `[LINAJE: UTHA Cap.1 §1.1.1]`, also 0.1, 1.2, 4.2, 5.1 and reino-compacto; formally a declared potential (Model 4.5); ledger: identity as a fixed point is an analogy (C0020), C0017, C0018; human identity UNRESOLVED
H-07	§1.1.2, p.15	`F_total = −grad(J_survival) − grad(V_identity)`; willpower raises k_{id}; the natural drift is "Hamiltonian Flow"	`GTPA v2.0`, `REINO`, `DROPPED`	a sum of gradients is a gradient and an added potential is a declared model (Model 4.5; CT-08 (a)); willpower as stiffness carried as analogy (1.2-identidad `[LINAJE: UTHA L322 — ANALOGÍA …]`); "Hamiltonian" not carried (U-11; §0.3)
H-08	§1.1.3, p.15	$B_{\text{effective}} = B_{\text{biological}} \cdot \text{Sensitivity}(q_{\text{self}})$; attention creates a "Phantom Gradient"	`GTPA v2.0`, `REINO`	a multiplicative coupling is a declared B (Definition 2.1); $\Omega = \|\nabla J_{\text{est}}\|^2 \cdot \text{Sensitivity}$ is a definition once `Sensitivity` is declared (Model 4.10); carried: reino-compacto, 1.2, 5.1 `[LINAJE: UTHA Cap.1 §1.1.3, sin veredicto: T-11]`; 5.6 F37
H-09	§1.2–§1.2.1, p.16; abstract §2	Axiom of Federalism; Kuramoto subsystems; dissociation for $K < K_{\text{critical}}$, agency above	`REINO`, `DROPPED`	carried: MR-K13 `[LINAJE: UTHA Cap.1 §1.2]` (phase toy), also 3.7, 4.1, 4.3, 5.1, 5.2, 5.3, 5.5, reino-compacto; 5.6 P2 (now pointing to this annex); the Kuramoto transition is a standard fact outside GTPA; dissociation and agency as states of a person: claims about people
H-10	§1.2.2, p.17	an Arbitration Core (mapped to the ACC) raises K above K_{critical}; $P_{\text{metabolic}} \sim K^2$	`REINO`, `DROPPED`	carried: $P_{\text{coupling}} \sim K^2$ in reino-compacto, 3.6, 5.1 `[LINAJE: UTHA Cap.1 §1.2.2]`; 5.6 F37; the ACC mapping is a claim about people (§0.1); metabolic power is physics
H-11	§1.2.3, pp.17–18	"Aliasing Theorem": the cortex (~40 Hz) oversamples the soma (~0.05 Hz); anxiety is aliasing noise; calm is a low-pass filter	`DROPPED`, `REINO`	no sampling model is declared; the frequencies and the anxiety reading are claims about people; carried as lineage and analogy: 5.4 D.2, 5.1, reino-compacto `[LINAJE: UTHA Cap.1 §1.2.3]`

#	UTHA v1.0 (§, p.)	Statement	Disposition	Where and why
H-12	§1.3–§1.3.2, pp.18–19	$P_{\text{sim}} \sim \beta_{\text{eff}} N^3$; recursion saturates at $L_{\max} \sim$ 2–3 on ~20 W; the opacity of the Self is a hardware safety feature	`GTPA v2.0`, `REINO`	dropped as law: the cubic barrier fails (D-06), the resolution capacity fails (D-07), Thm 2.1 is ill-posed (D-05), opacity as law is ill-posed (D-03); opacity kept as a declared proxy (Model 4.10); ledger REJECTED C0016, C0019, C0021; $L_{\max}$ stays illustrative in the Reino (5.6 D5; reino-compacto `[LINAJE: UTHA §1.3.2; ilustrativo, no medido]`; also 0.1, 3.4, 5.1)
H-13	abstract §2, p.4	willpower is the metabolic work that holds a phase-locked state ($K > K_{\text{critical}}$)	`DROPPED`	ill-posed: UTHA defines willpower in three incompatible ways (abstract §2; §1.1.2, raising k_{id}; §6.1, impedance inversion); metabolic work is physics
H-14	§2.1.1, p.20; abstract §4, p.5	perception as frequency tuning; at resonance reactive power → 0; understanding as conjugate impedances; a true belief as minimum dissipation	`GTPA v2.0`, `REINO`	fails: at a mode's frequency the minimum is the damping (B-08; EG-04.1); Z is a fixed matrix, not a frequency response; "equal frequencies" is ill-posed (CO-07; EG-04.2); survival is no proof of truth (D-04); kept as an [MR] reading (3.1; 5.2 `[LINAJE: UTHA Cap.2 §2.1]`)
H-15	§2.2.1, p.21; index §2.2	$P_{\text{total}} = P_{\text{defense}} + P_{\text{repair}} + P_{\text{external}}$, finite and strictly conserved, mutually exclusive modes	`PART H`, `REINO`, `DROPPED`	Model H.9 (declared three-part budget; EG-07.c); carried: MR-K05 `[LINAJE: UTHA Cap.2 §2.2.1]`, also 0.1, 4.3, 5.1, 5.2, 5.5, reino-compacto; strict conservation of metabolic power is physics
H-16	§2.2.2, pp.21–22	Prob(Danger) gates the current between defence and repair; $\Delta P_{\text{repair}} = -\Delta P_{\text{defense}}$; damped relaxation; placebo and nocebo	`PART H`, `REINO`, `DROPPED`	the reallocation is arithmetic in Model H.9; the gate is carried as lineage (5.1, reino-compacto `[LINAJE: UTHA Cap.2 §2.2.1 y §2.2.2]`); placebo, the gate's "mass" and its damped curve are claims about people and physics
H-17	§2.3.1, p.22	Law of Narrative Continuity: the internal manifold must stay connected and simply connected; a causal gap is a puncture (dissociation, psychosis)	`DROPPED`, `REINO` (proposed)	ill-posed: no internal manifold with that topology is declared; ledger REJECTED C0016, C0019; proposed: 3.3-artista §4 (Confabulación)
H-18	§2.3.2, p.22	`Cost(Continuity_Break) >> Cost(Factual_Error)`; the interpreter confabulates	`DROPPED`, `REINO` (proposed)	a claim about people with no model; in a declared model a miscalibrated self-model can make the decision worse (CONS-08; Result H.4); proposed as H-17
H-19	§3.1.1, p.24	$\beta_{\text{eff}} = \beta_{\min} \cdot Z$, $\beta_{\min} = k_B T \ln 2$; $Z \to 1$ is "superconducting"; Z grows exponentially with $\|\varepsilon\|$; ego depletion as inefficiency	`GTPA v2.0`, `REINO`	the Landauer floor is dropped with the physics (Ax 4.1, Def 4.1; §5); β is a declared weight (Definition 2.9); $Z \succeq \lambda_{\text{effort}} I$ is a matrix (Definition 2.4), so the product needs a declared scalar reading (5.6 D6); carried literally: 5.1, 5.2, 5.5, reino-compacto `[LINAJE: UTHA Cap.3 §3.1.1, literal; sin veredicto, T-11]`, also 0.0, 4.2
H-20	§3.2.1, p.25	$d\tau/dt = k\|d\text{State}/dt\|$; time dilates in danger and collapses in flow	`DROPPED`, `REINO` (proposed)	the base has no clock (Definition 1.5; U-13 (c)); the phenomenology is a claim about people; proposed: 2.3 §3 (Relatividad Temporal)

#	UTHA v1.0 (§, p.)	Statement	Disposition	Where and why
H-21	§3.3.1, p.26	$P(\text{Survival}_t) = \exp(-\int_0^t h)$, with h depending on Σ_{env}	`DROPPED`	a standard identity of survival analysis, outside GTPA; a human hazard rate is an empirical claim; no GTPA object is needed (CT-15's discounted compensation is tangential)
H-22	§3.3.2, Thm 3.1, p.27	hyperbolic discounting is the correct Bayesian response to a non-ergodic environment	`DROPPED`	ill-posed as written: no proof, and the hazard's uncertainty is not typed; it could re-enter as a declared model with a proof (a known route is an uncertain constant hazard, Sozou 1998: a pointer, not checked here); the horizon stays a declared parameter (Model 4.8)
H-23	§4.1.1, p.28	flow: $F_{\text{correction}} \to 0$, $Z \to 1$, $\beta_{\text{eff}} \to \beta_{\text{min}}$, subjective time collapses, the Self goes offline; mastery reshapes G_{model}	`GTPA v2.0`, `REINO`	zero demand gives zero action (Result 2.5); $u = 0$ recovers the base exactly (Definition 2.1); "$Z \to 1$" fails ($Z \succeq \lambda_{\text{effort}} I$; B-08); reshaping a metric is the optional Model 4.9; carried: 4.1 ($Z \to 1$ as [MR]), MR-K15 and 5.6 D7 (Mode I)
H-24	§4.2.1, p.29	"Zombie Mode": the Arbitration Core runs Mode I ~95% of the time; routine in Mode II would exceed P_{max}	`DROPPED`, `REINO`	an empirical claim about people; the regime is carried with Mode I (MR-K15 `[LINAJE: UTHA §3 Mode I-III]`; 5.6 D7)
H-25	§4.2.2, Thm 4.1, p.30	Logic = Projection(Intuition) + $\varepsilon_{\text{quantization}}$ ($d \sim 10^5 \to 1$); trusting the gut is physically rational	`GTPA v2.0`, `PART H`, `DROPPED`	kernel: a lower-dimensional continuous proxy merges states (Model 4.10; D-03); exact code losses in declared models (MEM-2–MEM-4; Result H.4); the dimensions and the "gut" claim are claims about people
H-26	§5.1, p.31	addiction: reward decouples from survival; a dissipative vortex; past an "event horizon" escape needs external energy	`GTPA v2.0`, `REINO`, `DROPPED`	without noise, leaving a well needs added control (CT-08 (b), Model 4.5); with noise, the first-order base leaves the declared well of Model 4.5 without control, the mean time spent in it growing like $e^{\Delta J/\varepsilon}$ on the logarithmic scale, $\varepsilon \ln E[\tau_D] \to \Delta J$ as $\varepsilon \to 0$ (Model 4.5, since v2.2; AB-5 (b); U-13), so its kernel, escape needs added control, fails on the noisy base; a vortex on the base needs A or a mobility that does not commute with the Hessian (CT-08 (a), (c)); internal error → 0 with external divergence is CT-12 (b); carried: 5.4 `[LINAJE: UTHA §5.1]`, also 3.7, 5.1, 5.2, reino-compacto; ATP and the horizon are physics
H-27	§5.2, p.32	compulsion is a stable limit cycle; a low signal-to-noise ratio keeps the stop signal below threshold	`GTPA v2.0`, `REINO`, `DROPPED`	without noise and with $A = 0$, J decreases strictly wherever $G\nabla J \neq 0$, so the base has no nonconstant periodic orbit (from U-04; cycles need A); a stopping rule is declared (Def 10.3; Model 4.7); carried: 5.4 `[LINAJE: UTHA §5.2]`; serotonin is a claim about people
H-28	§5.3, pp.33–34	ludomania: chasing reward prediction error, not value; a hijacked Genesis Operator pursuing dK/dt in noise	`REINO`, `GTPA v2.0`, `DROPPED`	carried: 5.4 `[LINAJE: UTHA §5.3]`; Kolmogorov complexity is not computable, so a model declares a measurable substitute (Model 4.10); the RPE reading is a claim about people

#	UTHA v1.0 (§, p.)	Statement	Disposition	Where and why
H-29	§6.1.1, p.35; App A.1, p.71	$u_{\text{opt}} = Z^{-1}F_{\text{wish}}$ with $Z \sim B^{\top}B$; an ill-conditioned B makes Z^{-1} huge	`GTPA v2.0`, `REINO`	replaced by the control law $u^{*} = Z^{-1}B^{\top}G_{\text{reg}}F_{\text{req}}$, $Z = B^{\top}G_{\text{reg}}B + \lambda_{\text{effort}}\,I$ (Definition 2.4, Result 2.5; CO-01, CT-05); UTHA's form mismatches dimensions unless $k = n$ and omits λ_{effort}; v1.2's printed forms fail (EG-02.1, EG-03.8); carried: MR-K09 `[LINAJE: UTHA Cap.6 §6.1.1]`, also 5.1, 5.2, reino-compacto
H-30	§6.1.2, p.36	tuning protocol: minimize $\|Z(\omega)\|$; at resonance reactive impedance drops to zero; tap the pendulum at the apex	`GTPA v2.0`, `REINO`	fails: the minimum is the damping, and Z has no frequency (B-08; EG-04.1); the "pause" has a model form: no interval of zero action under quadratic effort (CT-06), push, zero, reverse under an L1 cost with a cap (CT-07; Model 4.4); carried as analogy in MR-K09
H-31	§6.2.1, p.37; abstract Mode II, p.5	ego depletion: B loses rank, $\det Z \to 0$, u diverges, cost $\to \infty$	`GTPA v2.0`, `REINO`	fails: $Z \succeq \lambda_{\text{effort}}\,I$ is never singular and $\|u^{*}\| \leq \|B^{\top}G_{\text{reg}}F_{\text{req}}\|/\lambda_{\text{effort}}$ (Result 2.5; §5; EG-02.1, EG-03.8); without λ_{effort} the least-norm solution is finite (Result 2.6); carried with this correction: 5.4 D.1 `[LINAJE: UTHA §6.2]`, MR-K09
H-32	§6.2.2, p.37	thermal fuse: if cost > $P_{\max}$, a hard interrupt sets $u \to 0$ and reverts to Mode I; diets fail at night	`PART H`, `REINO`, `DROPPED`	the declared brake of Model H.5 (iii); in Model 4.7 it feeds the constitution's drift (Result H.7; EG-06.b); Def 10.3 is a declared rule with no safety guarantee (Model 4.7); carried: 5.4 D.1 `[LINAJE: UTHA §6.2.2]`; diets and the prefrontal cortex are claims about people
H-33	§7.1, pp.39–40	$T_{\text{soc}} \sim \Sigma M_{\text{cog}}\, v\, v^{\top}$; the hero senses social curvature ($\|\text{grad}\, V_{\text{social}}\| >> \|\text{grad}\, V_{\text{survival}}\|$) and blows as a fuse that lowers collective entropy	`GTPA v2.0`, `REINO`, `DROPPED`	T_{soc} is ill-posed (EG-03.10); M_{cog} is not a mass (EG-02.4); a social term is a declared potential (Model 4.5); T_{soc} carried (5.1, 5.2, 5.5 `[LINAJE: UTHA Cap.7 §7.1.1]`); heroism proposed: 4.4-mundo-exterior; entropy and the fuse are physics
H-34	§7.2, p.41	faith: at `grad(J) ~ 0`, $F_{\text{net}} = 0 \Rightarrow dq/dt = 0$ (Buridan); a virtual target creates a gradient as a metabolic loan, repaid if a real gradient is found	`GTPA v2.0`, `REINO` (proposed)	the deadlock fails on the noisy base, where noise moves the state (U-05, U-06); only the noiseless base stops at critical points (U-04); a virtual target is a declared demand or potential (Definitions 2.1, 2.4; Model 4.5); acting in order to learn: Model 4.11, ACT-1–ACT-4 (Result H.4); the loan is a Reino hypothesis (architecture record) not carried before Part H; proposed: 1.1-rol-del-soberano §2
H-35	§8.1.1, p.42	$J_{\text{expansion}} = \text{Minimize}(E_{\text{pred}}) + \text{Maximize}(dK(\text{Stream})/dt)$; boredom triggers a risk injection	`GTPA v2.0`, `REINO`, `DROPPED`	ill-posed as written (a minimum and a maximum with no weight); K is not computable (Model 4.10; Constraint 11.1 kept as a declared model); dimensions are fixed (Definition 1.2); carried: 5.2, 5.4 `[LINAJE: UTHA Cap.8 §8.1]`
H-36	§8.1.2, p.43	`Cost_Inflation ≈ β_eff((N+1)³ − N³) ≈ 3β_eff N²`; stress kills curiosity	`GTPA v2.0`, `REINO`	the premise fails (no cubic law: D-06; §5); the arithmetic $(N+1)^3 - N^3 = 3N^2 + 3N + 1$ is right (level 2); the Reino keeps a fenced cubic reading (decision O-1; MR-K05; 3.4, 4.3)

#	UTHA v1.0 (§, p.)	Statement	Disposition	Where and why
H-37	§8.2, p.44	play is a subsidized state, $\Gamma_{\text{risk}} \to 0$; `Cost_Play_Total = Cost_Internal + Cost_External(Sponsor)`	`REINO` (proposed)	no GTPA object (Model 4.10's budget is tangential); proposed: 4.1-jardines §B (Juego Puro)
H-38	§8.3, pp.44–45	sublimation: `G_external = Project(G_internal_chaos)` stabilizes G_{model}	`REINO`	carried: 5.2 `[LINAJE: UTHA §8.3]` (5.1, 5.4, 5.5 also name it); no GTPA object (an agent boundary is a declared rule, Model 4.10)
H-39	§9.1.1, p.46	sleep: REM injects T_{info} to escape local minima ($B \to 0$); slow-wave sleep minimizes $E_{\text{pred}} + \lambda \cdot \text{Complexity}$	`GTPA v2.0`, `REINO`, `DROPPED`	noise-driven exit times (U-13); a noise schedule is declared and lags (Definition 2.9; B-09); $B \to 0$ is the ablation (Definition 2.1); complexity is a declared measure (Model 4.10); carried: 2.3, 4.1 `[LINAJE: UTHA §9.1.1]`; T_{info} is not a temperature; the biology is a claim about people
H-40	§9.2, p.47	awe: $\text{Scale}(O) >> \text{Scale}(I)$ collapses the recursive stack to L0 and resets W_{ego} to zero	`REINO` (proposed), `DROPPED`	the recursion-depth premise falls (D-06, D-07; H-12); 1.2-identidad already cites W_{ego} (UTHA §9.2) as nearest lineage; proposed there
H-41	§9.3, p.48	grief: Ricci flow around a deleted attractor; $J_{\text{grief}} = \int(\beta\|dG/dt\|^2 + \alpha\|PE\|)$; an exponential decay is optimal	`GTPA v2.0`, `REINO` (proposed), `DROPPED`	Ricci flow is the optional Model 4.9 (CO-02 (c), CO-03); it can end in finite time and does not flatten in general (CO-06); J_{grief} is ill-posed (PE is not a declared function of G) and its exponential optimum is not derived; grief stays in the Reino (architecture record): 4.3-taller (Dique Seco) mentioned "proceso de duelo" without a tag; proposed there (tagged on 2026-10-06)
H-42	§10.1.1, p.50	trauma: $g_{rr} \sim 1/(\|q - q_{\text{trauma}}\| - R_s)$; infinite geodesic distance at a "Schwarzschild Radius"; flashbacks frozen at $t = 0$; dissociation as lensing	`DROPPED`, `REINO`	a physics analogy (CLOSED); and the printed metric gives a finite distance, $2\sqrt{r_0}$ from R_s to $R_s + r_0$ (level 2), so "infinite distance" does not follow (divergence needs $g_{rr} \sim (r - R_s)^{-2}$); carried literally: 3.7, 5.1, 5.4, reino-compacto `[LINAJE: UTHA §10.1]`; a note was proposed (added on 2026-10-06, except in 3.7)
H-43	§10.1.2, pp.50–51	topological hysteresis: `Work_Healing >> Work_Damage`; healing is Ricci flow, driven by therapy, back to a Euclidean baseline	`GTPA v2.0`, `REINO`	Ricci flow does not relax to a flat metric in general (CO-06); exit times are asymmetric (U-13); the work inequality has no object; v1.2's hysteresis is another object (EG-05.c, lab HOLD); carried: 4.3, 5.1, 5.4 `[LINAJE: … UTHA §10.1.2]`
H-44	§10.2.1, Thm 10.1, p.51	Noether: a vetoed impulse cannot vanish (constraint forces do no work) and turns into L_{spin}, the Shadow	`GTPA v2.0`, `REINO`	fails: momentum is dropped with the physics (§5; U-11); conservation is a condition, not a default (U-14); in a damped declared model Ψ decreases at rate $\zeta\|\dot{x}\|^2$ (CT-08 (b)); $u = 0$ removes a drift term and recovers the base exactly (Definition 2.1): it is not a constraint force; carried: 0.1, 3.7, 5.1, 5.4, reino-compacto `[LINAJE: UTHA §10.2]`
H-45	§10.2.2, p.52	psychic torque $\tau_{\text{shadow}} = r \times F_{\text{repressed}}$: precession (slips), projection (tidal locking onto a scapegoat), integration	`DROPPED`, `REINO`	physics and claims about people; carried literally: 5.1 `[LINAJE: UTHA Cap.10 §10.2.1 y §10.2.2, literal]`

#	UTHA v1.0 (§, p.)	Statement	Disposition	Where and why
H-46	§11.1.1, Ax 11.1, p.54; abstract §5	language as an ADC, $s = \text{Quantize}(\text{Project}(q))$; $f_{\text{speech}} << 2 \cdot BW_{\text{thought}}$, so aliasing and $\varepsilon_{\text{quant}}$ are irreducible	`GTPA v2.0`, `PART H`, `REINO`, `DROPPED`	kernel: Model 4.10's opacity; exact code losses (MEM-2–MEM-4; Result H.4); the numbers (10^{14} synapses, 39 bit/s) are empirical; carried: 3.6, 5.1, 5.2, reino-compacto `[LINAJE: UTHA Cap.11 §11.1]`, with the numbers marked as external data
H-47	§11.1.2, p.55	`v_received = Parallel_Transport(v_sent, G_i -> G_j)`: meaning rotates with the difference in curvature	`DROPPED`, `REINO` (proposed)	ill-typed: parallel transport runs inside one connection, not between two metrics; in R^n with a constant metric it is the identity (CROSSWALK §3, Ax 9.1); proposed: 3.6 §3
H-48	§11.2, pp.55–56; abstract §5	ritual as forced oscillation: entrainment $\nu_i \to \nu_{\text{driver}}$, Hebbian alignment of G_{model}, zero quantization noise	`REINO`, `DROPPED`	ritual stays in the Reino (architecture record): carried as an analog bypass in 3.6, 5.1, reino-compacto `[LINAJE: UTHA Cap.11 §11.1 y §11.2.1]`; entrainment by a phase that does not respond is a standard fact, worked for a declared pair in Model 4.6; "zero quantization noise" is a claim
H-49	§11.3, pp.56–57	laughter is a broadcast cooling signal, $d(T_{\text{social}})/dt << 0$	`DROPPED`, `REINO` (proposed)	physics (a temperature) and a claim about people; proposed: 3.6 §3
H-50	§12.1.1, p.58	Curvature Transfer Law `R_B(X) ~ coupling_coeff · R_A(X)`: desire is induced	`REINO`, `GTPA v2.0`	carried: 5.2 `[LINAJE: UTHA Cap.12 §12.1]`; an influence term is a declared potential (Model 4.5)
H-51	§12.1.2, Ax 12.1, p.59; abstract §5	love = parallel cores and orthogonal actions; `Vol_pair = Vol_A ∧ Vol_B`; the dyad uses the cubic law	`GTPA v2.0`, `REINO`	at a fixed total the effort is least when the intents are equal (CT-09); full correction needs intents that positively span, so some pair with $C_{ij} < 0$ (CT-10), and an orthogonal pair does not positively span; the cubic leverage fails (D-06); the Reino has reopened the question (5.6 P10)
H-52	§12.2.1, pp.58–59	Social Zeno Effect: being observed collapses drift into a persona eigenstate	`DROPPED`, `REINO`	quantum physics is CLOSED; ledger: a mask may be described as a second-order model of how another agent models the focal agent (C0022, ADMITTED MET); carried: 5.2 `[LINAJE: UTHA Cap.12 §12.2]`
H-53	§12.2.2, p.60; App A.4, p.73	$W_{\text{social}} = T_{\text{info}} \cdot D_K L(P\|Q)$; the introvert's tax; privacy as cooling; burnout under total observation	`REINO`, `DROPPED`	carried: MR-K17 `[ANALOGÍA; LINAJE: UTHA Cap.12 §12.2.2]`, also 0.1, 3.6, 5.1, 5.2, 5.4; a rival form on HOLD: mask cost proportional to the squared mismatch (C0023, EMP-HYP); $D_K L \geq 0$ is standard; T_{info} is not a temperature (Definition 2.9); introversion and burnout are claims about people
H-54	§13.1, Thm 13.1, p.62	$P_{\text{total}} = \Sigma\, \beta_{\text{eff}}\, \dim(M_j)^3 \leq P_{\max}$ gives $N_{\max} \approx 150$; out-group agents compressed to scalars	`GTPA v2.0`, `DROPPED`	the derivation fails (D-06, D-07); the largest affordable size is the inversion of a declared cost (Model 4.10; D-05); 150 is an empirical input, not a result; dehumanization is a claim about people
H-55	§13.2, Mech. 13.1, pp.63–64	the scapegoat: all project their shadow momentum onto one victim; its elimination drops S_{total} discontinuously	`DROPPED`	physics (momentum, entropy) and a claim about groups; no Reino home is proposed (it would read harm as a function)

#	UTHA v1.0 (§, p.)	Statement	Disposition	Where and why
H-56	§13.3, p.64	cruelty as an entropy pump: $dS_{\text{victim}}/dt >>> 0$ cools the sadist	`DROPPED`	physics and a claim about people; no Reino home is proposed (same reason)
H-57	§13.4, Def 13.1, p.64	forgiveness as active damping, $W_{\text{forgive}} = ½\, m\, v^2_{\text{attack}}$	`DROPPED`, `REINO` (proposed)	mass and kinetic energy are dropped (§5; EG-02.4); declared damping is CT-08 (b); the notes' forgiveness reading (LA-17) stays separate; proposed: 3.6 §3
H-58	§13.5, p.65; abstract §5	Kuramoto social phases: gas ($K < K_c$), crystal ($K \to \infty$, $C_{ij} \to 1$, zero resilience), fluid ($K \approx K_c$, an excited state)	`GTPA v2.0`, `REINO`	formal analog: full coherence gives one ray and a blind spot (CT-10); the maximizers of Eff·Res lie below full coherence under a sufficient condition (CT-11); v1.2's Thm 9.2, 9.3 ill-posed and "shatters instantly" withdrawn (EG-03.11, EG-03.2; §5); K_c is a standard fact outside GTPA; T_{social} is physics; carried: 4.4, 5.1, 5.2, 5.6, reino-compacto `[LINAJE: UTHA Cap.13 §13.5]`
H-59	§14.1, Mech. 14.1, pp.67–68; index, p.13; App A.2, p.72	the Class A agent is a thermodynamic diode that harvests plastic work and maximizes the Rectification Ratio η_{rect}	`DROPPED`	physics; and ill-posed: η_{rect} has three incompatible definitions (`Work_Absorbed / Entropy_Generated`, p.13; `Work_Absorbed_from_Stress / Structural_Complexity_Generated`, p.68; $W_{\text{absorbed}}/K_{\text{generated}}$, p.72)
H-60	§14.2, Ax 14.1, p.68	Darwinian in hardware, Lamarckian in topology: artifacts impose $G_{\text{child}} \to G_{\text{template}}$	`DROPPED`, `REINO` (proposed)	no object; a claim about evolution; proposed: 4.2-gran-obra
H-61	§14.3.1, p.69; abstract	$d(S_{\text{environment}})/dt \geq k_{\text{dissipation}} \cdot d(K_{\text{internal}})/dt$: intelligence must export disorder	`DROPPED`	thermodynamics is CLOSED; K is not computable (Model 4.10)
H-62	abstract, p.3	one variational principle: maximize η_{rect} subject to the Entropy Exclusion Principle	`DROPPED`	ill-posed (H-59, H-61); a different error from v1.2's "maximizing" S_{total} (an erratum, CT-12 (e))
H-63	abstract §3, p.4; App A.3, p.72	Mode I: $F_{\text{net}} \approx 0$, $Z(\omega) \to 1$, $u = 0$, $dq/dt = X_H(q)$ (pure Hamiltonian drift), ~95% of existence	`GTPA v2.0`, `PART H`, `REINO`	$u = 0$ is the ablation, which recovers the base exactly (Definition 2.1; C0107); the drift is $(A - G)\nabla J$ with noise, not X_H (§0.3; U-11); $Z(\omega) \to 1$ fails (B-08); the brake of Model H.5 (iii) is this ablation on an interval; carried: MR-K15, 5.6 D7 `[LINAJE: UTHA §3, Modos I–III]`
H-64	abstract §3, p.5; App A.3, p.73	Mode II: $Bu = F_{\text{wish}}$, $u_{\text{opt}} = Z^{-1}F_{\text{wish}}$ by SVD, cost $\int \beta_{\text{eff}}\|u\|^2$; $\det Z \to 0$ trips the fuse	`GTPA v2.0`, `REINO`	the law, the effort metric and the SVD: Definition 2.4, Results 2.5, 2.6; the singularity fails (Result 2.5); the fuse is a declared rule (Def 10.3; Model H.5 (iii)); carried: MR-K09, MR-K15
H-65	abstract §3, p.5; App A.3, p.73	Mode III: inflation $M_{\text{new}} = M_{\text{old}} \oplus R^1$, $P_{\text{inflation}} \sim \beta_{\text{eff}} N^3$, Ricci flow $dG/dt = -2R$, rectification $\Delta G = \eta_{\text{rect}} \int F_{\text{shock}} \cdot dq$	`GTPA v2.0`, `REINO`, `DROPPED`	the cubic cost fails (D-06); Ricci flow is the optional Model 4.9; inflation has no object (dimensions are fixed, Definition 1.2); rectification is ill-posed (H-59); carried: MR-K15, 5.6 D7

#	UTHA v1.0 (§, p.)	Statement	Disposition	Where and why
H-66	App A.1–A.2, pp.71–72	glossary: q, p, G, $B(q)$, $Z \approx B^{\top}GB$, T_{soc}; β_{eff}, $\beta_{\min} = k_B\,T\,\ln 2$, K_{couple}, K_{comp}, η_{rect}, T_{info} (a temperature), $D_{\text{met}} = P_{\text{reactive}}$, τ_{soma}; with physical units	`GTPA v2.0`, `DROPPED`	q: the base state (U-01); p: not carried (U-11); G: the mobility, a model metric is h (§0.3); B: Definition 2.1; Z: Definition 2.4, with $+\lambda_{\text{effort}}\,I$; T_{soc}: ill-posed (EG-03.10); β_{eff}, $\beta_{\min}$: dropped (β a declared weight); K_{couple}: Reino (MR-K13); K_{comp}: Model 4.10; η_{rect}: dropped (H-59); T_{info}: the schedule $\varepsilon_{\text{cog}}(t)$, not a temperature (Definition 2.9; B-09); D_{met}: a declared mismatch cost (Model 4.1; C0005); τ_{soma}: no object; units: none (U-01)
H-67	App A.4, p.73	Kuramoto synchronization; `Vol_pair = Vol_A ∧ Vol_B`; the KL tax $W_{\text{social}} = T_{\text{info}}\,\Sigma\,P\,\log(P/Q)$	`REINO`, `GTPA v2.0`	as H-09, H-51 and H-53: MR-K13, MR-K17, 3.6 `[LINAJE: UTHA glosario A.2/A.4]`; Model 4.6
H-68	abstract, conclusion, p.6; §14.3.2, pp.69–70	human "flaws" are safety valves and rectification circuits; humans are antifragile phase-transition engines and the most expensive objects in the universe	`REINO` (proposed), `DROPPED`	non-formal surplus (architecture record): proposed: 0.0-manifiesto; "the most expensive objects" and the heat death are physics

Counts. Each tag is counted once per row that carries it: `PART H` 11, `GTPA v2.0` 35, `REINO` 52 (40 carried, 12 proposed), `DROPPED` 45. By the first disposition of each row: 7, 29, 12 and 20. Eight rows are dropped with nothing else (H-13, H-21, H-22, H-55, H-56, H-59, H-61, H-62).

Reino moves, executed on 2026-10-06. The moves that this annex proposed were made in `modelo-del-reino/` on 2026-10-06 (research `22058b5`, merged at `6c0d41a`), for the theory lead and before the v2.1 release. They are text edits only: lineage tags and notes were added, and nothing was removed or renumbered. Each locator is UTHA v1.0's section and page (H.0.1). One independent read of the edits found 1 real error (an overstated status) and 4 minor points, all fixed; not re-read. The Reino keeps its own register of the moves in the section «Parte H de GTPA v2.1 — movimientos aplicados» of `05-ANEXOS/5.6-pendientes-teoricos.md`. Where they landed:

1. New lineage tags, for UTHA statements that the Reino used without a tag or not at all:
   - faith (H-34): 01-SOBERANO/1.1-rol-del-soberano.md §2.B (the Freno), a new entry, `[LINAJE: UTHA Cap.7 §7.2, p.41] [HIPÓTESIS]`;
   - grief (H-41): 04-ESPACIOS/4.3-taller.md §2 (Dique Seco), on "proceso de duelo", `[LINAJE: UTHA Cap.9 §9.3, p.48] [HIPÓTESIS]`;
   - play (H-37): 04-ESPACIOS/4.1-jardines.md §3.B (Juego Puro), a new sentence, `[LINAJE: UTHA Cap.8 §8.2, p.44] [ANALOGÍA]`;
   - awe (H-40): 01-SOBERANO/1.2-identidad.md §5.B (Parpadeo del Ego), a new entry, `[LINAJE: UTHA Cap.9 §9.2, p.47] [ANALOGÍA]`;
   - heroism (H-33): 04-ESPACIOS/4.4-mundo-exterior.md §2, a new entry, `[LINAJE: UTHA Cap.7 §7.1, pp.39–40] [ANALOGÍA]`;
   - narrative continuity and confabulation (H-17, H-18): 03-CONSEJO/3.3-artista.md §4, a new sentence, `[LINAJE: UTHA Cap.2 §2.3, p.22] [ANALOGÍA]`;
   - subjective time (H-20): 02-TERRITORIO/2.3-cronobiologia-y-tiempo.md §3, a new sentence, `[LINAJE: UTHA Cap.3 §3.2.1, p.25] [ANALOGÍA]`;
   - misunderstanding as a metric mismatch, laughter and forgiveness (H-47, H-49, H-57): 03-CONSEJO/3.6-embajador.md §3, a new entry among the lineage pieces, `[LINAJE: UTHA`

Cap.11 §11.1.2, p.55, §11.3, pp.56–57; Cap.13 §13.4, p.64] [ANALOGÍA];

- the extended phenotype (H-60): 04-ESPACIOS/4.2-gran-obra.md §2, a new sentence, [LINAJE: UTHA Cap.14 §14.2, p.68] [ANALOGÍA];
- the conclusions (H-68): 00-FUNDAMENTOS/0.0-manifiesto.md §6, item 3, a new sentence, [LINAJE: UTHA, conclusión, p.6] [HIPÓTESIS].

2. Notes on lineage the Reino already carried, each dated 2026-10-06:
   - trauma (H-42), in 5.1 §D.5, 5.4 §C.1 and reino-compacto §8: the printed metric gives a finite distance ($2\sqrt{r_0}$), so the "infinite distance" is not in the formula. In 3.7 the UTHA §10.1 lineage tags a passage that does not speak of distance, and it has no note;
   - the thermal fuse and the Freno (H-32), in 5.4 §D.1 and 1.1 §2.B: in the declared Model 4.7, holding $u = 0$ feeds the constitution's drift and, held long enough, turns recovery into collapse (Result H.7);
   - the three-part budget (H-15, H-16), in 5.1 §D.9 and reino-compacto §8: Model H.9 is its declared form; strict conservation is not carried; the source is UTHA v1.0 p.21. The card MR-K05 was not touched;
   - 5.6 P2 ("Parte H de UTHA", T-11) now points to this annex.
3. Nothing in the Reino was removed. The scapegoat and cruelty readings (H-55, H-56) are not in the Reino and were not brought into it.

Annex. Errata = CROSSWALK-v1.2.md

The errata annex of GTPA v2 is gtpa-v2/CROSSWALK-v1.2.md, cited, not copied. §1 answers the 29 (+2) questions on v1.2 in 34 verdict rows (16 ILL_POSED_AS_WRITTEN, 15 FAILS_WITH_COUNTEREXAMPLE, 3 HOLDS_UNDER_STATED_HYPOTHESES) plus one convention row, each with its check level; §2 is the v1.2 proof inventory; §3 gives each of the 71 numbered v1.2 statements its disposition; §4 gives short verdicts D-01..D-10 to the twelve statements that no question asked.

References

The proofs cited here carry their references in the block files: UFD §9, Block B §6, Block C theorems §6, Block C objects §6 and the crosswalk's "References and sources for §3 and §4". Cited directly in the core (§0–§5):

- M. Li, P. Vitányi, *An Introduction to Kolmogorov Complexity and Its Applications*, 3rd ed., Springer, 2008 — §2.3.
- A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002 — §2.B.
- S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed., Westview Press, 2015 — §9.3; §4.5 and §8.6 for Model 4.6 (§4.5, "Fireflies", checked against the book's table of contents on 2026-10-13; §8.6 not checked).
- I. Karatzas, S. E. Shreve, *Brownian Motion and Stochastic Calculus*, 2nd ed., Springer, 1991 — Thm 5.2.5.
- S. Karlin, H. M. Taylor, *A Second Course in Stochastic Processes*, Academic Press, 1981 — ch. 15 ("Diffusion Processes"; the chapter title checked on 2026-10-13, the section not given).

- N. Berglund, "Kramers' law: validity, derivations and generalisations", *Markov Processes and Related Fields* 19 (2013), 459–490 (arXiv:1106.5799; its bibliographic record checked on 2026-10-13, and eqs. (1.8), (3.15) against that arXiv version on 2026-10-14).
- K. J. Åström, B. Wittenmark, *Adaptive Control*, 2nd ed., Addison-Wesley, 1995.
- W. M. Wonham, "Some applications of stochastic differential equations to optimal nonlinear filtering", *J. SIAM Control Ser. A* 2 (1965), 347–369.
- R. S. Liptser, A. N. Shiryaev, *Statistics of Random Processes I: General Theory*, 2nd ed., Springer, 2001 — ch. 9.

Cited directly in Part H:

- P. Billingsley, *Convergence of Probability Measures*, 2nd ed., Wiley, 1999 — Thm 1.2 and its proof (a finite measure on a metric space is determined by its integrals of bounded Lipschitz functions; used in Result H.14).
- X. Mao, C. Yuan, *Stochastic Differential Equations with Markovian Switching*, Imperial College Press, 2006 — Ch. 1 (the generalized Itô formula for a diffusion whose coefficients switch with a Markov chain; used in Result H.14; the theorem number was not checked here).
- D. Revuz, M. Yor, *Continuous Martingales and Brownian Motion*, 3rd ed., Springer, 1999 — Ch. IV, Prop. 1.26 (a continuous local martingale converges on $\{\langle M\rangle_\infty < \infty\}$).
- H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, 2002 — Lemma 8.2 (Barbalat's lemma).
- P. D. Sozou, "On hyperbolic discounting and uncertain hazard rates", *Proc. R. Soc. Lond. B* 265 (1998), 2015–2020 — a pointer for H-22 only, not checked here (level 4).

Research notes used here (revisable notes, not part of this text and not ledger claims): `gtpa-v2/notas/informacion-y-margen-de-decision.md`, §§1–4 (delivered at `85d5b4bc36548ada44e1a06eb10495d0dc79fcfd`), for Model 4.11; `gtpa-v2/notas/libre-albedrio.md`, §§6–7 (delivered at `5379e72072eb4911398284ef62a257a611c012d9`), for the additions after Definition 2.1 and Result 2.5; `gtpa-v2/notas/asociacion-potencial-y-vida.md` (delivered at `af5cf3fc96258a2902bbeddcb28b3dde2e1af352`, adjudicated on 2026-10-05), for the research-note paragraph at the end of §4. For the additions of 2026-10-13, each adjudicated result by result by an independent AI context (record: `handoffs/integration/2026-10-13-v2.2-adjudicacion.md`) and read at research `1be97970c69f37f820666a5383742849c25f66fd`: `gtpa-v2/notas/reflexiones/disenar-el-terreno.md`, TER-1, TER-2, TER-3 (without discount), TER-4 (a) and TER-7 (a), for Model 4.1; `gtpa-v2/notas/reflexiones/atraccion-frente-a-borde.md`, AB-1, AB-2 (b) and AB-3 to AB-7, for Model 4.5; `gtpa-v2/notas/informacion-y-accion-colectiva.md`, IAC-4, `gtpa-v2/notas/fronteras/automodelo-colectivo.md`, AC-5, AC-7 and AC-9, and `gtpa-v2/notas/reflexiones/acoplamiento-entre-dos.md`, AC-1 to AC-7, for Model 4.6; `gtpa-v2/notas/informacion-y-accion-colectiva.md`, IAC-0 to IAC-3, `gtpa-v2/notas/conocer-y-cambiar-la-interfaz-de-accion.md`, CCI-1 to CCI-3, CCI-5 (with the witness of CCI-6) and CCI-7 to CCI-10, and `gtpa-v2/notas/reflexiones/calibrado-y-ganancia.md`, CG-1 to CG-5 (c) and CG-8, for Model 4.11. The notes automodelo-colectivo and acoplamiento-entre-dos both number their statements AC-n; the IDs are not renumbered, and this text cites each with its note. The five notes on which "`PROVED` in the note" rests (those cited above for Models 4.1, 4.5 and 4.6) are deposited with the GTPA v2.2.1 record as `RESEARCH-NOTES-v2.2.1.pdf`: public copies as the adjudication read them, at that commit, without their provenance line, their pointers to unpublished working files and, in one note, a closing remark on internal use; their status lines, check levels and what they say GTPA contains are those of the notes before the adjudication (their base is GTPA v2.1), and where this text states a result with other hypotheses, wording or level, this text governs.

Sources read for the core: research main `ff91d6f` (`ufd/UFD-NUCLEO.md`; the three block files; `gtpa-lab/THEORY-STATE.md`, N0–N3; `gtpa-lab/CLAIM-LEDGER.md` for GTPA-C0001, C0005, C0061,

C0067, C0068, C0075, C0079, C0086 and C0107–C0136, read only; `gtpa-lab/source-audits/GTPA-MR-T1-TYPED-BINDING-SOUNDNESS-RESULT-01.md` §4, read only; `formal-framework/governance/math-owner-binding-record-n0-v1.md`); branch `research/gtpa-v2/core/claude` for the crosswalk's §3–§4; house `modulos/formacion/fuentes/GTPA-v1.2.txt` (blob 32787369) for the v1.2 locators; house `modulos/teoria/gtpa/GTPA-OPERATIVA.md` and FORMACION's course files for what the consumers use.

Sources read for Part H: research `origin/main` at `380f5c4` (`gtpa-v2/GTPA-v2.md`, the three block files, `CROSSWALK-v1.2.md`, `gtpa-v2/notas/` for the notes memoria-automodelo-y-control, aprender-actuando-y-referencia-del-juicio, consciencia-y-calibracion-del-propio-mapa and libre-albedrio, `gtpa-lab/CLAIM-LEDGER.md` read only for C0016–C0023, C0060, C0107, C0111, C0126, GTPA-CENSUS-HUMAN-IDENTITY, GTPA-ZO-11 and GTPA-ZO-12, `modelo-del-reino/` read only (05-ANEXOS, projections/MR-P5, reino-compacto and the chapters named in Annex H-A), `legacy/UTHA.md` (blob `bafda475`), `handoffs/governance/2026-09-25-theory-architecture-proposal-v1.md`, `handoffs/governance/2026-09-25-formacion-eg01-eg10-eg19-triage-01.md` and .json, `handoffs/gtpa-lab/2026-08-02-utha-reabsorption-proposal.md`, `ROADMAP.md` row 16); the public UTHA v1.0 (H.0.1); house `modulos/formacion/fuentes/GTPA-v1.2.txt` (blob 32787369) for the v1.2 locators.